

Passive Network Designs for MmWave Wireless Transceivers

Taiyun Chi

May 2024

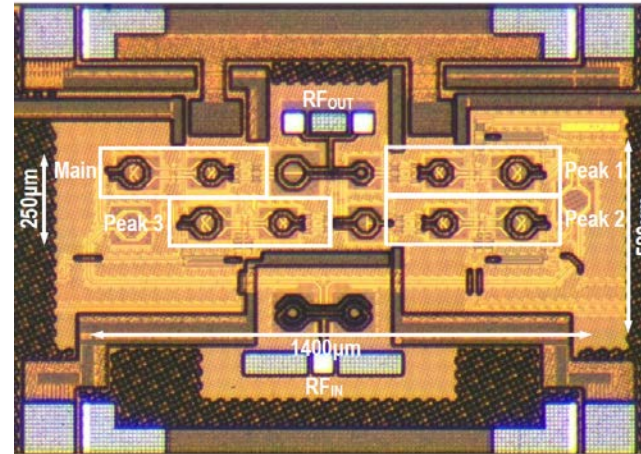


Rice Integrated Systems and Electromagnetics (RISE) Lab

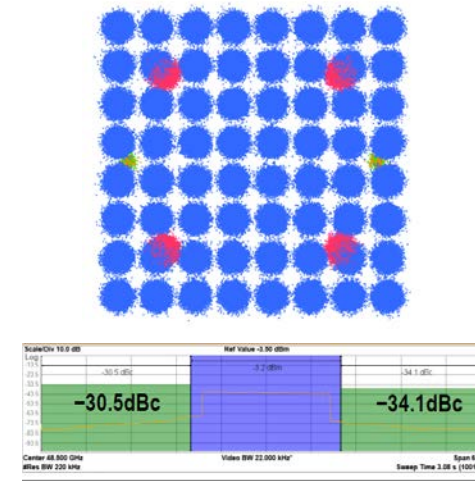


- Assistant Professor, Rice, 2019
- 8 Ph.D. Students + 1 Postdoc
- RF/mmWave IC for wireless
- Systems for neuro-engineering
- <https://chilab.info/>

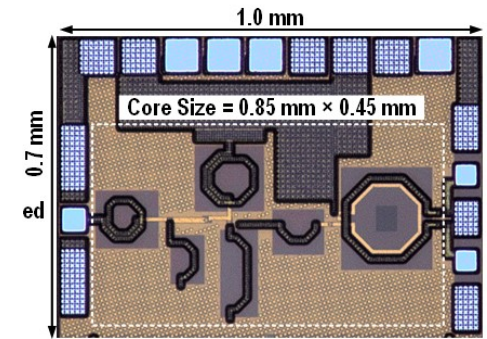
- 47 GHz 4-Way Doherty PA for 2000MHz 5G NR OFDM



2024 ISSCC

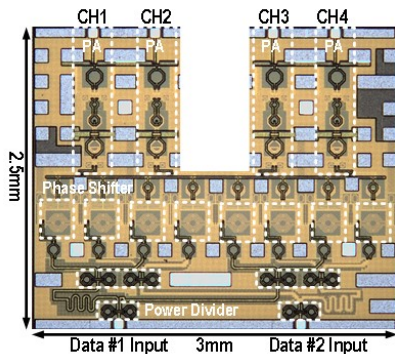


- 27-46 GHz LNA for Multi-Band 5G



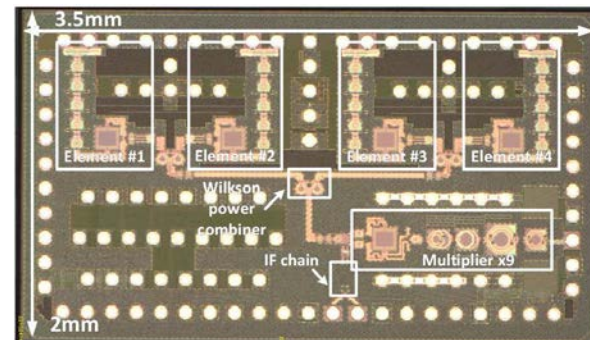
2021 IEEE IMS Advanced Practice Paper Award Finalist

- 28 GHz Hybrid Beamformer for Multi-User MIMO



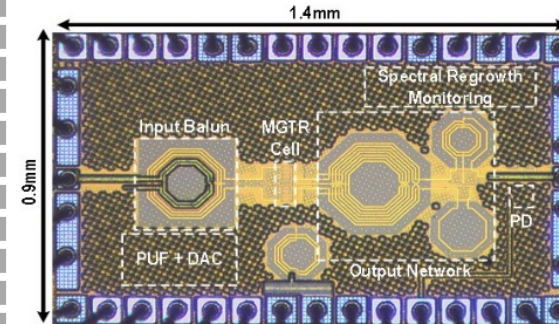
2021 IEEE CICC Best Student Paper Award

- 140 GHz Phased Array for Beyond 5G



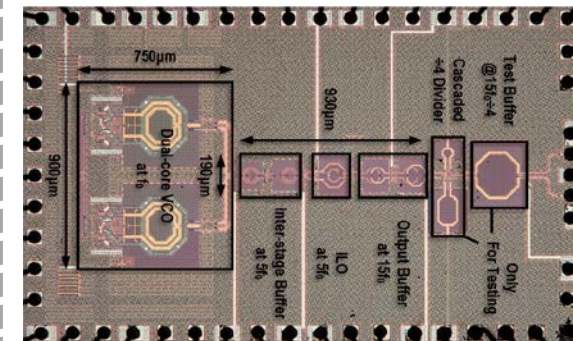
2023 IEEE RFIC Best Industry Paper Award Finalist

- RF Fingerprinting for Wireless Security



2021 ISSCC, 2024 T-MTT

- 100 GHz LO Generator with 191dBc/Hz FoM



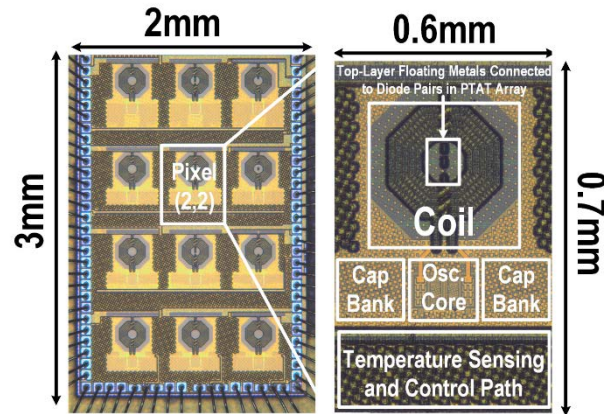
2023 CICC

Rice Integrated Systems and Electromagnetics (RISE) Lab



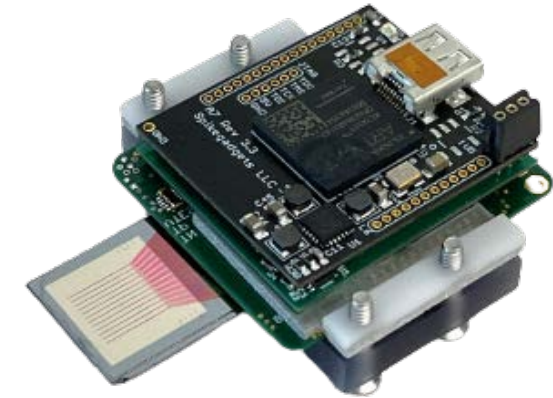
- Assistant Professor, Rice, 2019
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• In Vitro Magnetogenetics



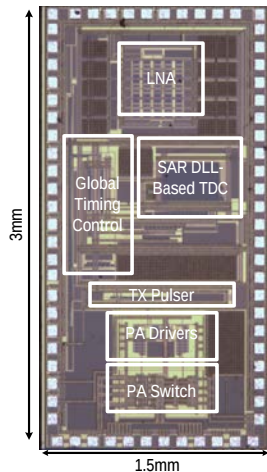
2021 ISSCC, T-BioCAS

• Large-Scale Neural Recording

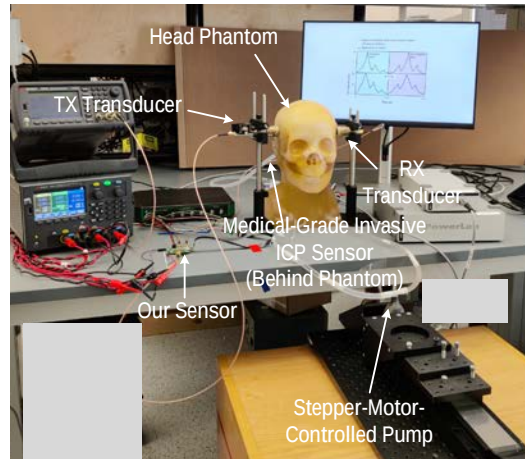


Ongoing

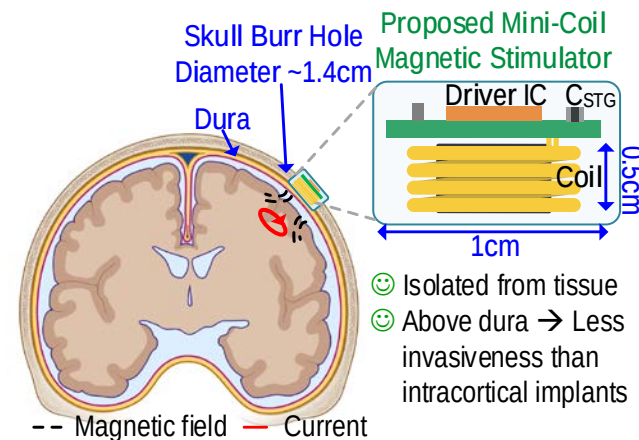
• Ultrasonic TRX for Intracranial Pressure Monitoring



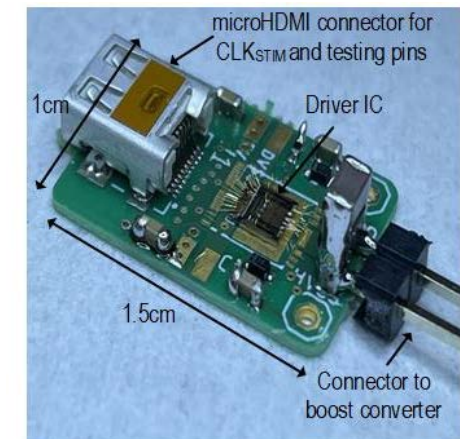
2024 ISSCC



• Miniaturized Magnetic Stimulator



2024 ISSCC



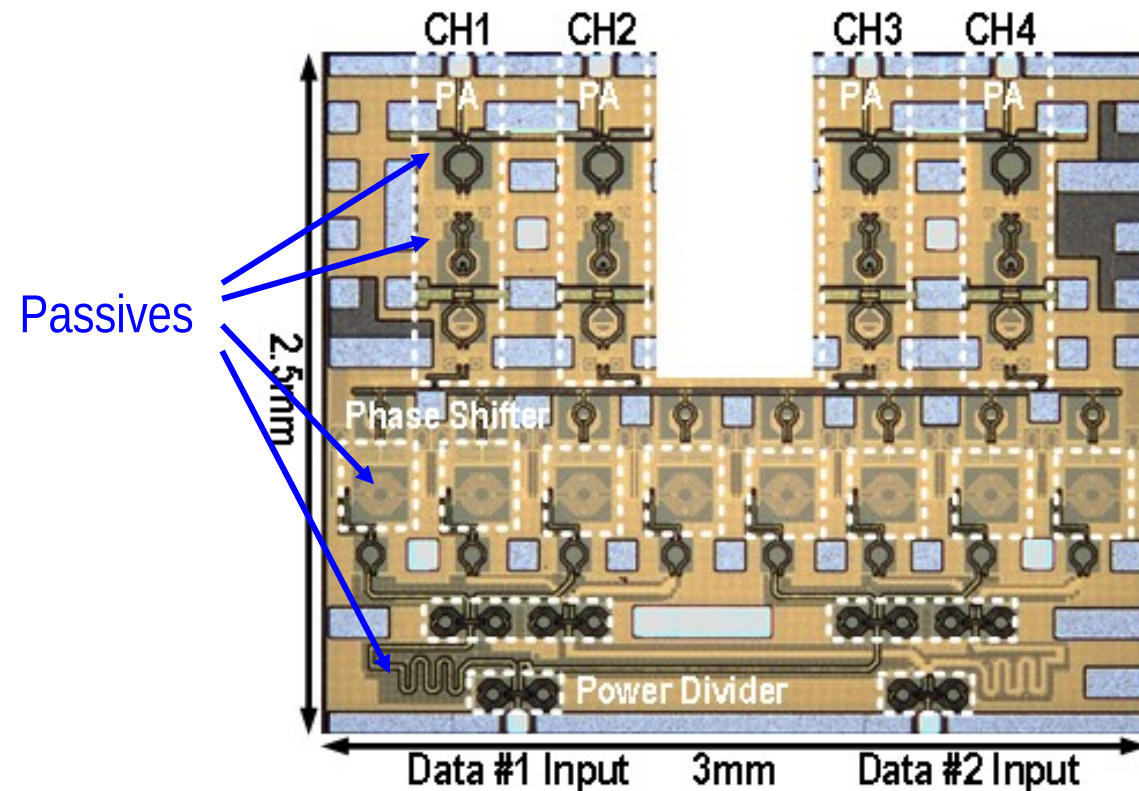


- Introduction to Passive Networks Design
- Three Transceiver Building Block Design Examples
 - LNA with an Octave Bandwidth
 - Doherty PA with Efficiency Enhancement up to 12-dB Back-off
 - 100-GHz LO Generator with >190 dBc/Hz FoM
- Conclusions

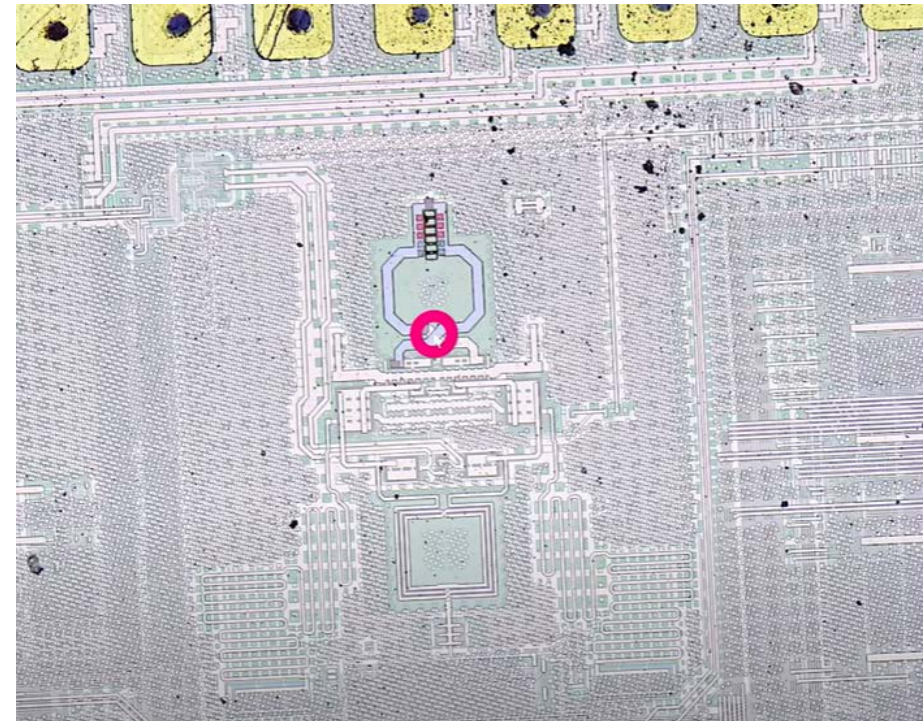
Introduction

- RF IC design is similar to analog → transistor sizing, biasing, feedback...
- Demand additional attention to passive networks design

[Die photo of a 28-GHz 4-channel beamformer](#)



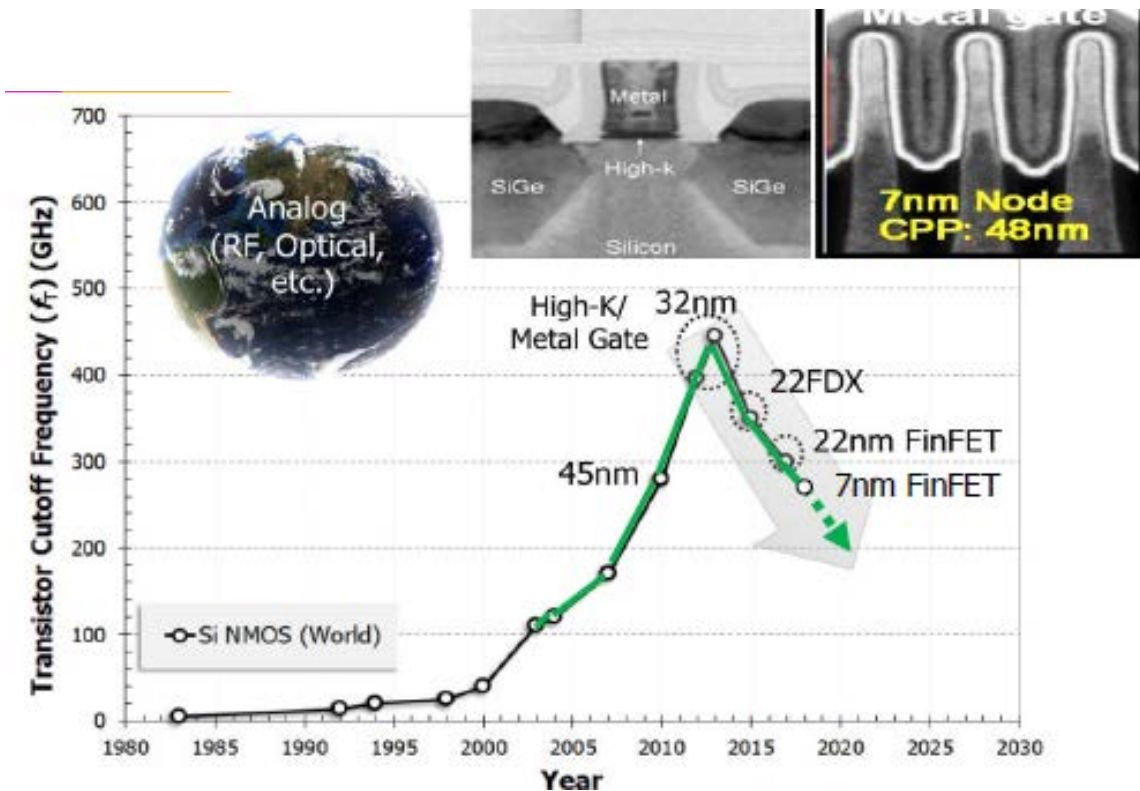
[VCO of TI's 77-GHz radar chip](#)



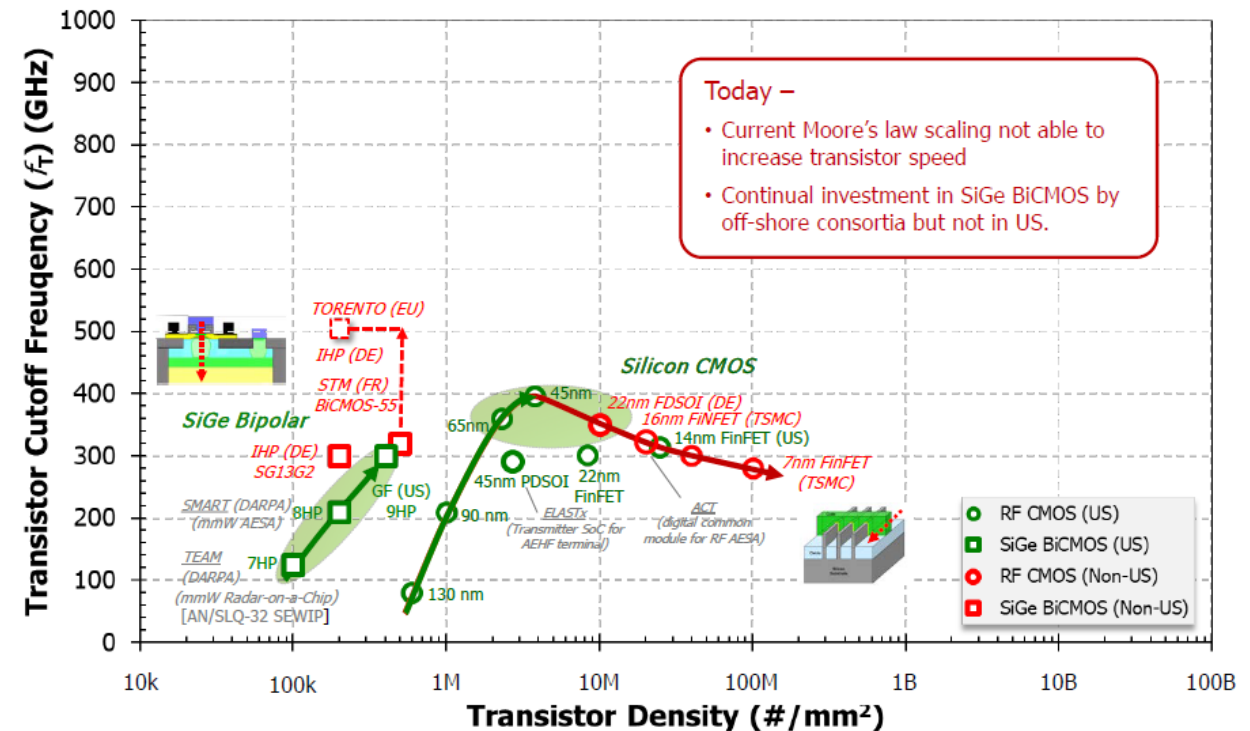


Introduction

- f_T and f_{max} no longer benefit from technology scaling
- Same as RF power handling capability
- Until new technologies or packaging become available (e.g., GaN or InP on CMOS)



source: DARPA 2019





Introduction

Functions to Achieve

- Impedance matching
- Filtering
- Voltage or current gain
- Power combining
- Phase shifting
- Harmonic shaping
- ...

Intuitive + Rigorous

End Goals

- Schematic
- Layout
- EM simulations
- ...

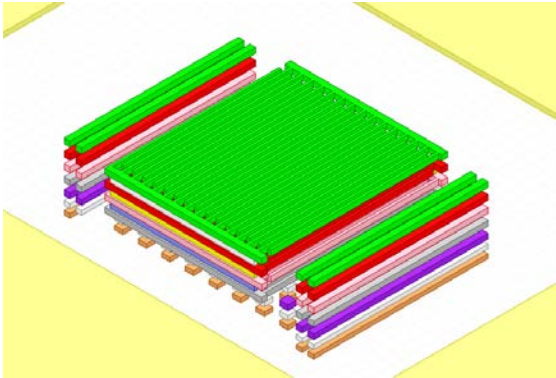
Trade-offs (Design Constraints)

- Size
- Passive efficiency (or signal loss)
- Complexity
- ...

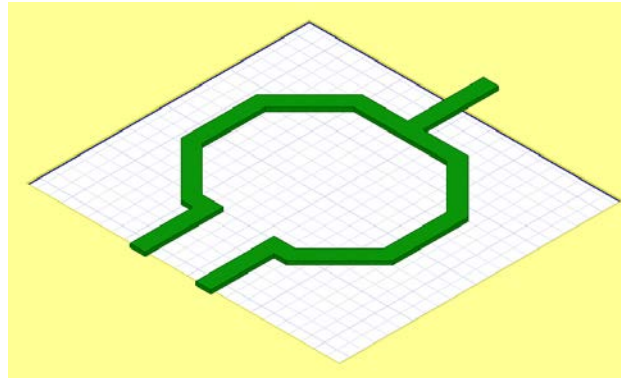
Introduction

- Circuit Elements to Construct Passive Networks

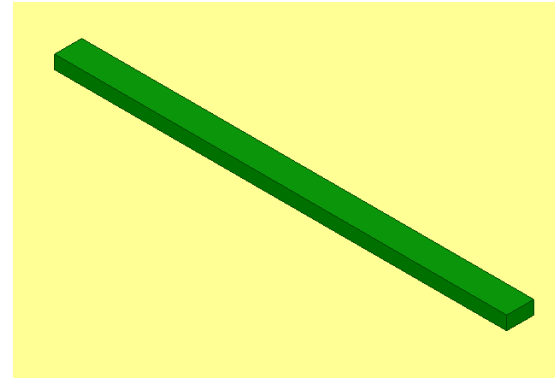
Capacitor



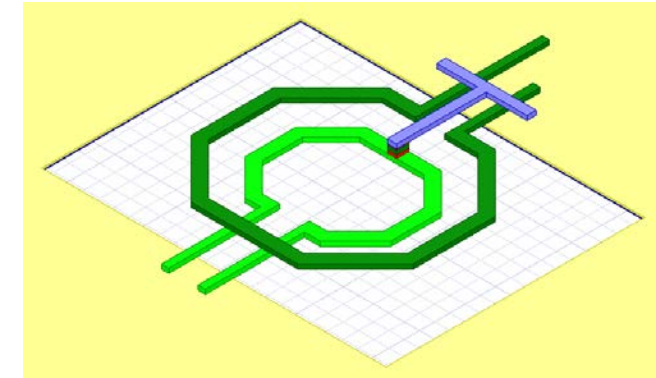
Inductor



Transmission Line



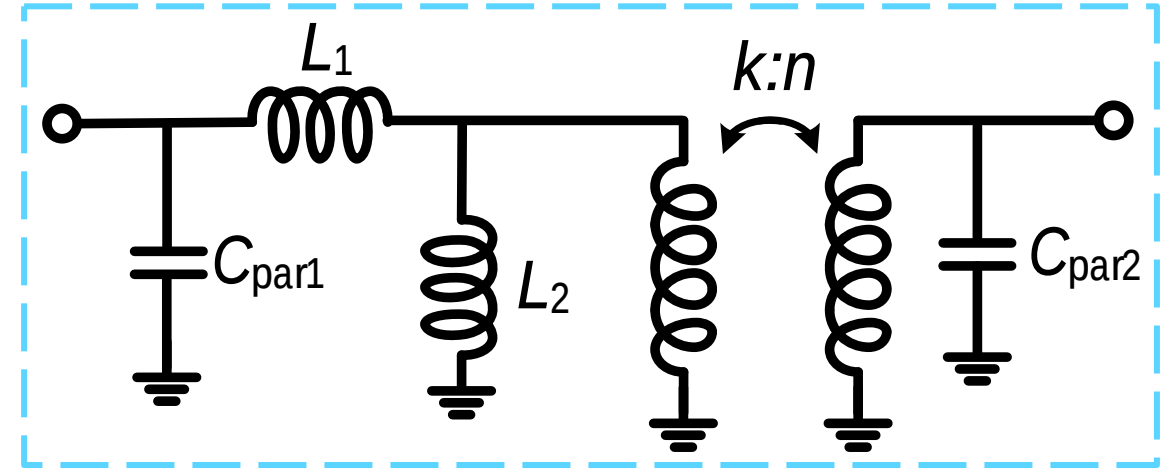
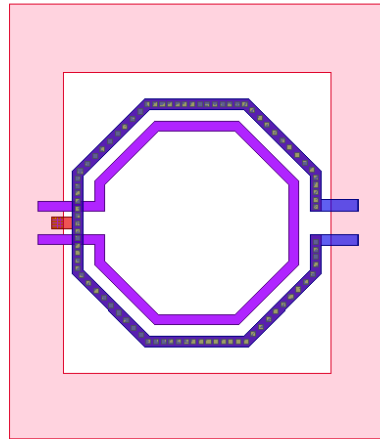
Transformer



- 
- DC Isolation
 - Conversion between single-ended to differential
 - Higher-order network in a reduced footprint

Introduction

- Lumped Circuit Model of On-chip Transformer



- k : magnetic coupling coefficient
- n : turn ratio
- $L_1 = (1-k^2)L_p$, leakage inductor
- $L_2 = k^2L_p$, magnetization inductor
- C_{par1} , C_{par2} : parasitic capacitance



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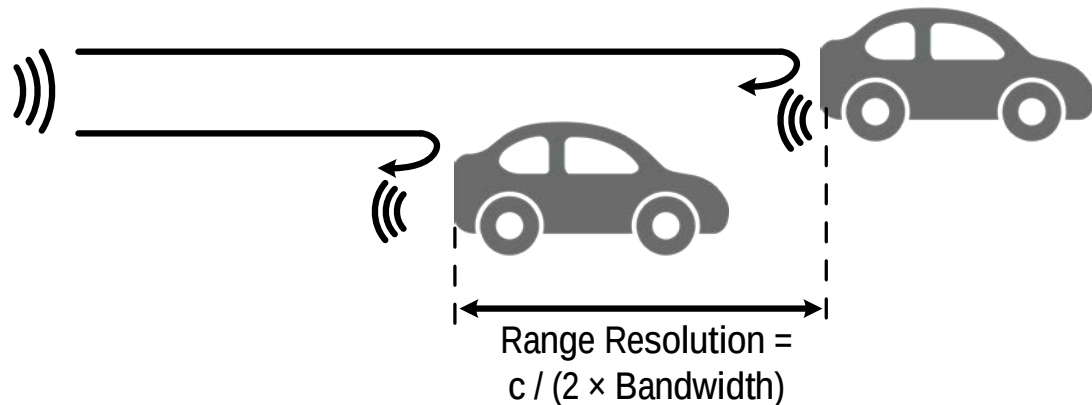
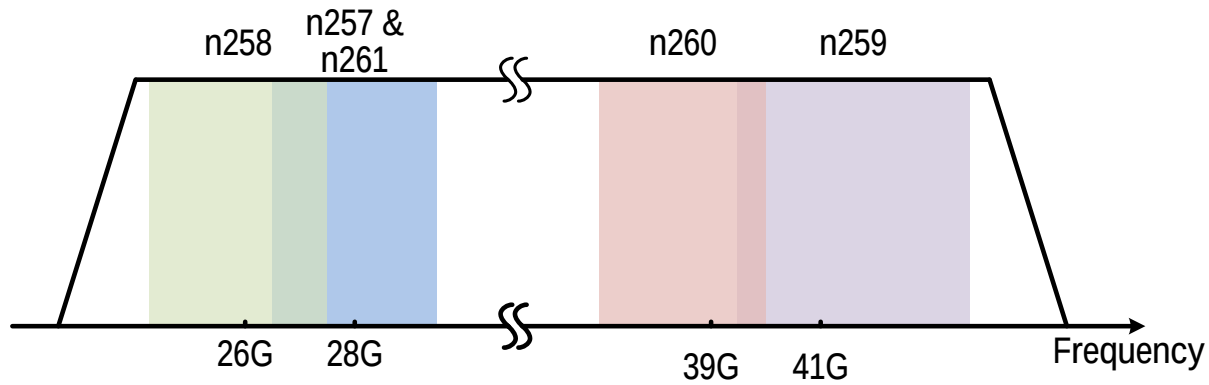
Yaolong Hu

Y. Hu and T. Chi, "[A 27.5–46.2-GHz Broadband Low Noise Amplifier With IP3 Enhancement](#)," MWTL 2023.

Y. Hu and T. Chi, "[A Systematic Approach to Designing Broadband Millimeter-Wave Cascode Common-Source With Inductive Degeneration Low Noise Amplifiers](#)," T-CAS I 2023.



Octave-Bandwidth LNA



- Full-band Coverage of 5G NR FR2

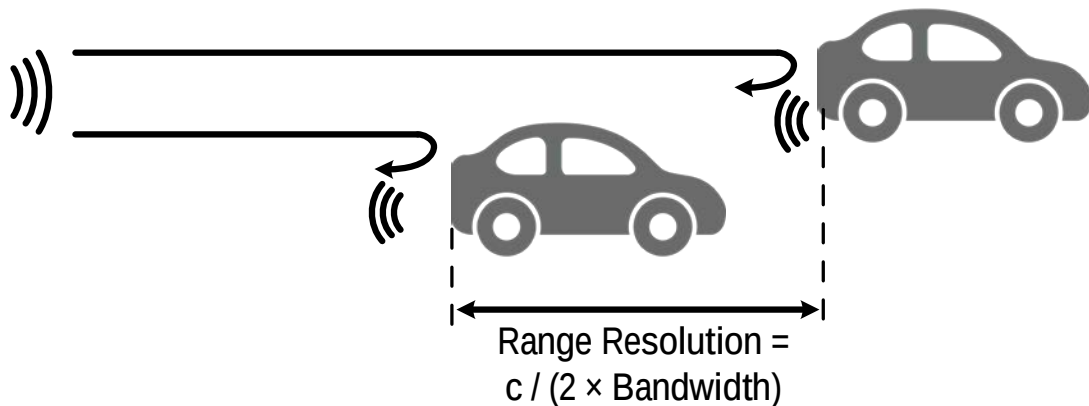
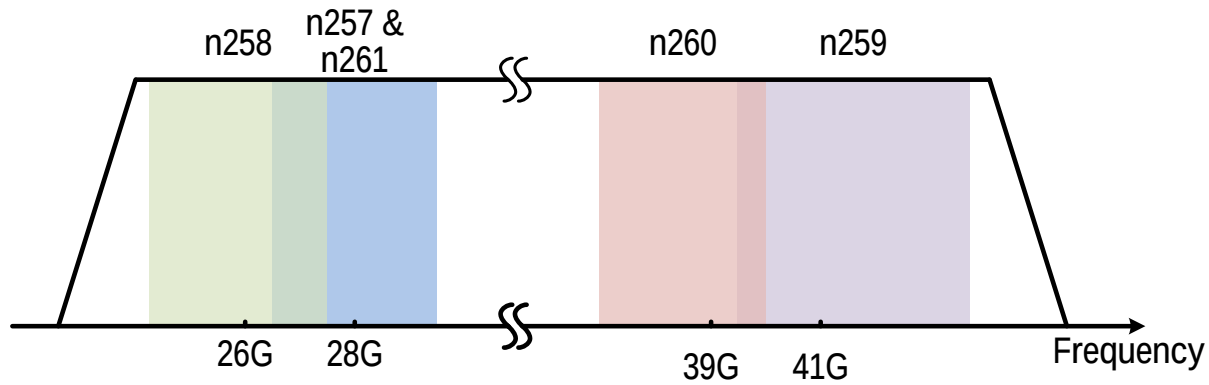
- Inter-band carrier aggregation
- Global multi-standard coverage
- Agile frequency hopping

- Wideband Wireless Sensing

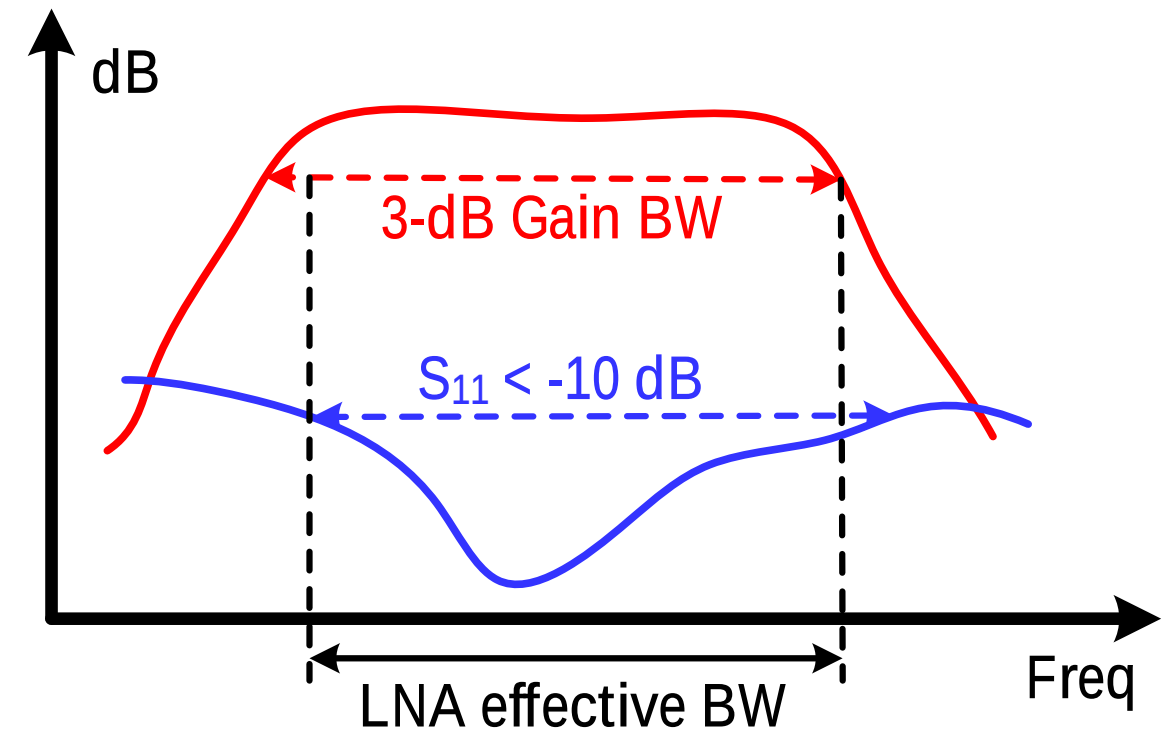
- Improved range resolution



Octave-Bandwidth LNA



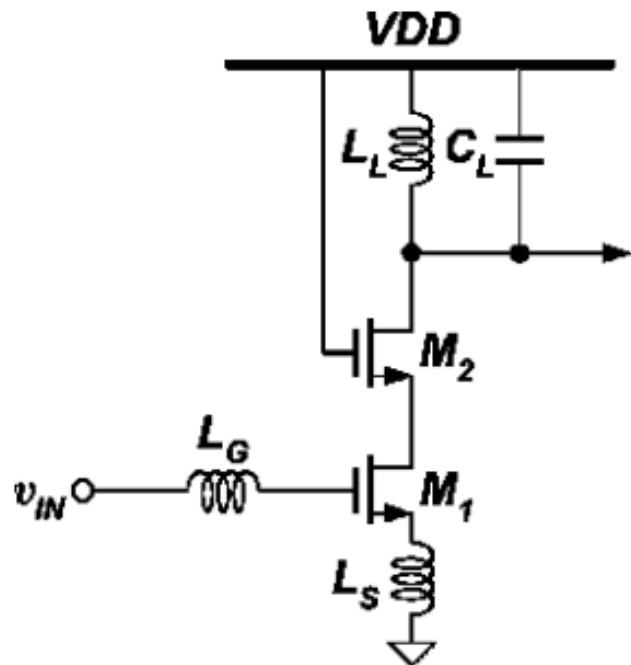
- LNA Effective BW



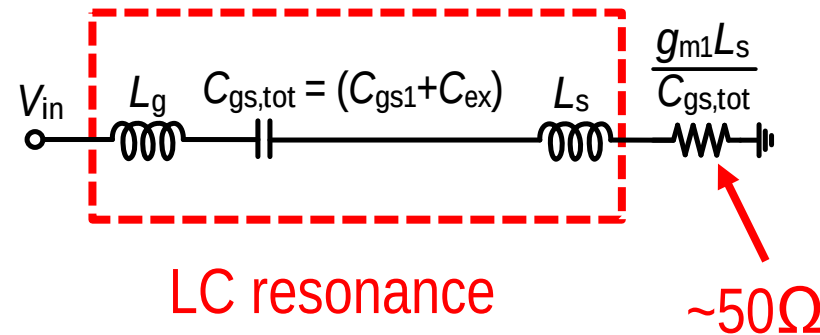


Octave-Bandwidth LNA

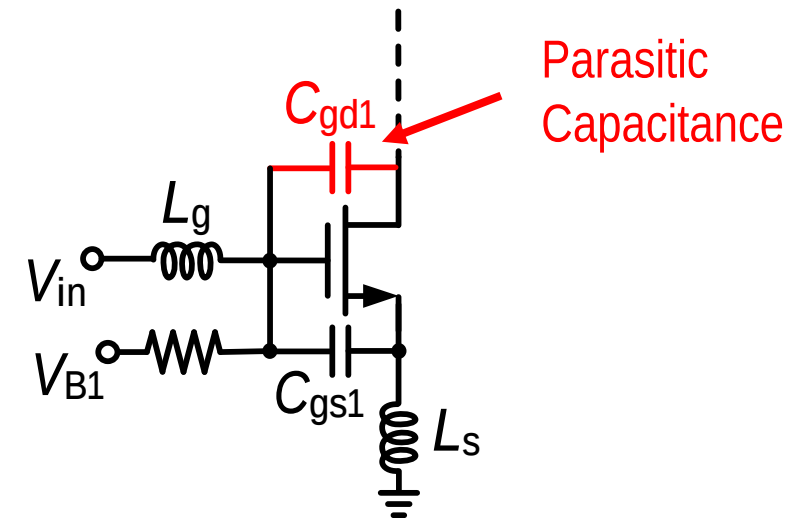
- Prevalent LNA Topology: Common Source with Inductive Degeneration



- Equivalent circuit for input matching



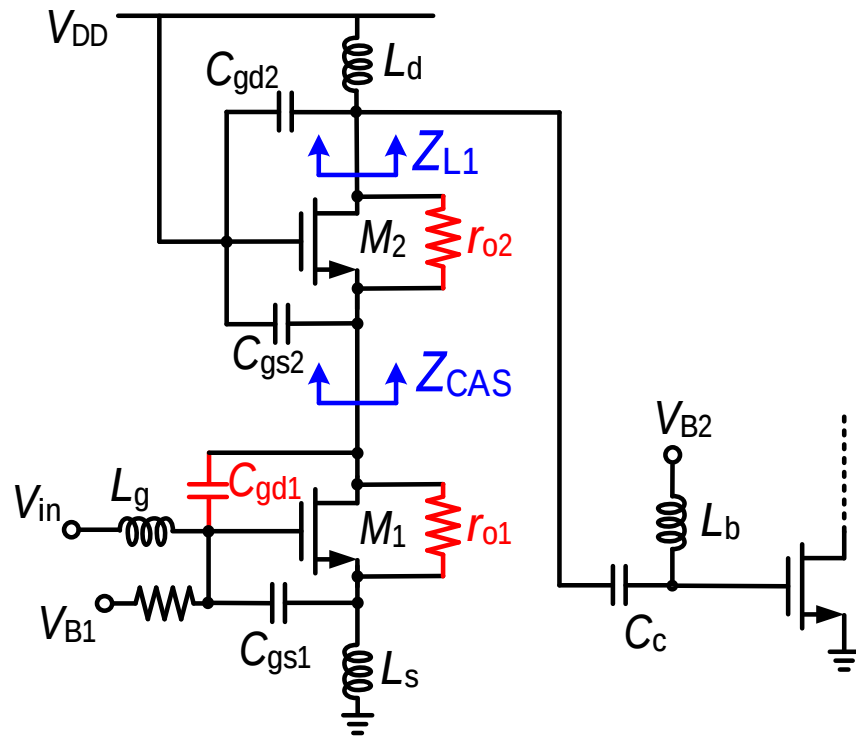
$$f_{res} = \frac{1}{2\pi\sqrt{C_{gs1,tot}(L_g + L_s)}}$$



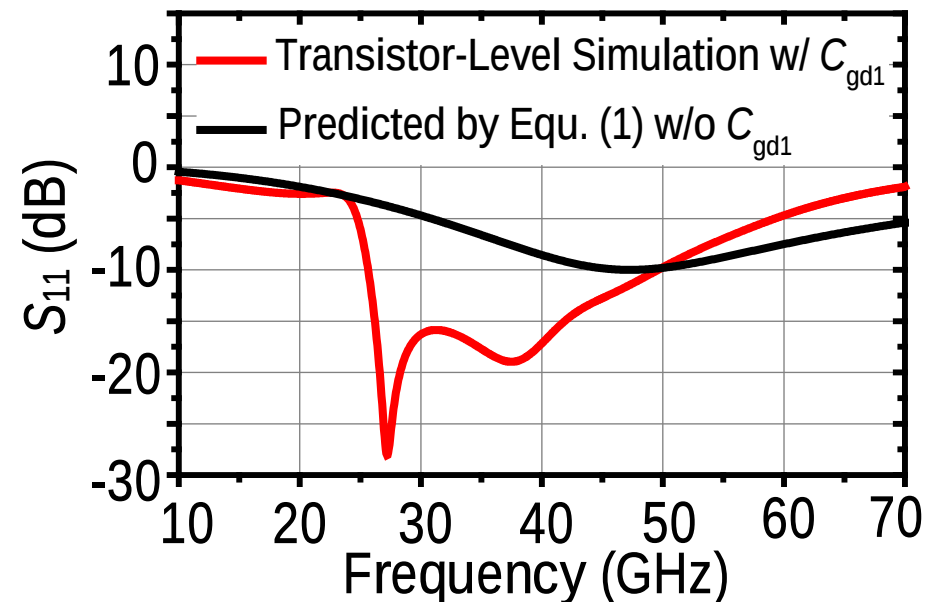
- $C_{gd1}:C_{gs1} \sim 1:2$
- Cannot simply neglect C_{gd1} at mmWave

Octave-Bandwidth LNA

- Prevalent LNA Topology: Common Source with Inductive Degeneration



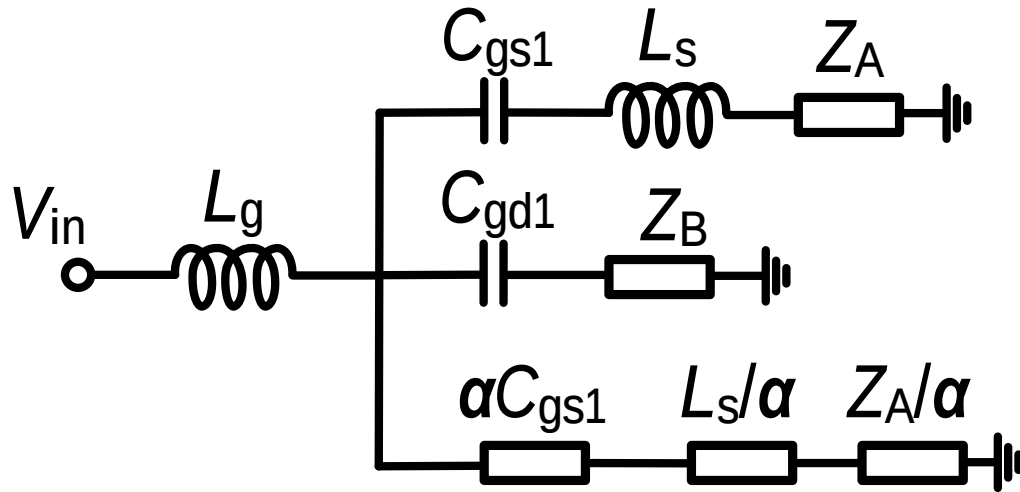
Question: Can we turn a conventional CS with inductive degeneration topology into a wideband design by properly designing component values?





Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Re-Derive Equivalent Circuit for Input Matching



$$Z_A = \frac{\omega^2 L_s + g_{m1} L_s r_{o1} / C_{gs1}}{Z_{CAS} + r_{o1} + j\omega L_s} \quad Z_B = \frac{r_{o1} + j\omega L_s}{Z_{CAS} + r_{o1} + j\omega L_s} Z_{CAS}$$

- 1st path: same as conventional CS with inductive degeneration
- 2nd path: modeling the feedforward current through C_{gd1}
- 3rd path: scaled version of 1st path with complex coefficient α

$$\alpha = \frac{\frac{g_{m1} Z_{CAS}}{j\omega C_{gs1}} - Z_{CAS} \frac{\frac{g_{m1} Z_{CAS}}{j\omega C_{gs1}} + j\omega L_s + \frac{g_{m1} L_s}{C_{gs1}}}{r_{o1} + j\omega L_s + Z_{CAS}}}{\frac{1}{j\omega C_{gd1}} + Z_{CAS} \frac{r_{o1} + j\omega L_s}{r_{o1} + j\omega L_s + Z_{CAS}}}$$

- transistor intrinsic parameters

$$(g_{m1}, C_{gs1}, C_{gd1}, r_{o1}) \text{ and } Z_{CAS}$$

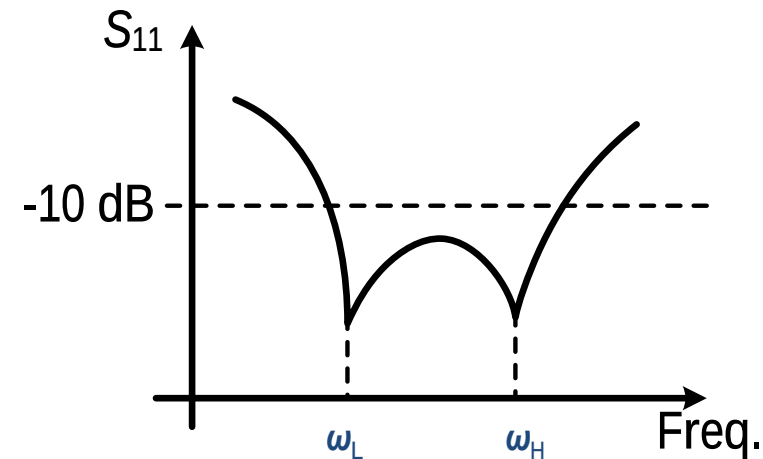
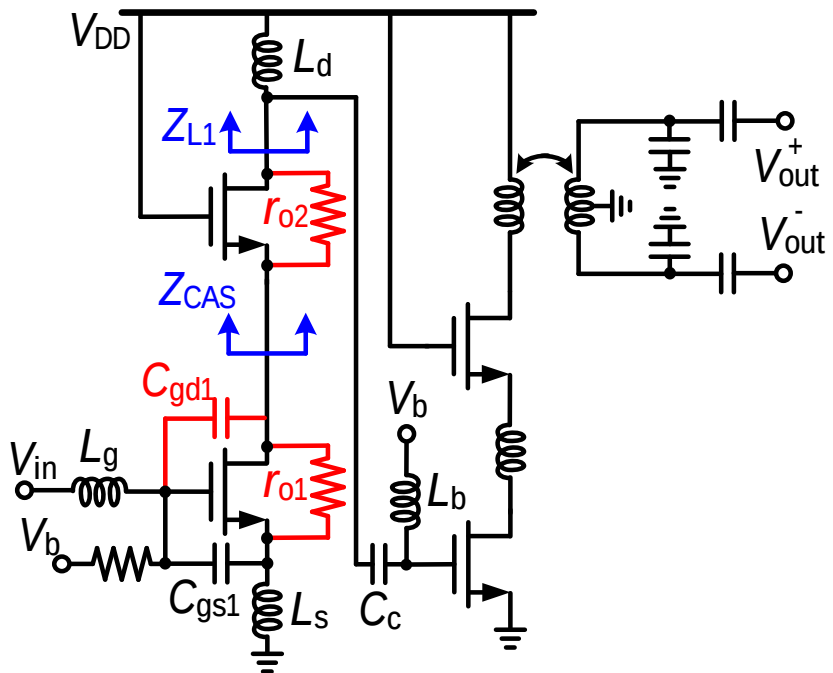
Frequency
independent

Frequency
dependent



Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Re-Derive Equivalent Circuit for Input Matching



$$Z_{CAS} \approx \frac{1}{g_{m2}} + \frac{Z_{L1}}{1 + g_{m2}r_{o2}}$$

$$\alpha = f(g_{m1}, C_{gs1}, C_{gd1}, r_{o1}, Z_{CAS})$$

Frequency
dependent

Key idea: leverage frequency dependence of Z_{CAS}

→ Z_{CAS} behaves as high impedance at ω_L but low impedance at ω_H

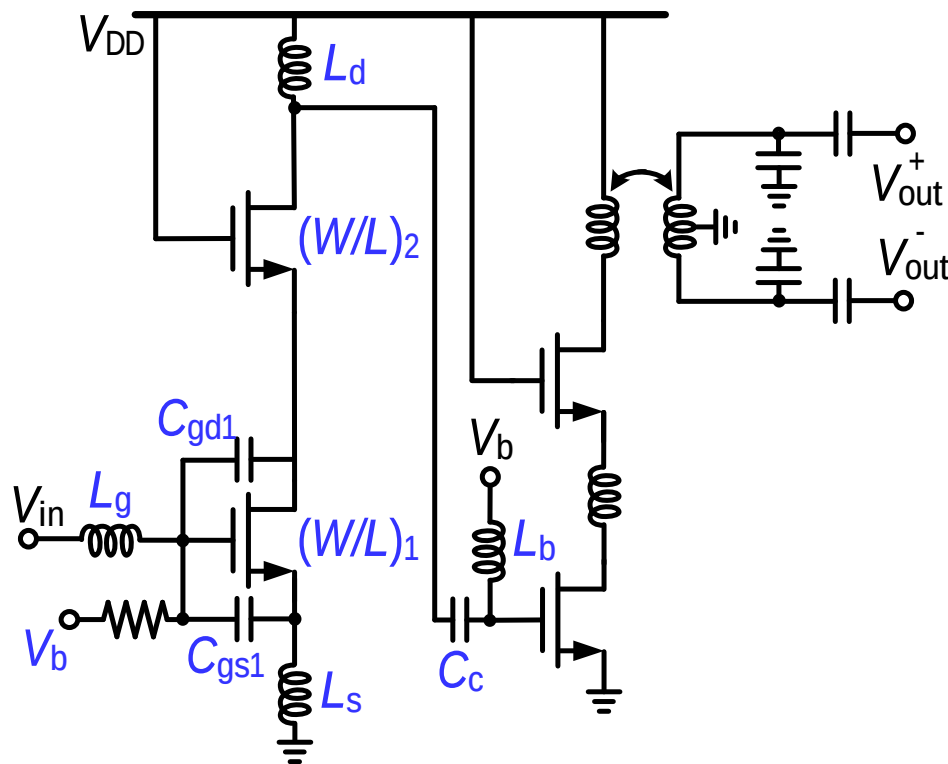
→ optimize two S_{11} resonances sequentially



Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Design Flowchart

- Parameters to be determined



Specify ω_L , ω_H , and DC power budget for the LNA

Step 1: Simulate NF_{min} and f_{max} vs. $J_D \rightarrow J_{D,opt}$
DC power budget $\rightarrow$ Determine $(W/L)_1 \rightarrow g_{m1}$, C_{gs1} , C_{gd1} , and r_{o1} are therefore decided

Step 2: Synthesize high-frequency S_{11} resonance $\rightarrow$ Determine $(W/L)_2$, g_{m2} , L_g , and L_s

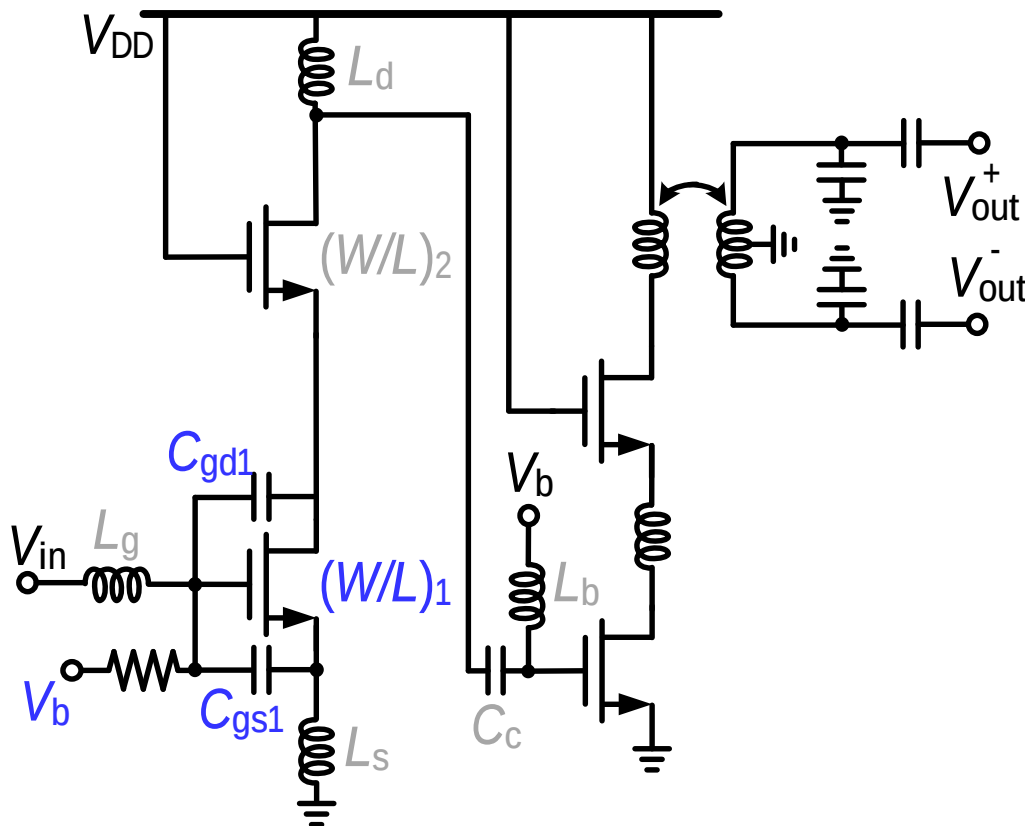
Step 3: Synthesize low-frequency S_{11} resonance $\rightarrow$ Determine L_d , L_b , and C_c

Y. Hu and T. Chi, "[A Systematic Approach to Designing Broadband Millimeter-Wave Cascode Common-Source With Inductive Degeneration Low Noise Amplifiers](#)," T-CAS I 2023.

Octave-Bandwidth LNA: Dual-Resonant Input Matching

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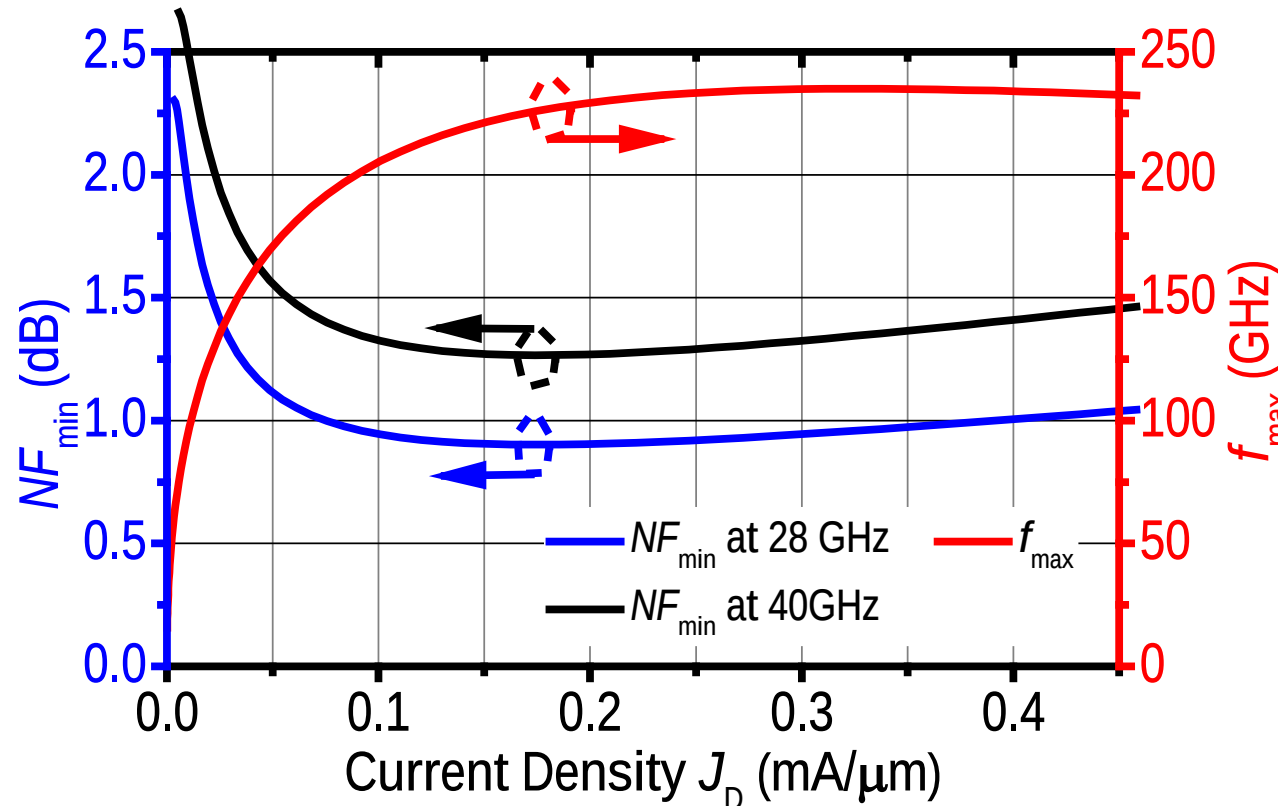
Step 1: Simulate $NF_{\min}$ and $f_{\max}$ vs. $J_D \rightarrow J_{D,\text{opt}}$
 DC power budget $\rightarrow$ Determine $(W/L)_1 \rightarrow g_{m1}$, C_{gs1} , C_{gd1} , and r_{o1} are therefore decided

Step 3: Synthesize low-frequency S_{11} resonance $\rightarrow$ Determine L_d , L_b , and C_c

Octave-Bandwidth LNA: Dual-Resonant Input Matching

• Design Procedure

Step 1: Simulated $NF_{\min}$ and $f_{\max}$ vs. J_D + DC power budget $\rightarrow J_{D, \text{opt}}$ and $(W/L)_1$
 $\rightarrow g_{m1}$, C_{gs1} , C_{gd1} , and r_{o1} are therefore determined



Design Example in GlobalFoundries 45-nm CMOS SOI

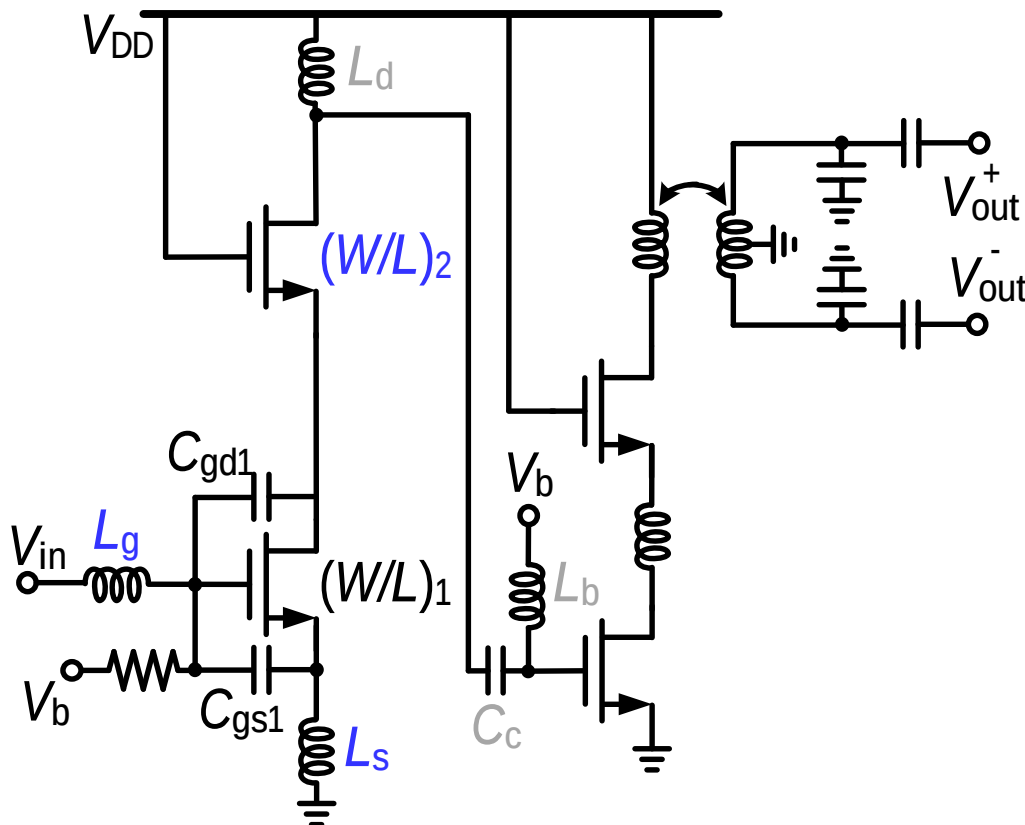
- $J_{D, \text{opt}} = 0.2$ mA/ μ m
- 11 mA DC current budget $\rightarrow$ choose $W = 55$ μ m
- $g_{m1} = 63.5$ mS, $C_{gs1} = 38$ fF, $C_{gd1} = 20$ fF, $r_{o1} = 84$ Ω



Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Design Flowchart

- Parameters to be determined



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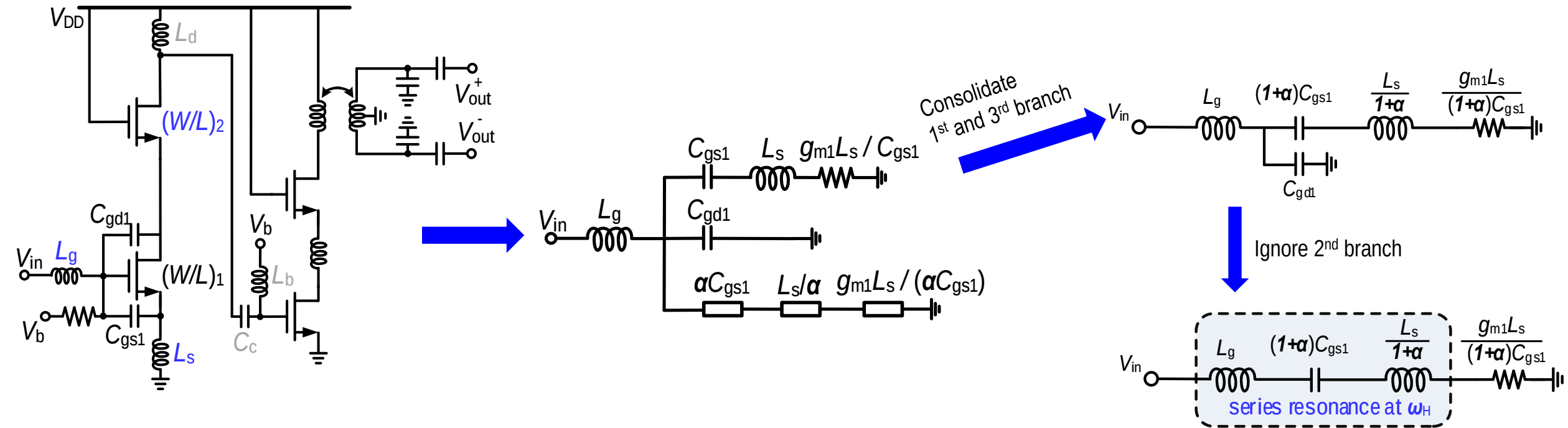


Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Design Procedure

Step 2: Z_{L1} is low impedance at $\omega_H \rightarrow Z_{CAS} \approx \frac{1}{g_{m2}}$ and $\alpha \approx \frac{g_{m1}C_{gd1}}{g_{m2}C_{gs1}}$ at ω_H

Already determined in step 1

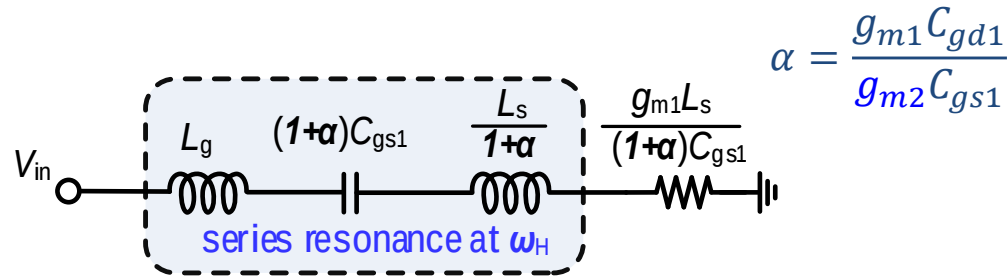


- Merge 1st and 3rd branches, and ignore 2nd branch to gain design intuition



Octave-Bandwidth LNA: Dual-Resonant Input Matching

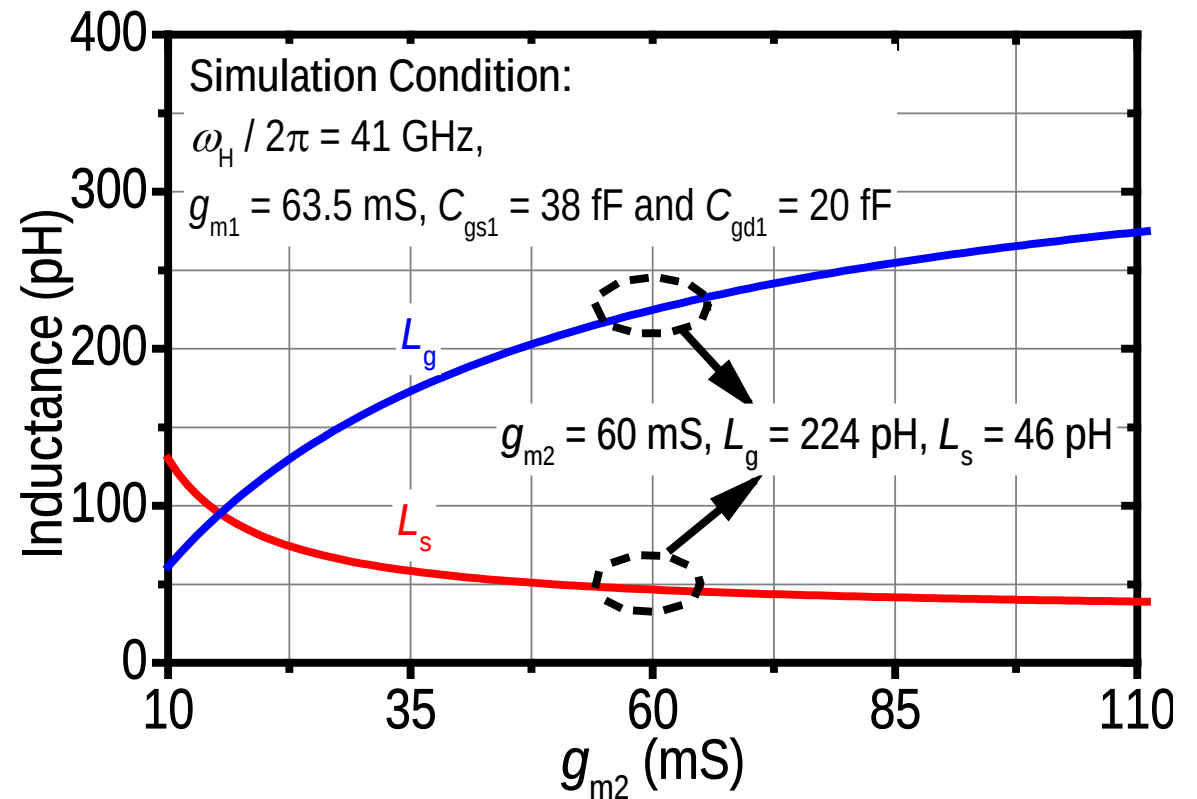
- Design Procedure: simplified approach



$$\text{Im}\{Z_{in}\} \sim 0 \text{ at } \omega_H$$

$$\text{Re}\{Z_{in}\} \sim 50 \Omega \text{ at } \omega_H$$

- 3 variables (L_g , L_s , and g_{m2}) and two equations $\rightarrow$ a family of solutions



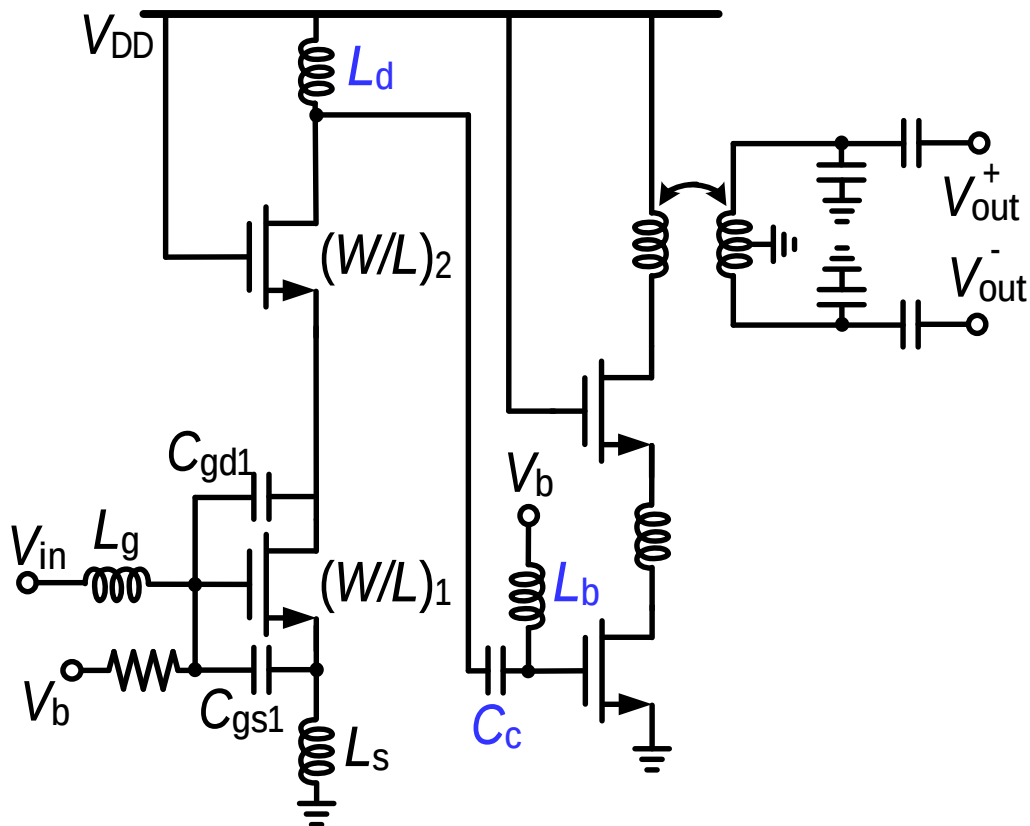
Design Example in GlobalFoundries 45-nm CMOS
SOI, assuming $\omega_H \sim 2\pi \times 41 \text{ GHz}$



Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Design Flowchart

- Parameters to be determined



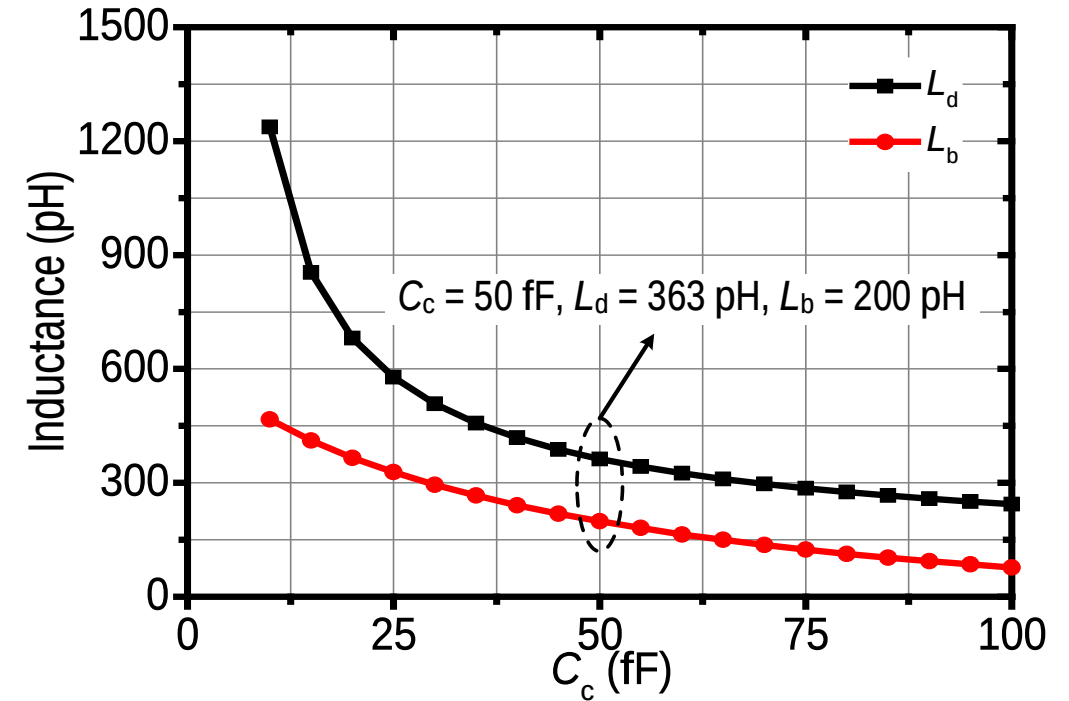
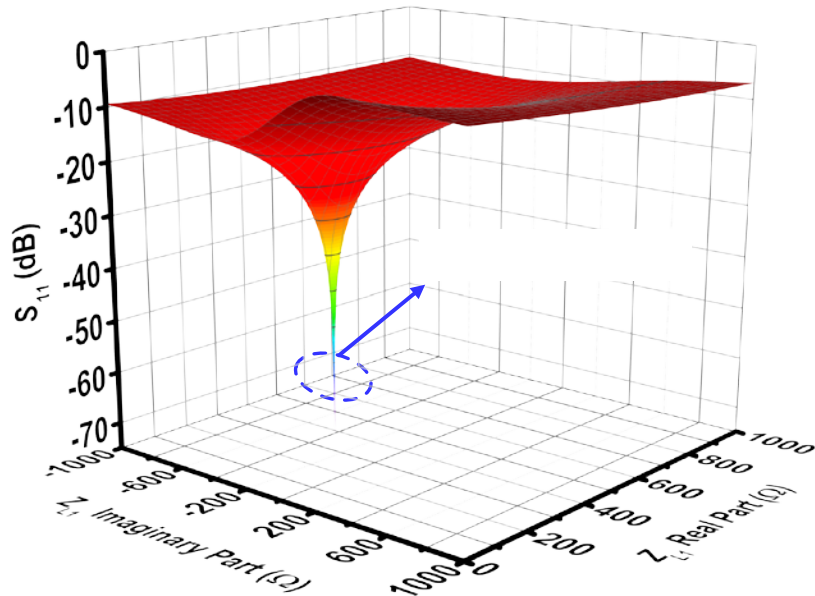
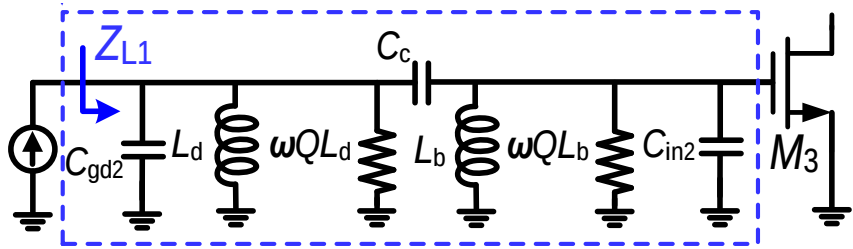
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Octave-Bandwidth LNA: Dual-Resonant Input Matching



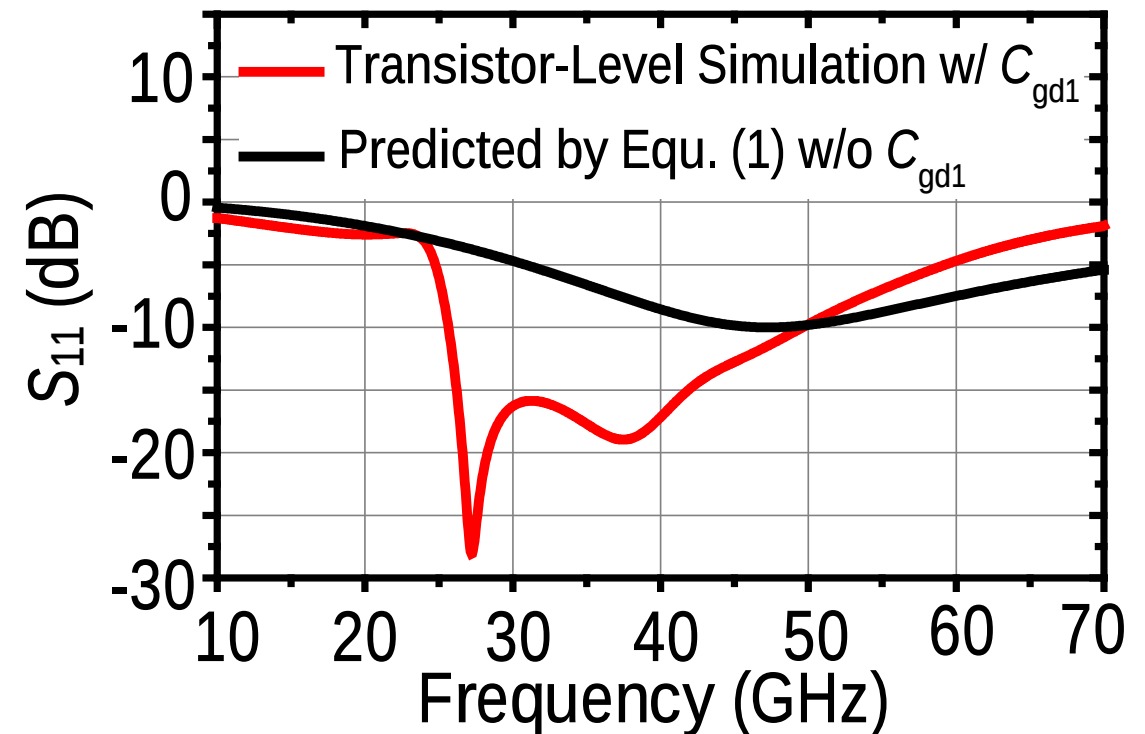
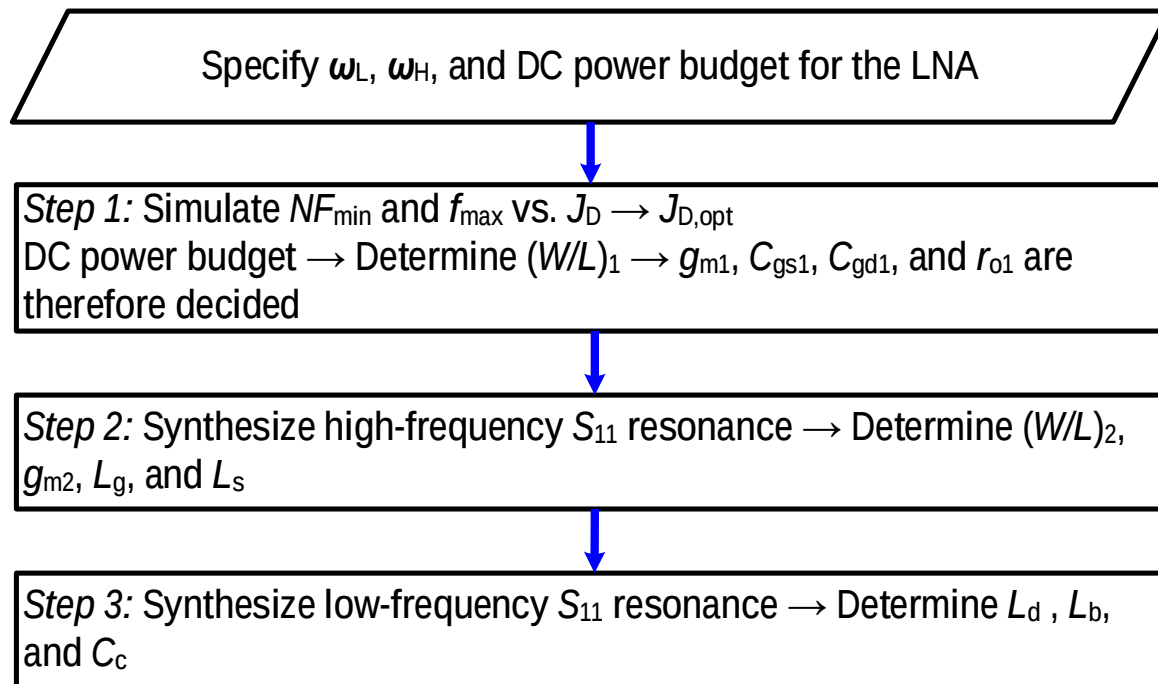
$$\text{Im}\{Z_{L1}\} \sim f(L_d, L_b, C_c) \text{ at } \omega_L \quad \text{Re}\{Z_{in}\} \sim f(L_d, L_b, C_c) \text{ at } \omega_L$$

- 3 variables (L_d, L_b, C_c) two equations
→ a family of solutions

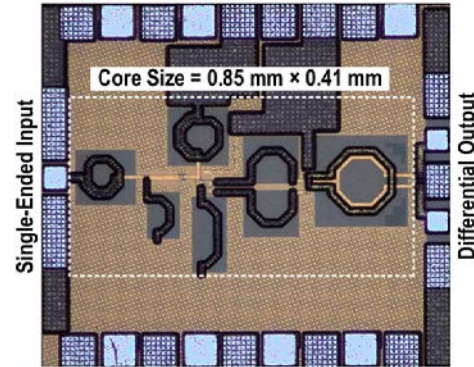
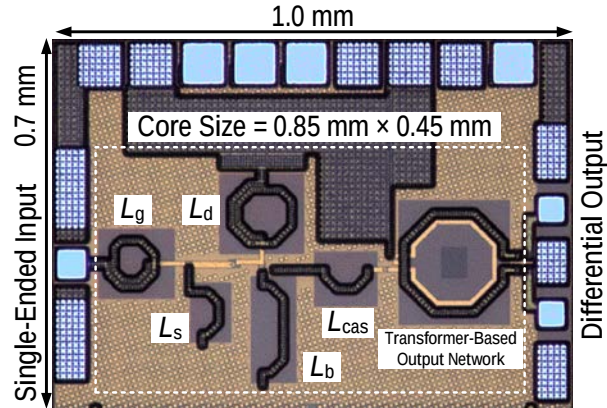


Octave-Bandwidth LNA: Dual-Resonant Input Matching

- Design Flow Chart



Octave-Bandwidth LNA: Measurement Results



Y. Hu and T. Chi, "[A 27.5–46.2-GHz Broadband Low Noise Amplifier With \$IP_3\$ Enhancement](#)," MWTL 2023.

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART BROADBAND MMWAVE LNAs

Reference	BW* (GHz)	3-dB BW (GHz)	Peak Gain (dB)	NF (dB)	IIP ₃ (dBm)	IP _{1dB} (dBm)	P _{DC} (mW)	Core Size (mm ²)	FoM**	Technology
This Work	27.5–46.2	26.3–46.2 ^a	19.8 ^a	2.1–4.4	-3.6 ^a	-14.6 ^a	25.5 ^a	0.35	1380 ^a	45-nm CMOS SOI
		26.1–46.2 ^b	20.3 ^b		-6.0 ^b	-15.4 ^b	27.4 ^b		828 ^b	
TCAS-I 2020 [5]	26.3–47	22–47	22.2	3.0–4.3	-13.8	-22.6	9.5	0.13	430	0.13- μ m SiGe
JSSC 2021 [6]	22.9–38.2	22.9–38.2	14.5	2.6–4.6	-3.6	-13.2	18.9	0.16	400	28-nm CMOS
MWCL 2022 [7]	20–44	18–44	19.5	2.6–3.5	-8.9 ^{††}	-18.5	17	0.16	667	65-nm CMOS
MWCL 2022 [8]	21.4–41.1	21.3–41.1	20.3	2.9–3.6	-6.1 ^{††}	-15.7	27.5	0.13	669	28-nm FDSOI
TCAS-I 2023 [9]	27–46	25.5–50	21.2	2.4–4.2	-9.5	-17.4	25.5	0.38	424	45-nm CMOS SOI

* Intersection of 3-dB gain BW and 10-dB return loss BW.

^{††} Estimated using IP_{1dB} + 9.6 dB.

^a PMOS M_6 in saturation and NMOS M_3 in triode.

^b PMOS M_6 in triode and NMOS M_3 in saturation.

** $FoM = \frac{10^3 \times Gain[\frac{W}{W}] \times BW_{eff}[GHz] \times IIP_3[mW]}{P_{DC}[mW] \times (NF[linear] - 1) \times f_c[GHz]}$, f_c is the geometric mean, peak gain and minimum NF are taken in FoM calculation.



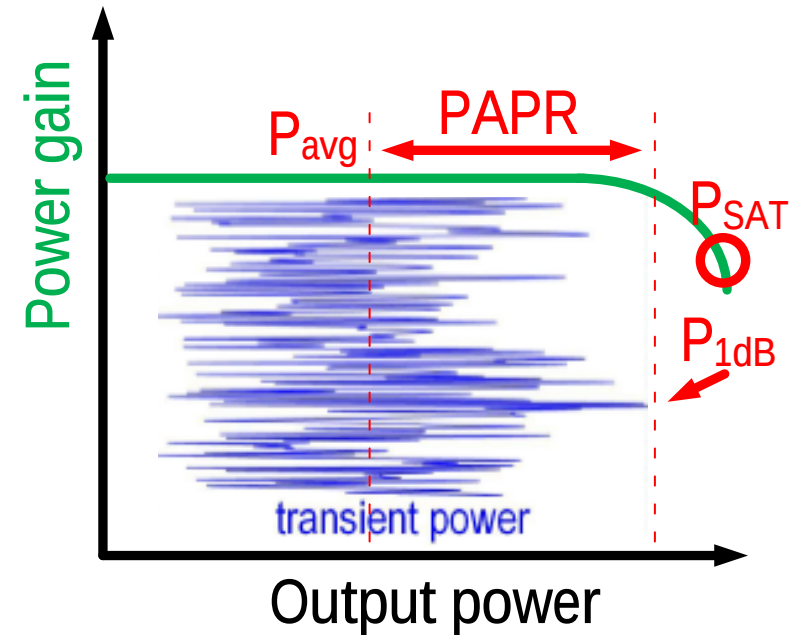
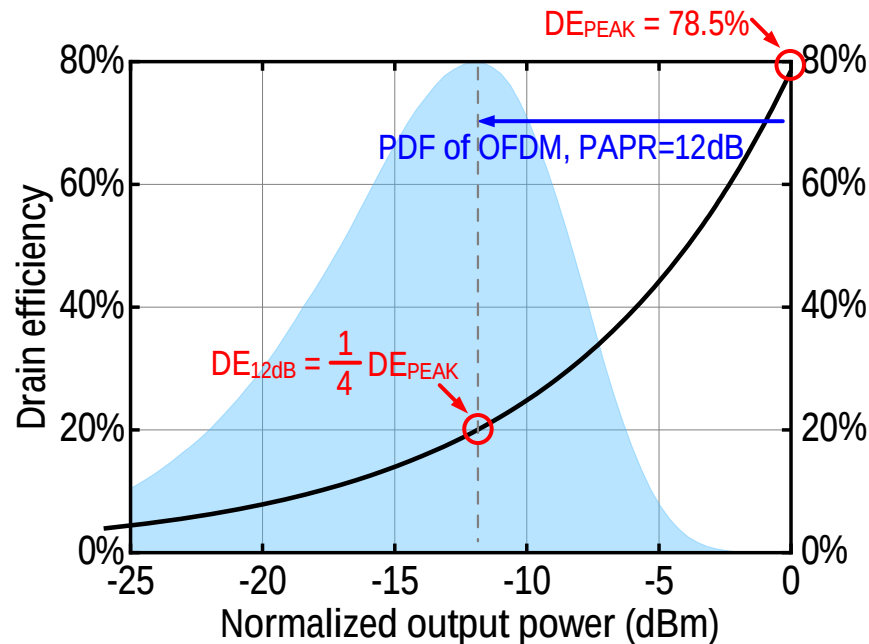
- Introduction to Passive Networks Design
- Three Transceiver Building Block Design Examples
 - LNA with an Octave Bandwidth
 - Doherty PA with Efficiency Enhancement up to 12-dB Back-off
 - 100-GHz LO Generator with >190 dBc/Hz FoM
- Conclusions



X. Zhang, H. Guo, and T. Chi, "[A 47GHz 4-way Doherty PA with 23.7dBm P1dB and 21.7% / 13.1% PAE at 6 / 12dB Back-off Supporting 2000MHz 5G NR 64-QAM OFDM](#)," ISSCC 2024.

Doherty PA: Introduction

- Spectrally efficient modulations (QAM), OFDM, and carrier aggregation (CA)
→ high peak-to-average power ratio (PAPR)



- Compared to P_{SAT} and PAE_{PEAK} , $PAE_{BACK-OFF}$ and linearity (P_{1dB} and AM-PM) are more relevant metrics for communication PAs



Doherty PA Fundamentals

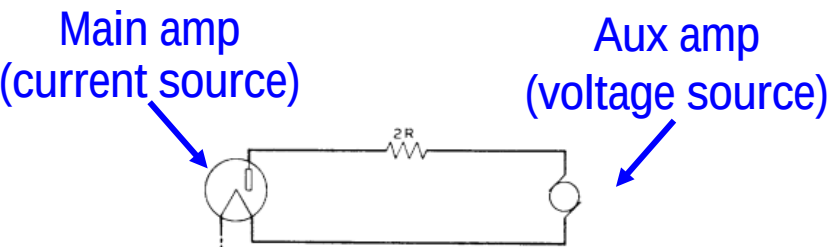


Fig. 2—Insertion of a hypothetical source of additional voltage.

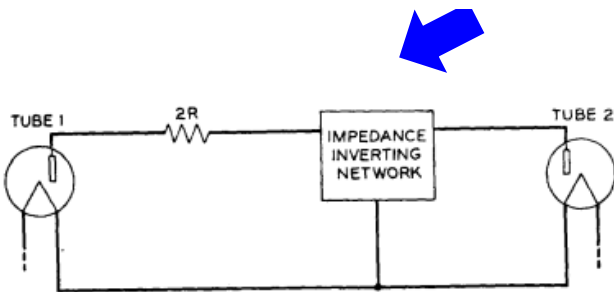
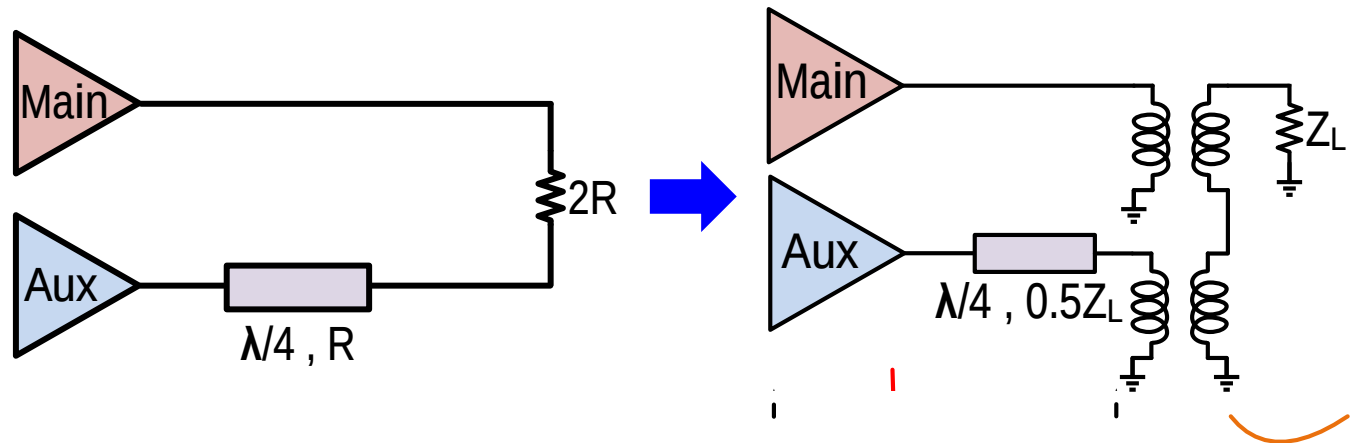


Fig. 3—Fundamental form of a high efficiency circuit.



Series Doherty



Doherty PA Fundamentals

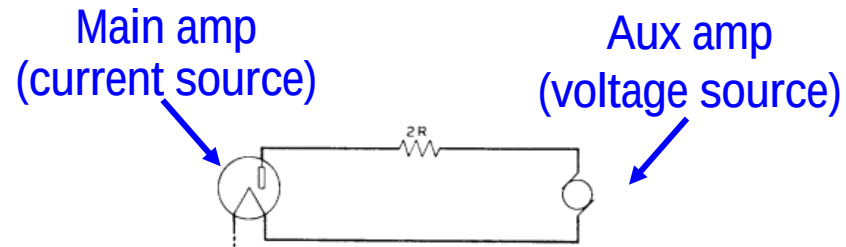


Fig. 2—Insertion of a hypothetical source of additional voltage.

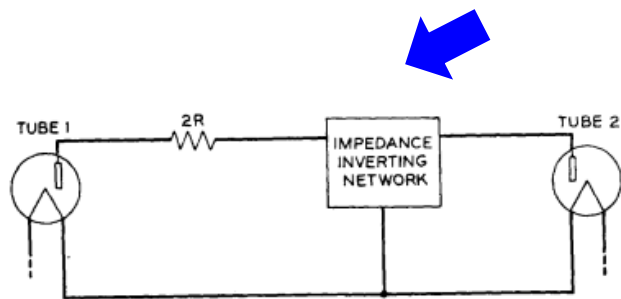


Fig. 3—Fundamental form of a high efficiency circuit.

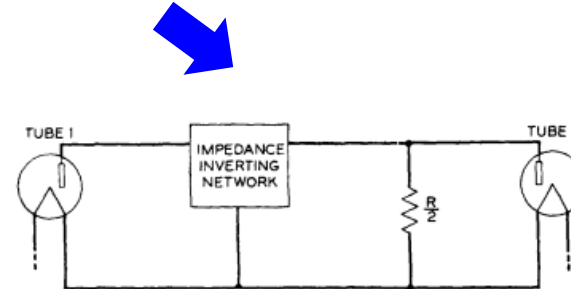
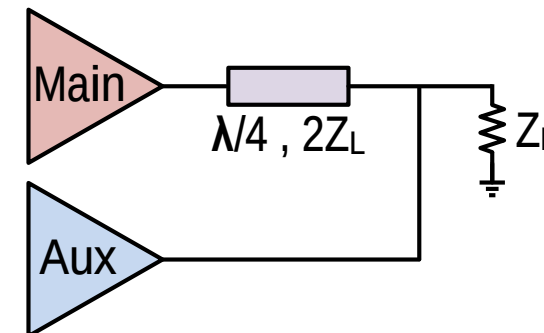


Fig. 4—Second fundamental form of high efficiency circuit.

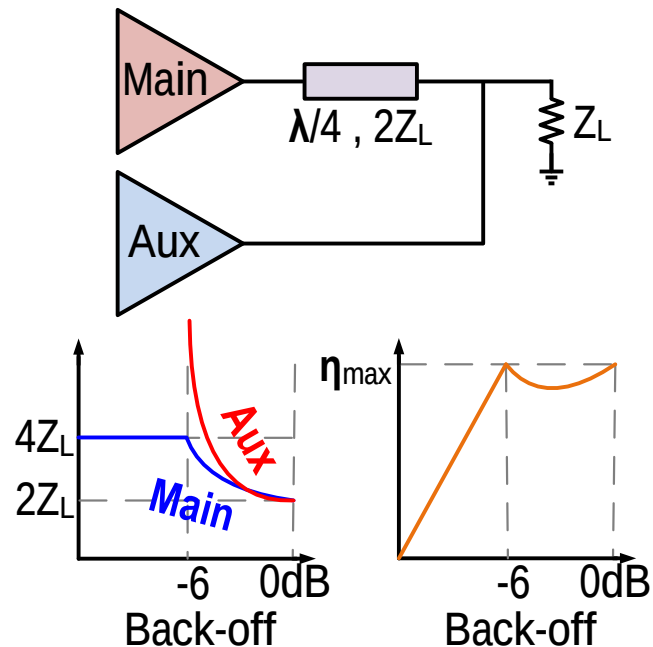
Parallel Doherty



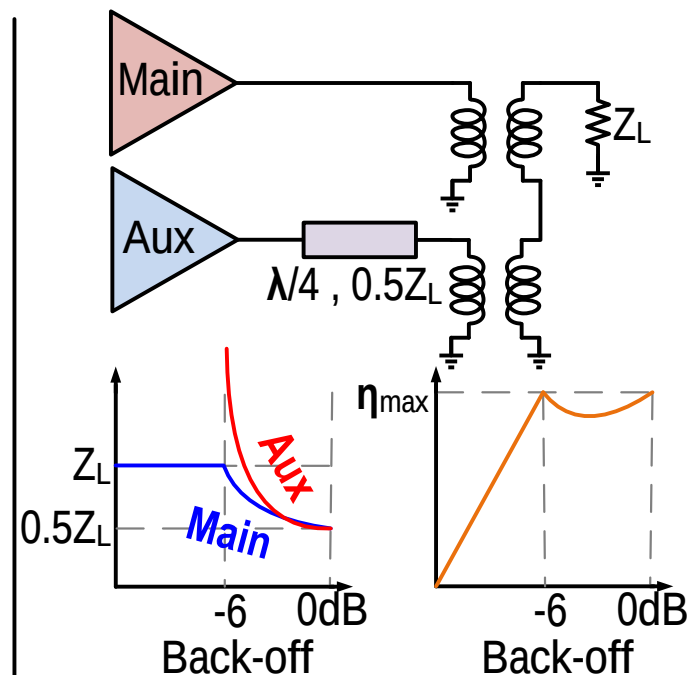


Doherty PA Fundamentals

- Parallel Doherty



- Series Doherty

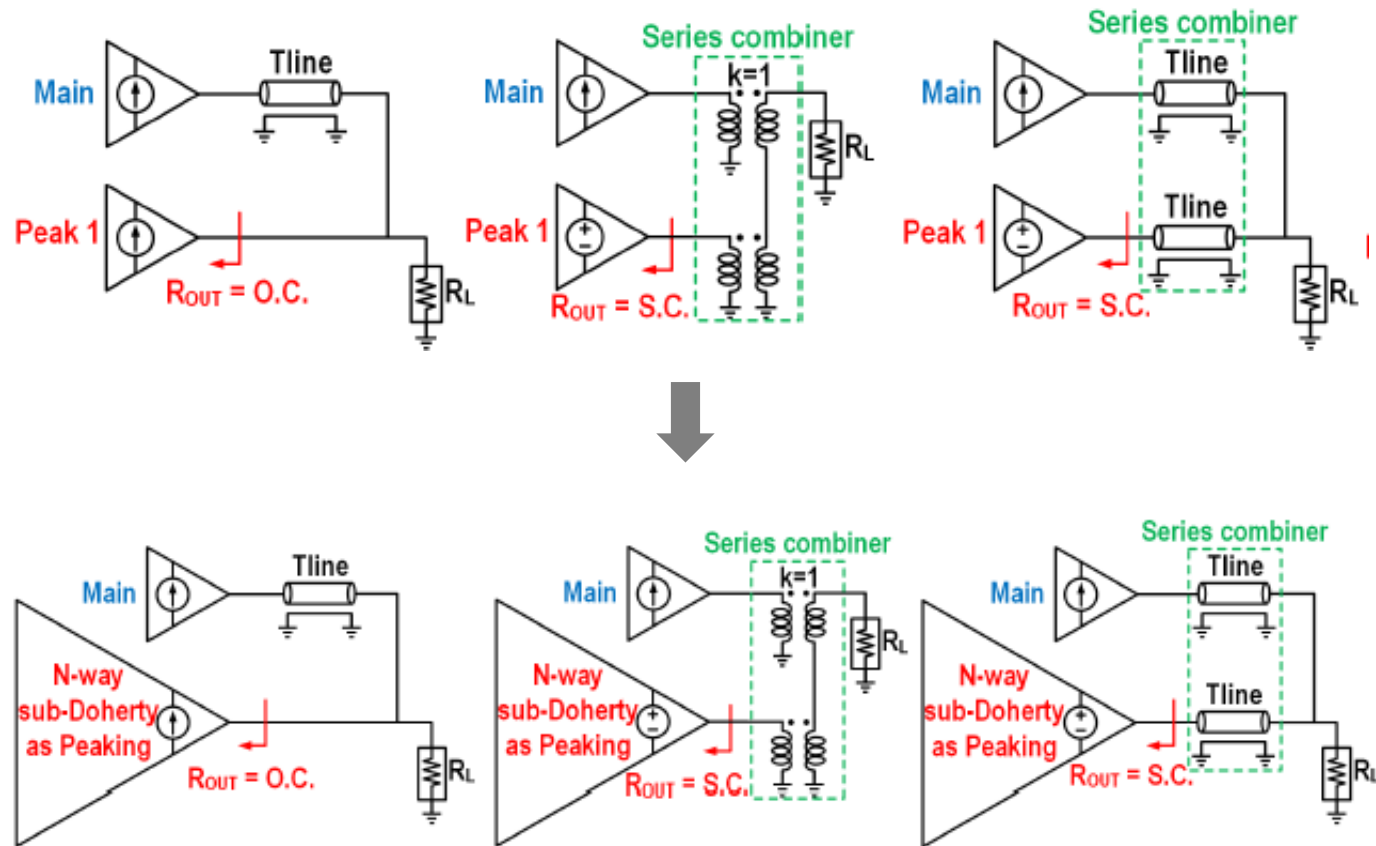


Key ingredients:

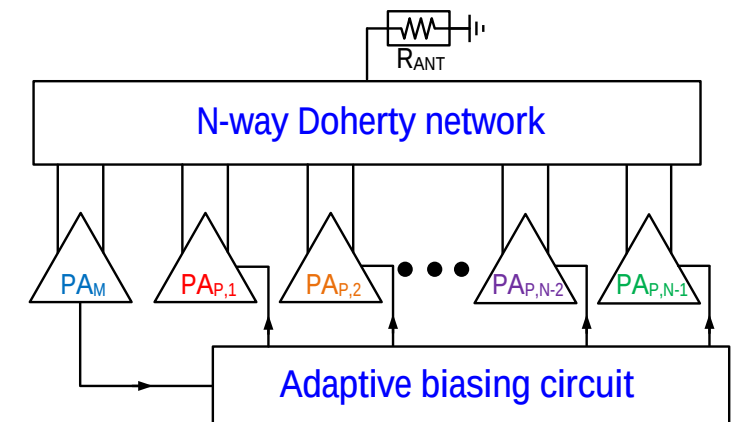
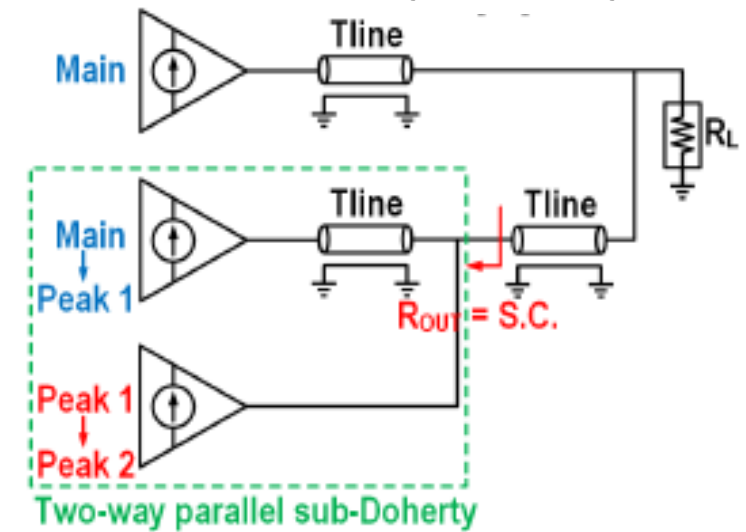
- Doherty network implementation: desired load modulation yet low loss
- Co-operation between Main and Aux: adaptive biasing



Extending Two-Way Doherty to N Ways



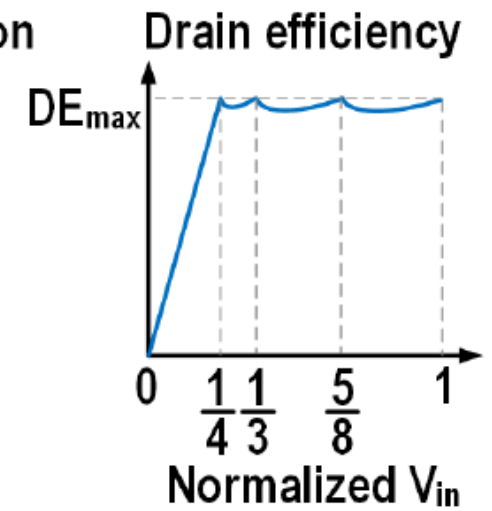
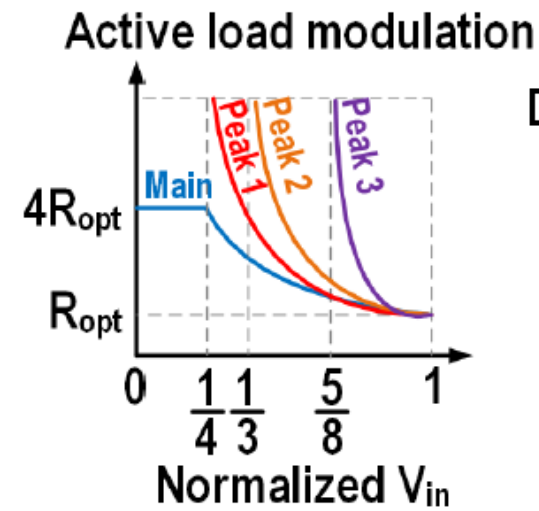
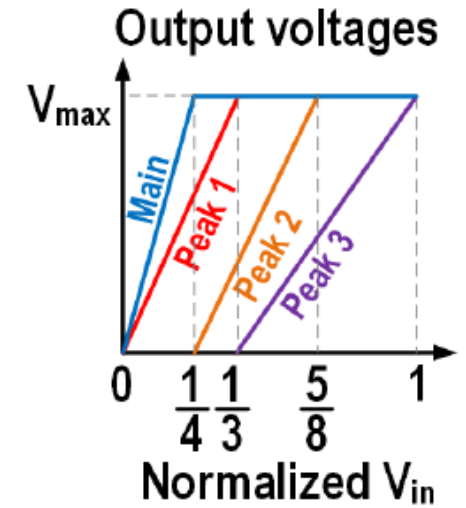
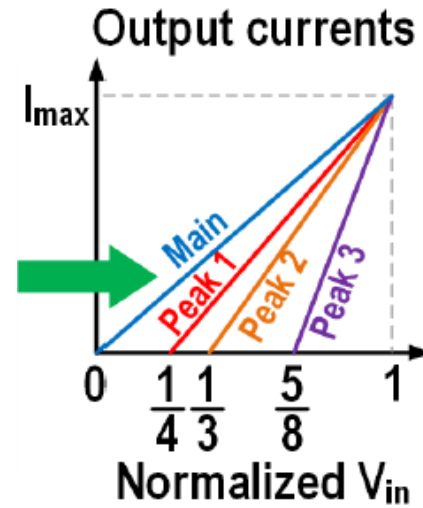
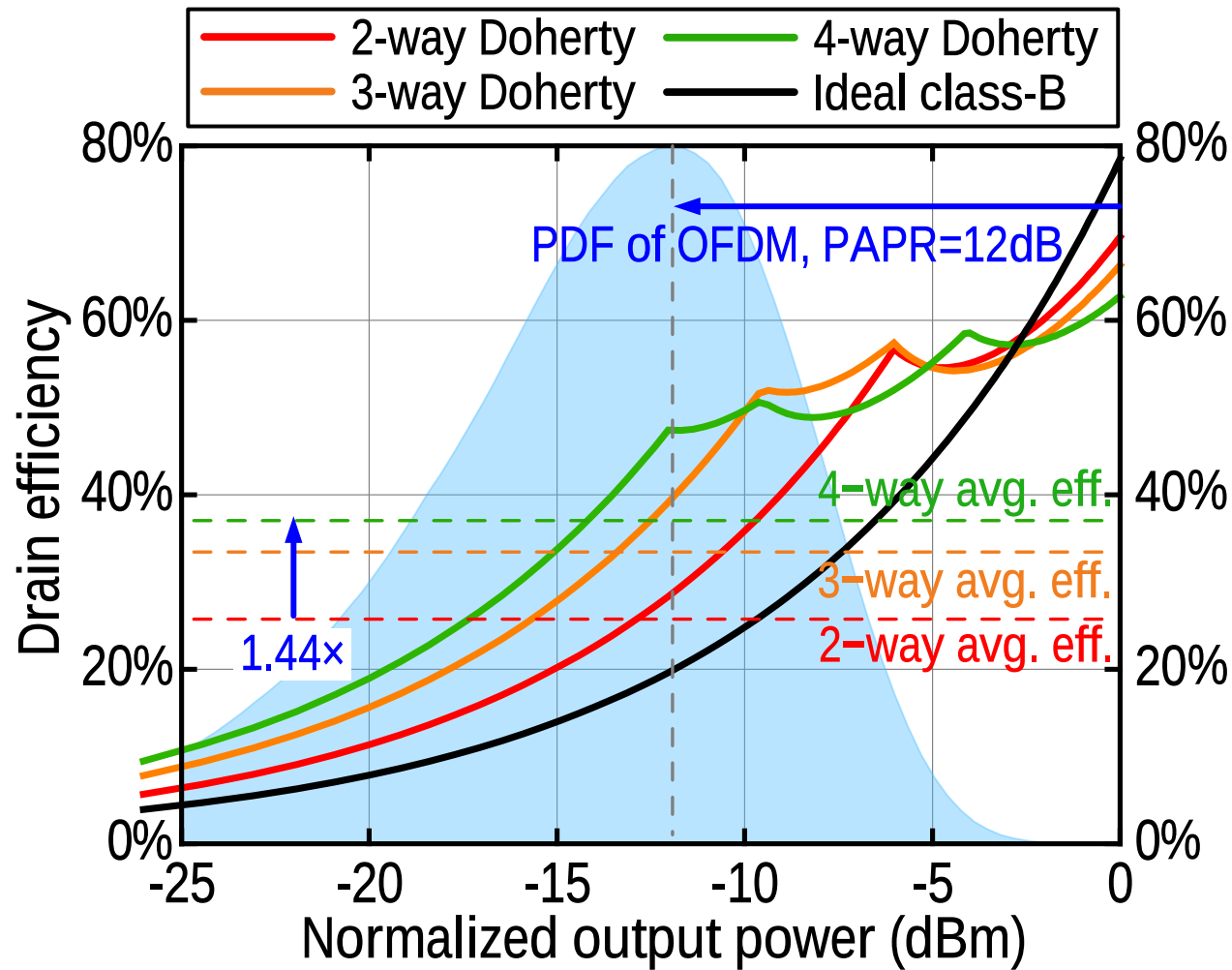
- Three-way Doherty



X. Zhang, H. Guo, and T. Chi, "A Millimeter-Wave Four-Way Doherty Power Amplifier with Over-GHz Modulation Bandwidth," JSSC (under review)

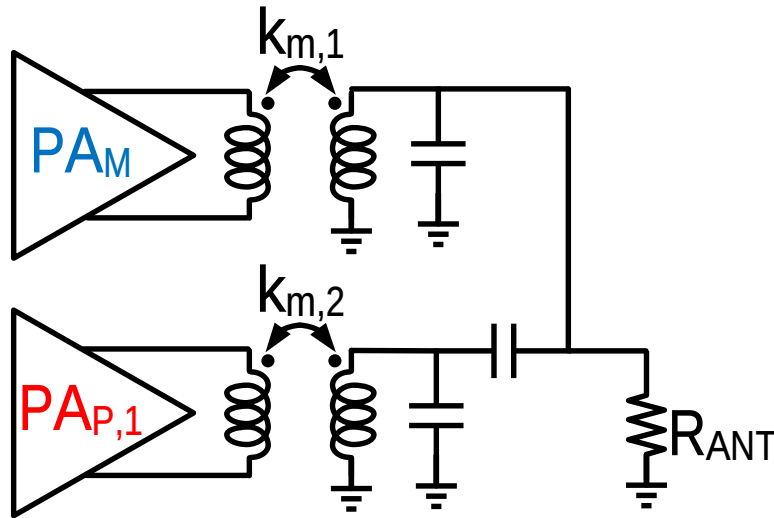


Four-Way Doherty PA

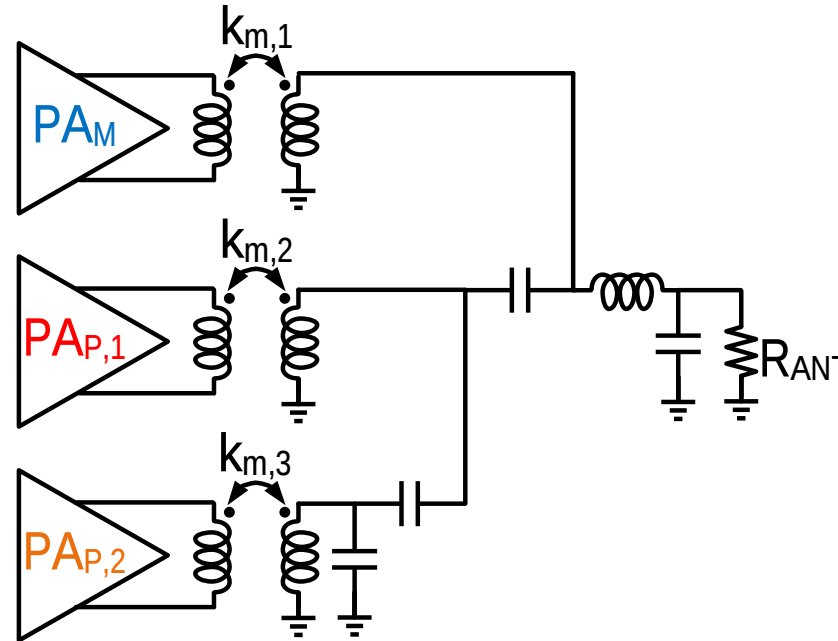


N-Way Doherty PA: Transformer-Based Network

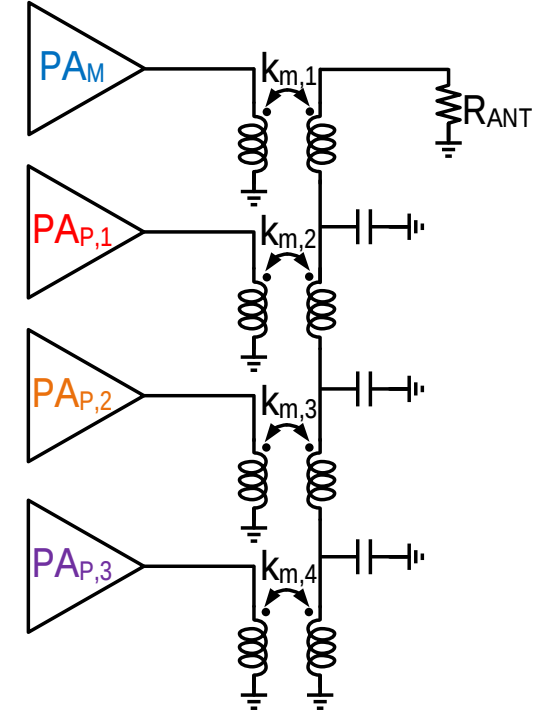
- Existing N-way Doherty networks require at least N transformers



2-way Doherty with
2 transformers
[S. Hu, JSSC'19]



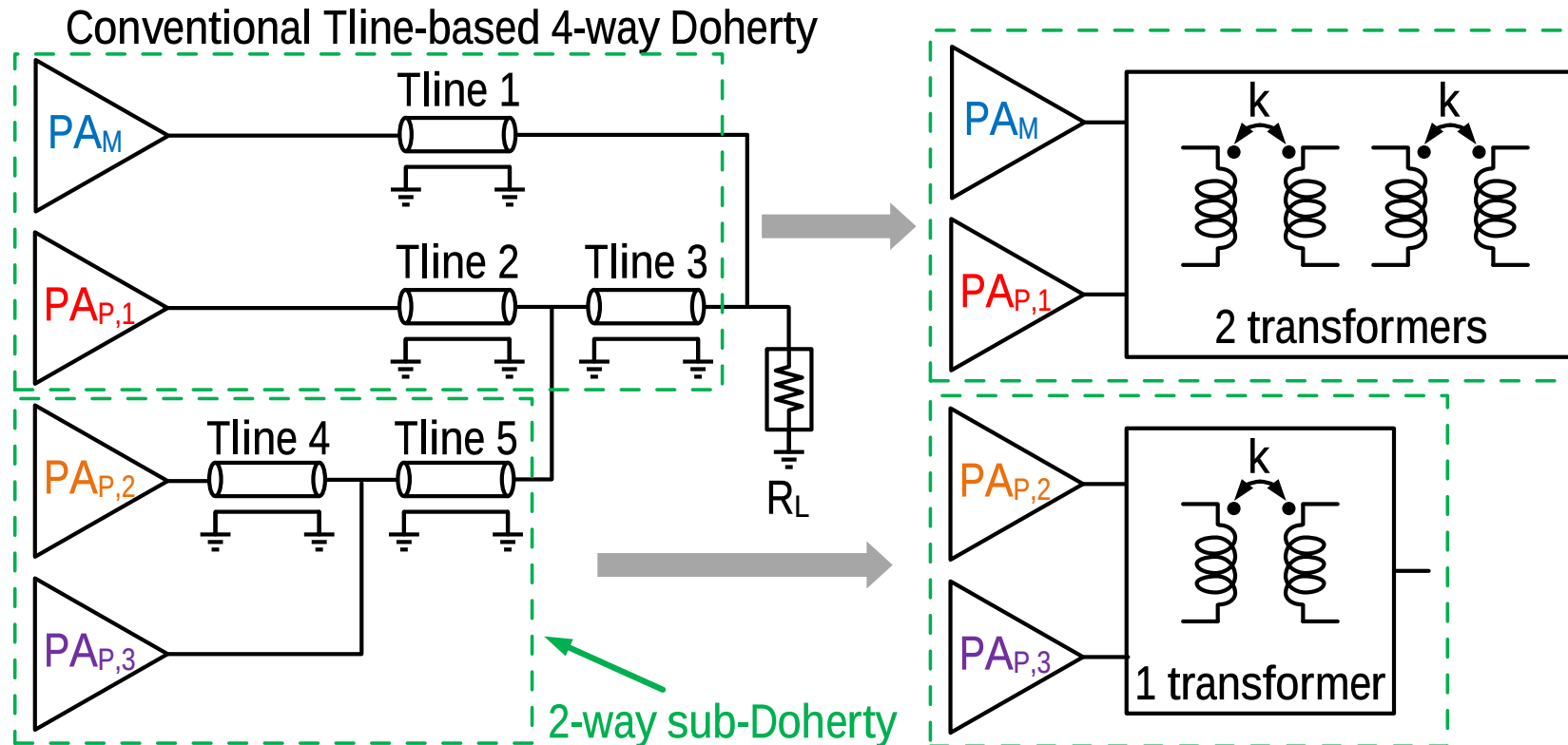
3-way Doherty with
3 transformers
[A. Kumaran, RFIC'23]



4-way Doherty with
4 transformers
[M. Mortazavi, JSSC'22]

N-Way Doherty Network Synthesis Flow

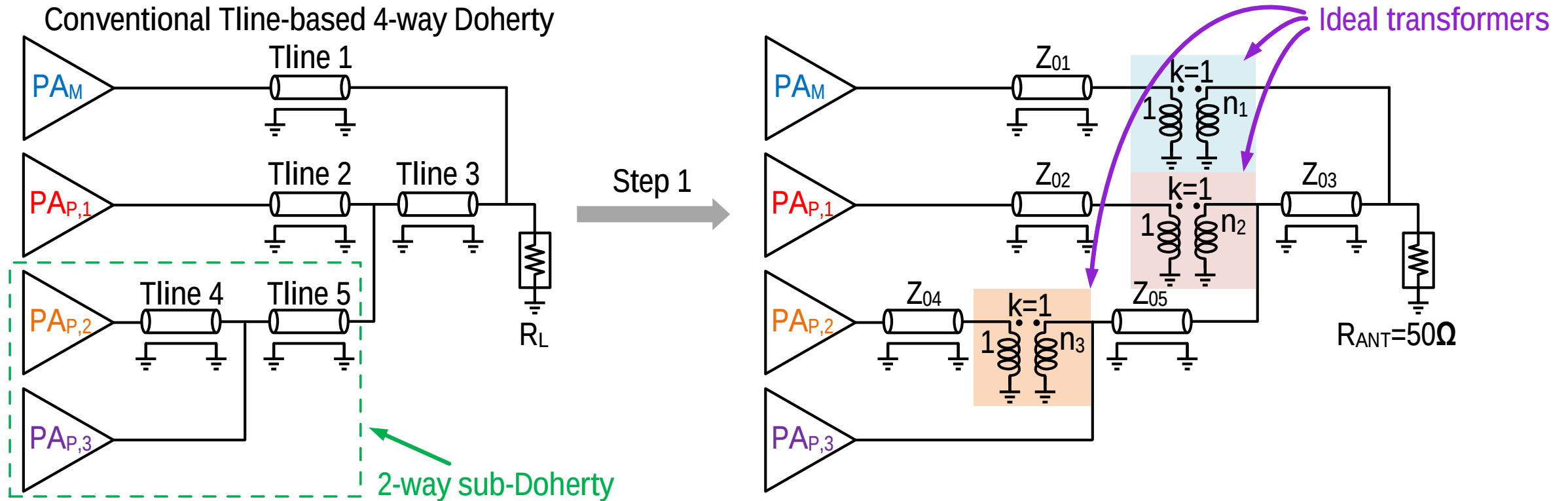
- Key insight: last 2-way sub-Doherty can be implemented using a single transformer



X. Zhang, H. Guo, and T. Chi, "[A 47GHz 4-way Doherty PA with 23.7dBm P1dB and 21.7% / 13.1% PAE at 6 / 12dB Back-off Supporting 2000MHz 5G NR 64-QAM OFDM](#)," ISSCC 2024.

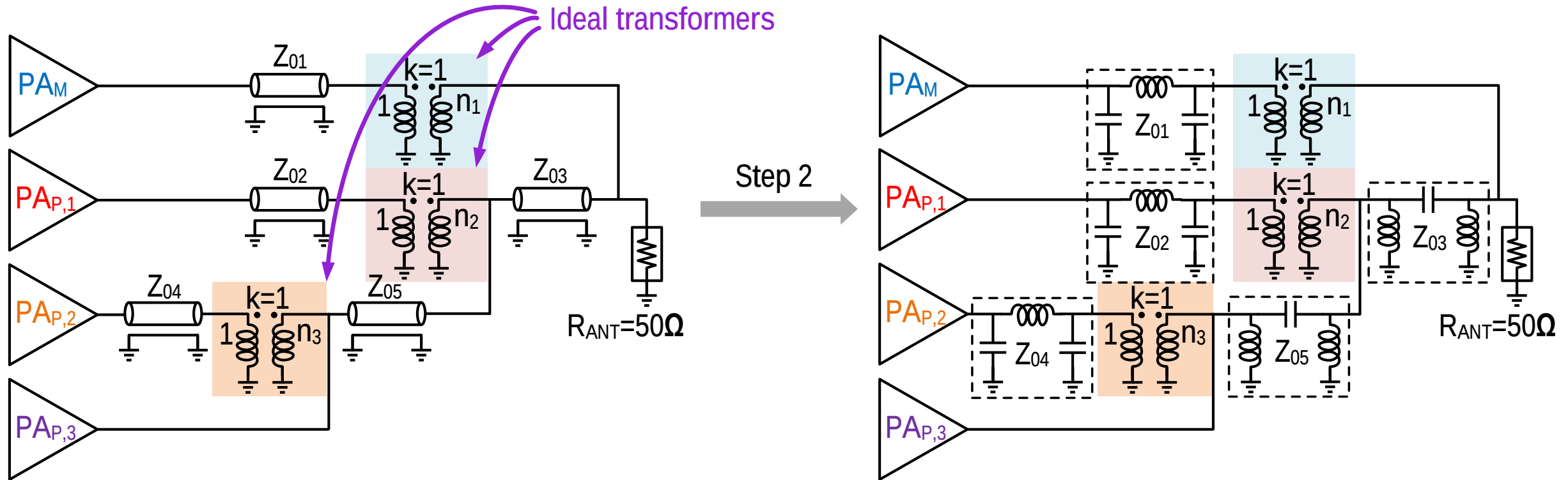
N-Way Doherty Network Synthesis Flow

- Step 1: insert 3 ideal transformers



N-Way Doherty Network Synthesis Flow

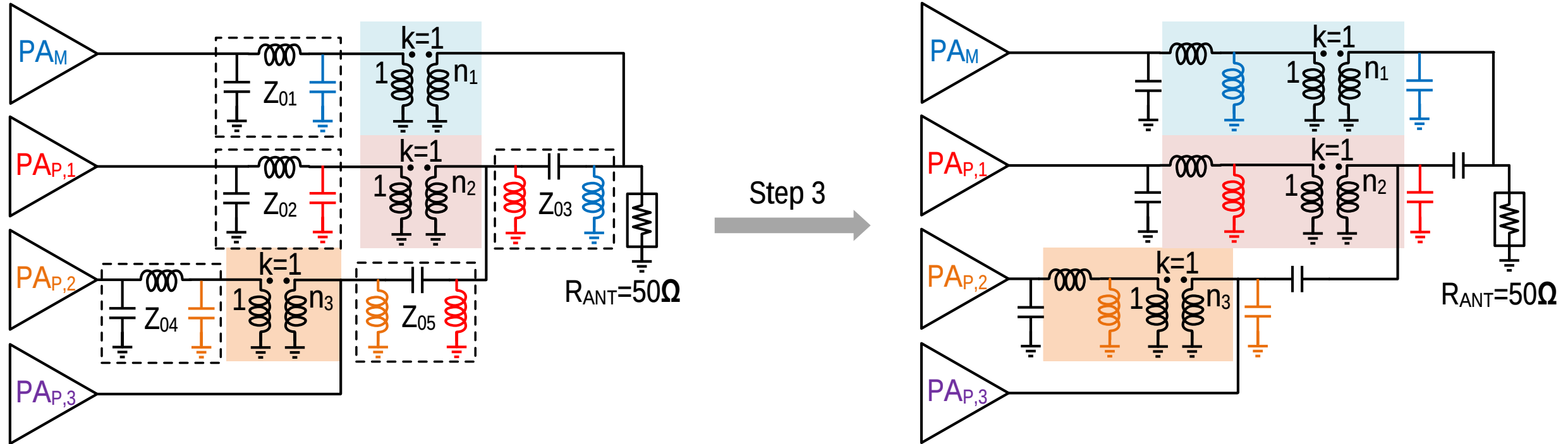
- Step 2: replace Tlines with π network





N-Way Doherty Network Synthesis Flow

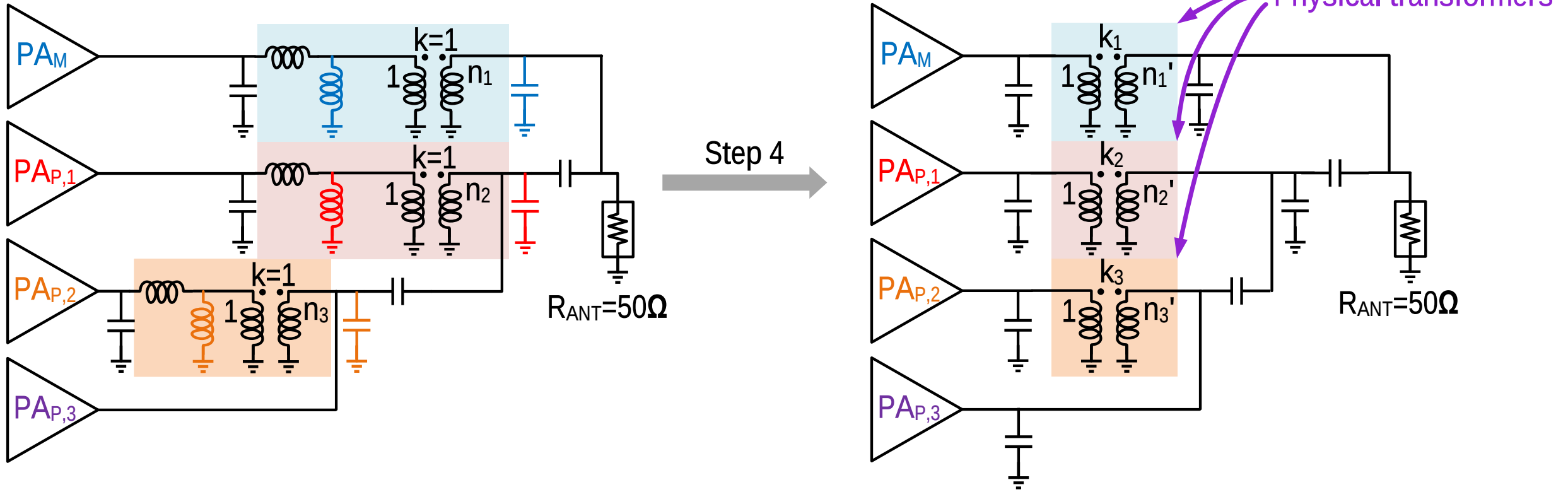
- Step 3: re-arrange components marked in colors





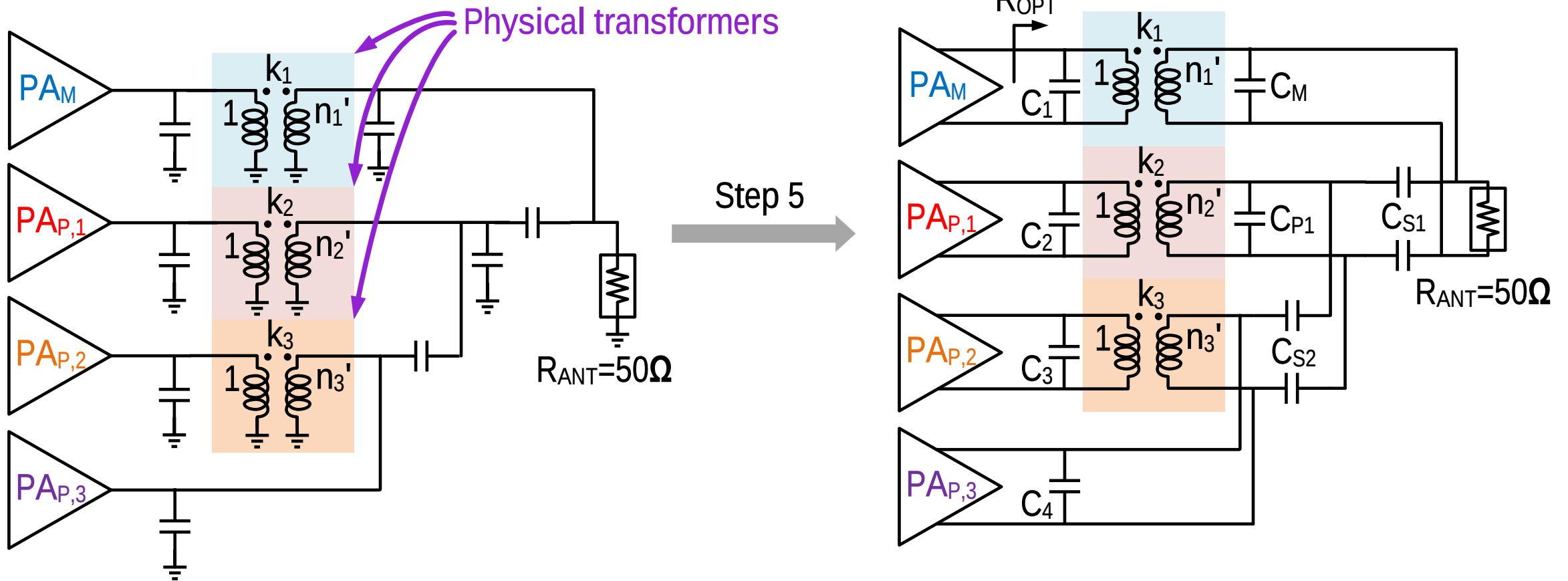
N-Way Doherty Network Synthesis Flow

- Step 4: consolidate into 3 physical transformers



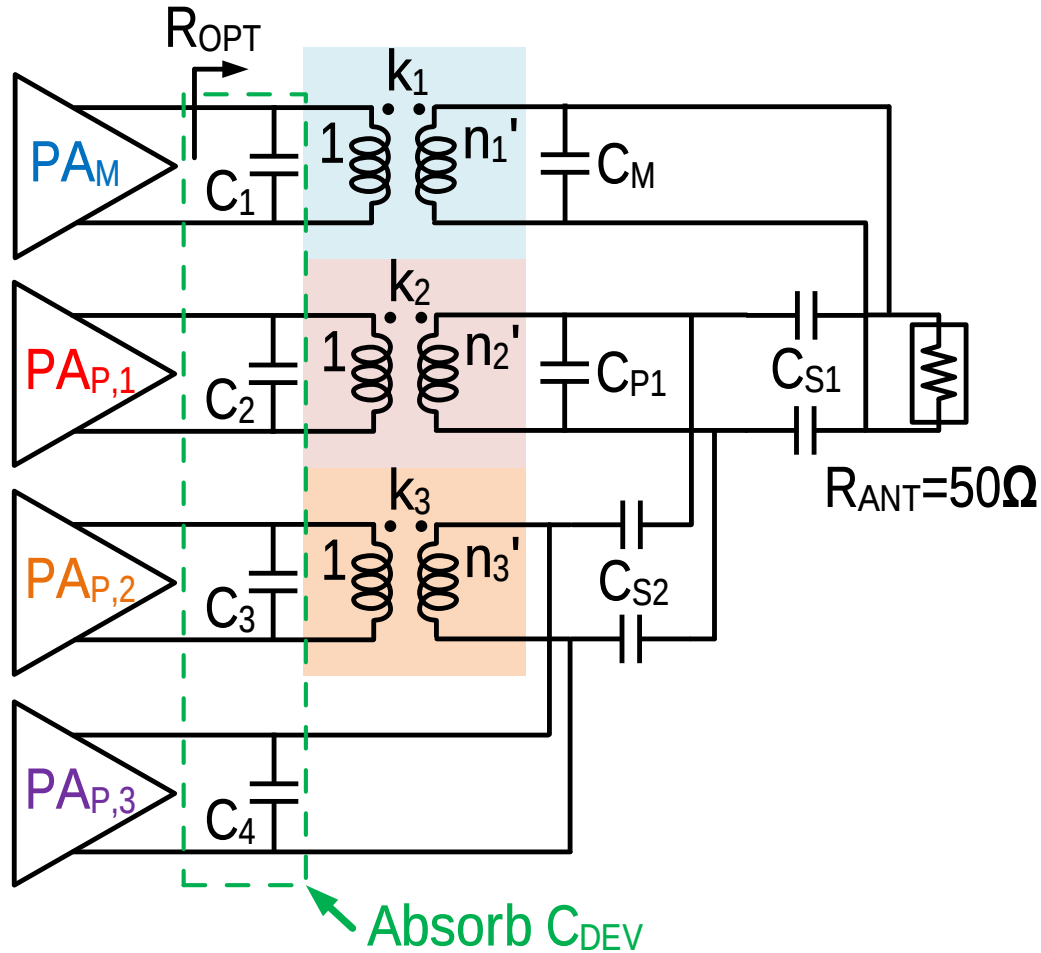
N-Way Doherty Network Synthesis Flow

- Step 5: convert to differential





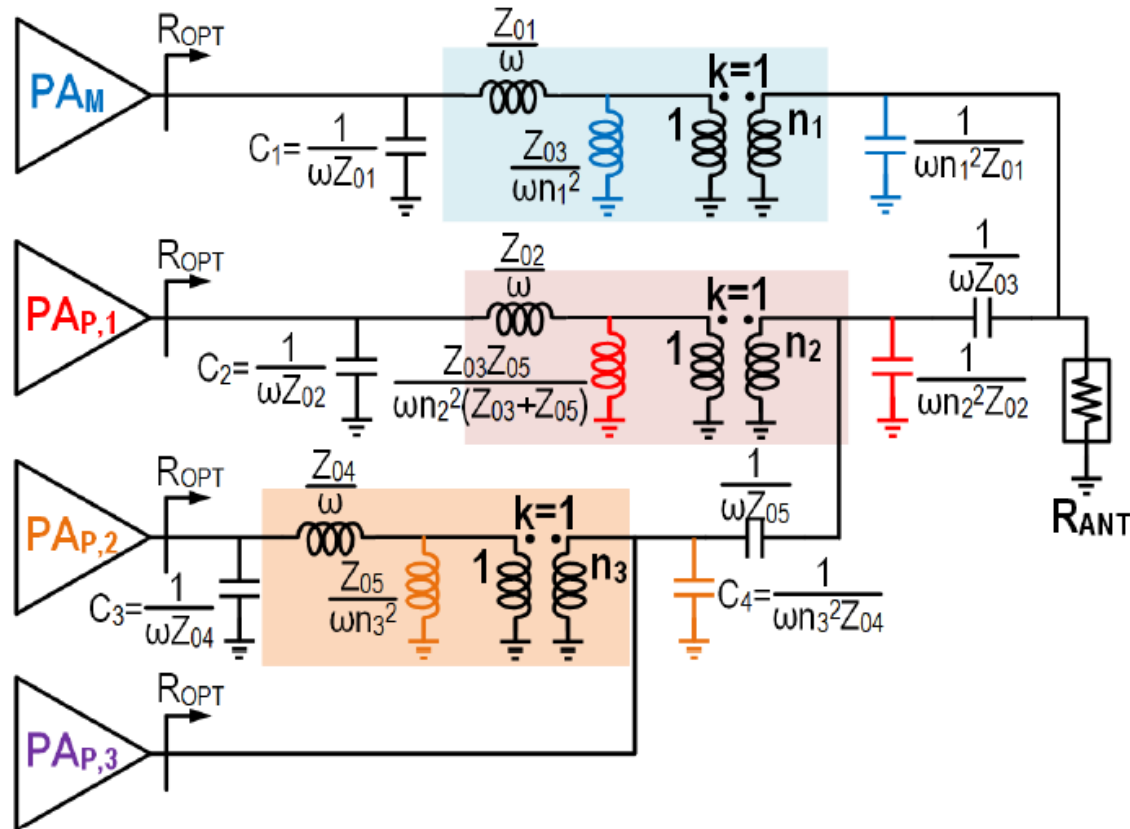
N-Way Doherty Network Synthesis Flow



- 4-way Doherty with only 3 transformers
- Interface with differential power cells
- Impedance transformation from R_{ANT} to R_{OPT}
- Absorb transistor device parasitic C_{DEV}



N-Way Doherty Network Synthesis Flow: Design Space



$Q_{\text{Loadpull}} = \omega R_{\text{OPT}} C_{\text{DEV}} = \text{constant for given technology and frequency}$

$C_1 - C_4$ fully absorb C_{DEV} :

$$C_{\text{DEV}} \leq C_1 \rightarrow Q_{\text{Loadpull}} \leq n_1 \sqrt{\frac{R_{\text{OPT}}}{2R_{\text{ANT}}}}$$

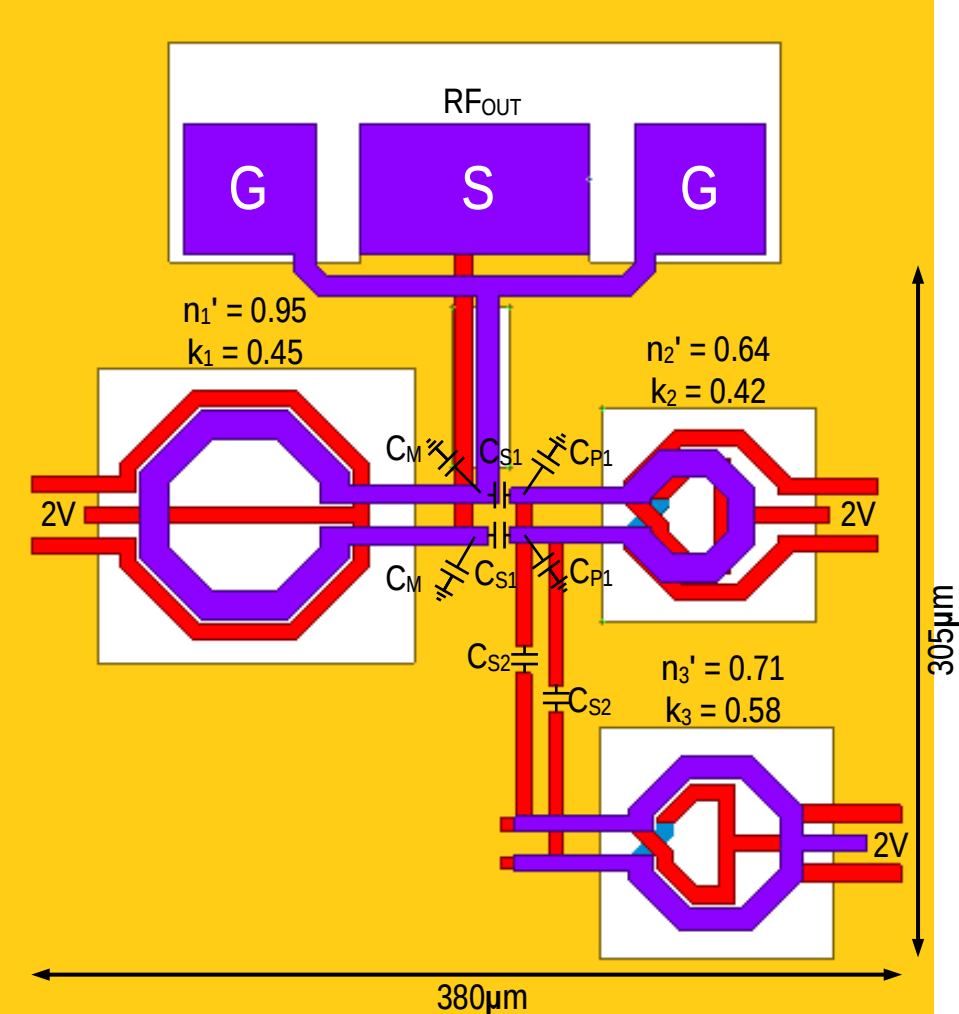
$$C_{\text{DEV}} \leq C_2 \rightarrow Q_{\text{Loadpull}} \leq \frac{2n_2}{3Z_{03}} \sqrt{2R_{\text{OPT}}R_{\text{ANT}}}$$

$$C_{\text{DEV}} \leq C_3 \rightarrow Q_{\text{Loadpull}} \leq n_3$$

$$C_{\text{DEV}} \leq C_4 \rightarrow Q_{\text{Loadpull}} \leq \frac{1}{n_3}$$



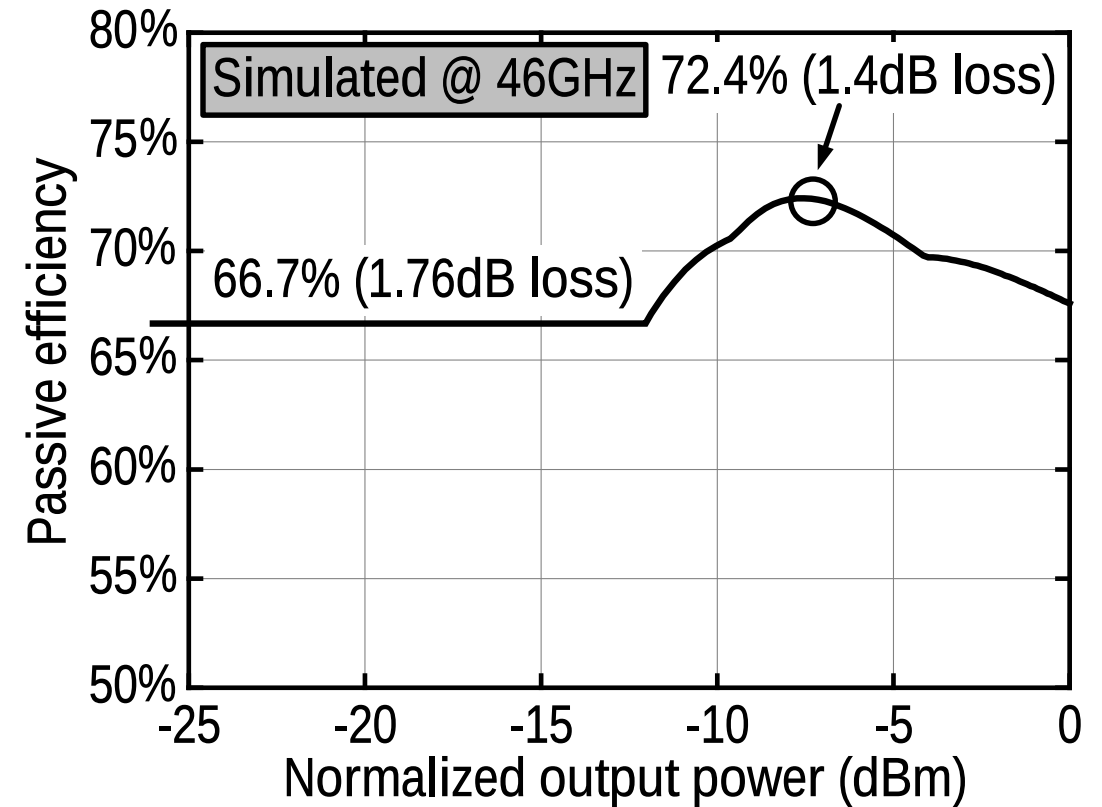
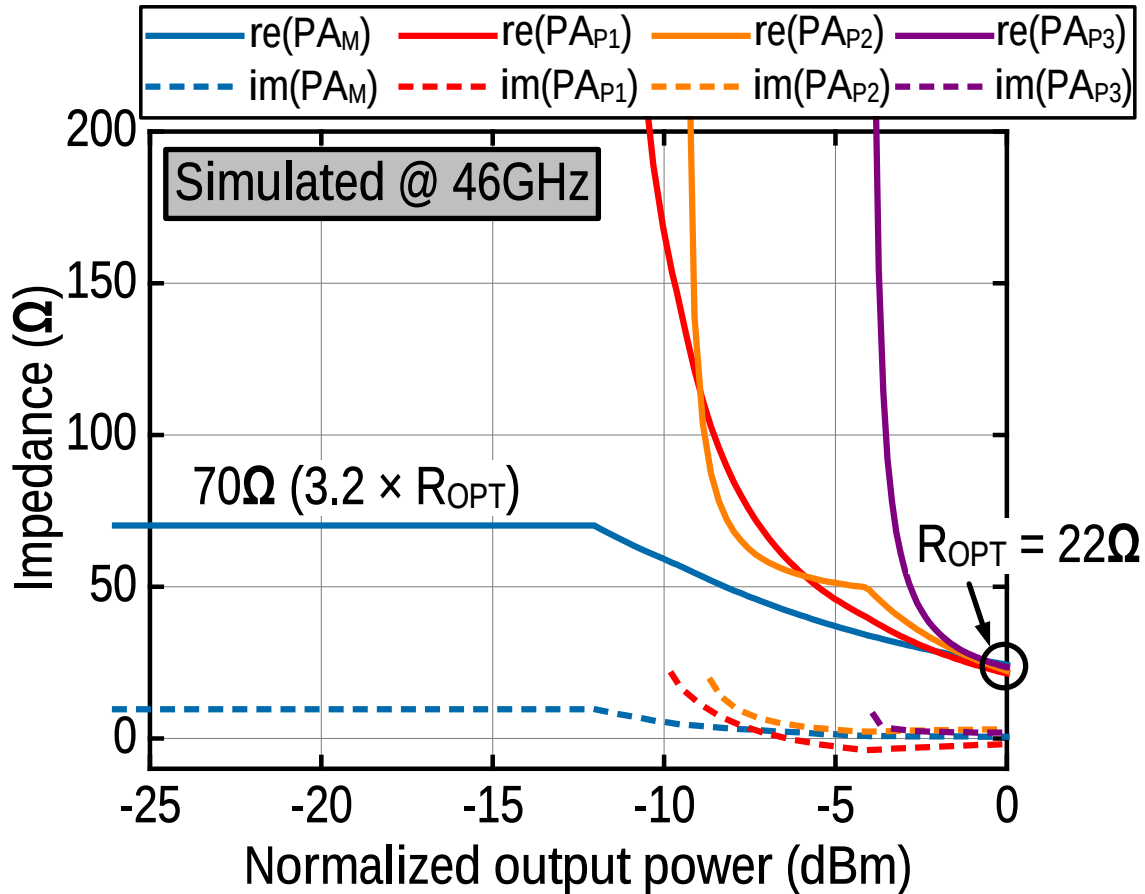
Transformer Design Parameters



Transformer Design Parameter	Design Equation	Theo. Value	EM Sim.
k_1	$\sqrt{\frac{Z_{03}}{Z_{03} + 2n_1\sqrt{2R_{OPT}R_{ANT}}}}$	0.44	0.45
n_1'	k_1n_1	0.92	0.95
L_1	$\frac{Z_{03}}{\omega n_1^2 k_1^2}$	197pH	169pH
k_2	$\sqrt{\frac{1}{1 + 2n_2(1 + 3/4\sqrt{2R_{OPT}/R_{ANT}})}}$	0.40	0.42
n_2'	k_2n_2	0.60	0.64
L_2	$\frac{Z_{03}}{\omega n_2^2 k_2^2 (1 + 4/3\sqrt{R_{ANT}/2R_{OPT}})}$	190pH	175pH
k_3	$\sqrt{\frac{Z_{03}}{Z_{03} + 4/3n_3\sqrt{2R_{OPT}R_{ANT}}}}$	0.65	0.58
n_3'	k_3n_3	0.65	0.71
L_3	$\frac{3/4 Z_{03}\sqrt{2R_{OPT}/R_{ANT}}}{\omega n_3^2 k_3^2}$	278pH	221pH

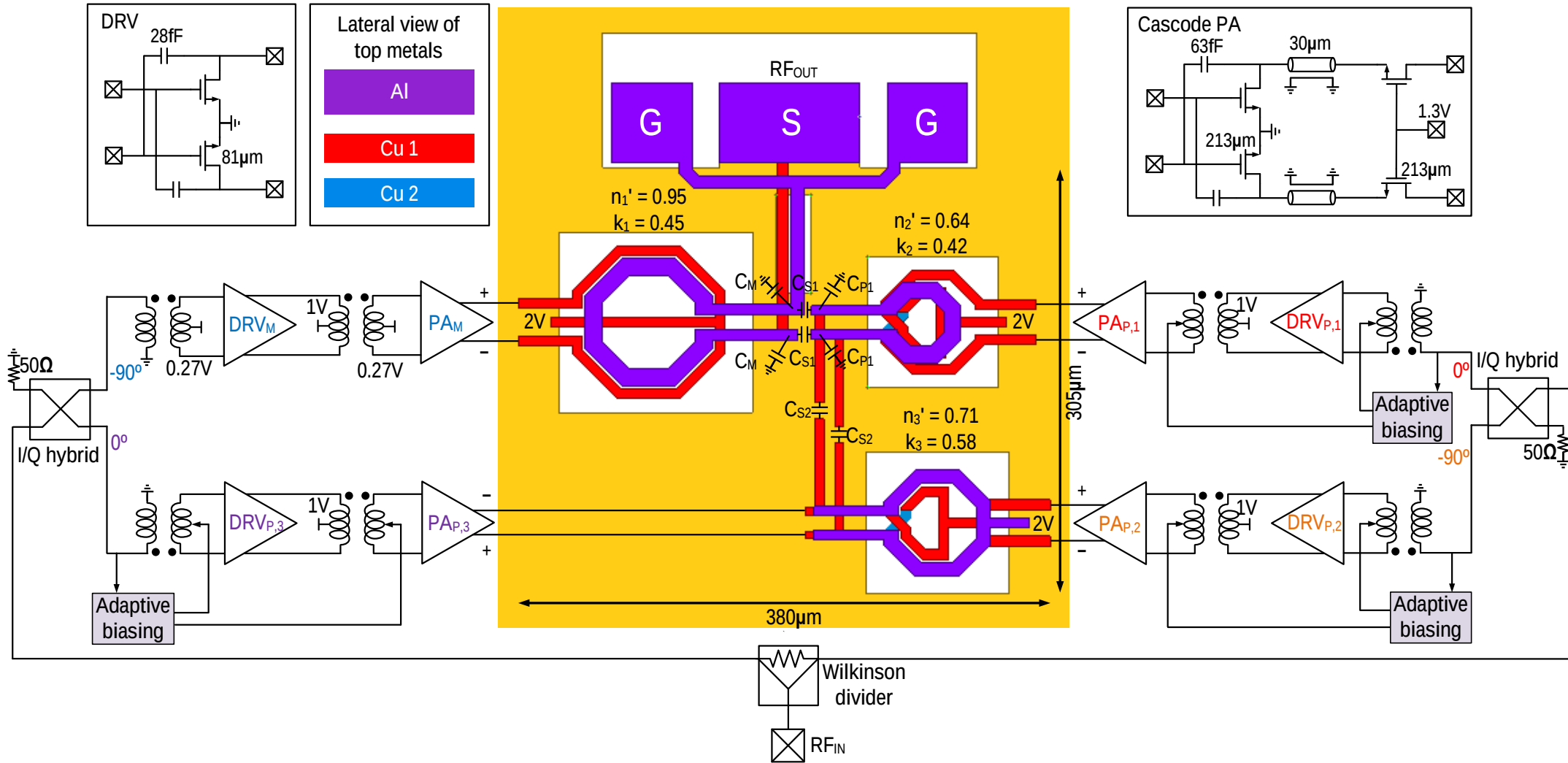


EM-Simulated Doherty Load Modulation



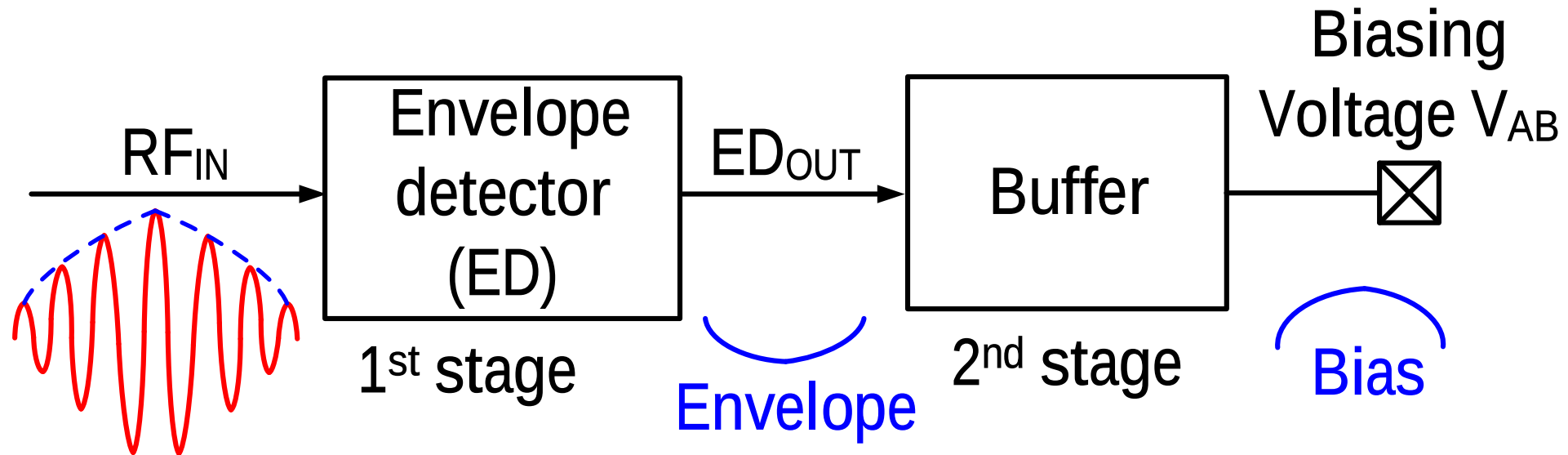
- Low loss due to reduced number of transformers

Four-Way Doherty PA: Top-Level Schematic





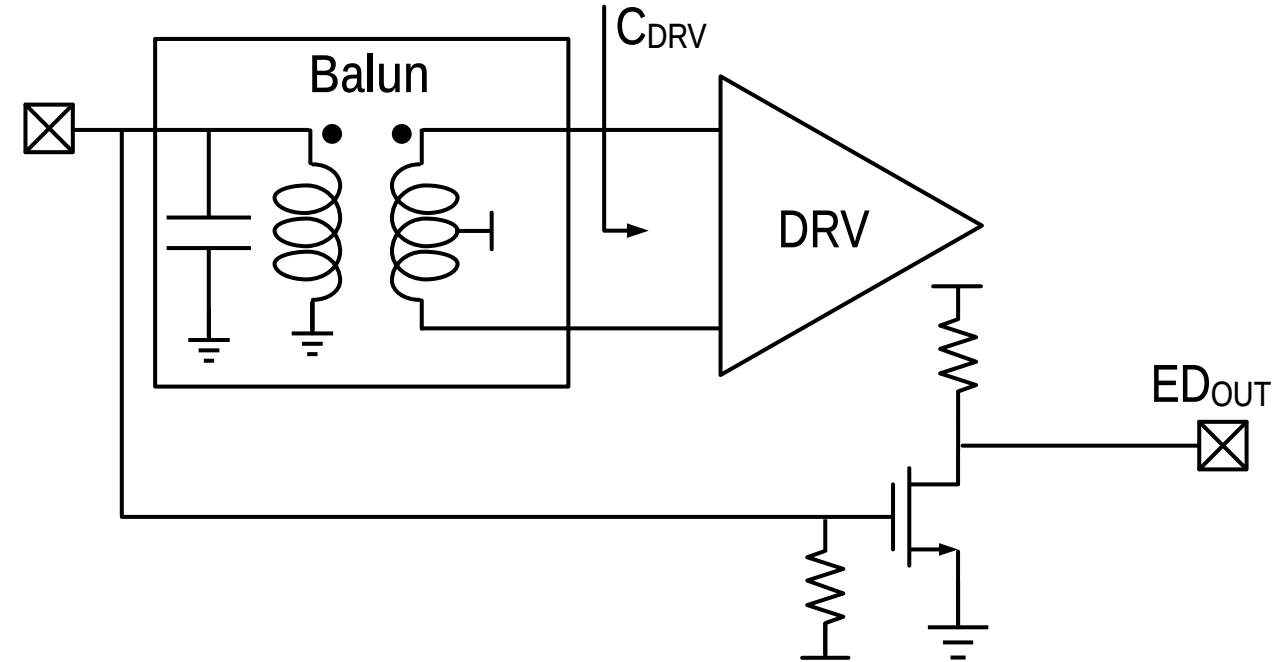
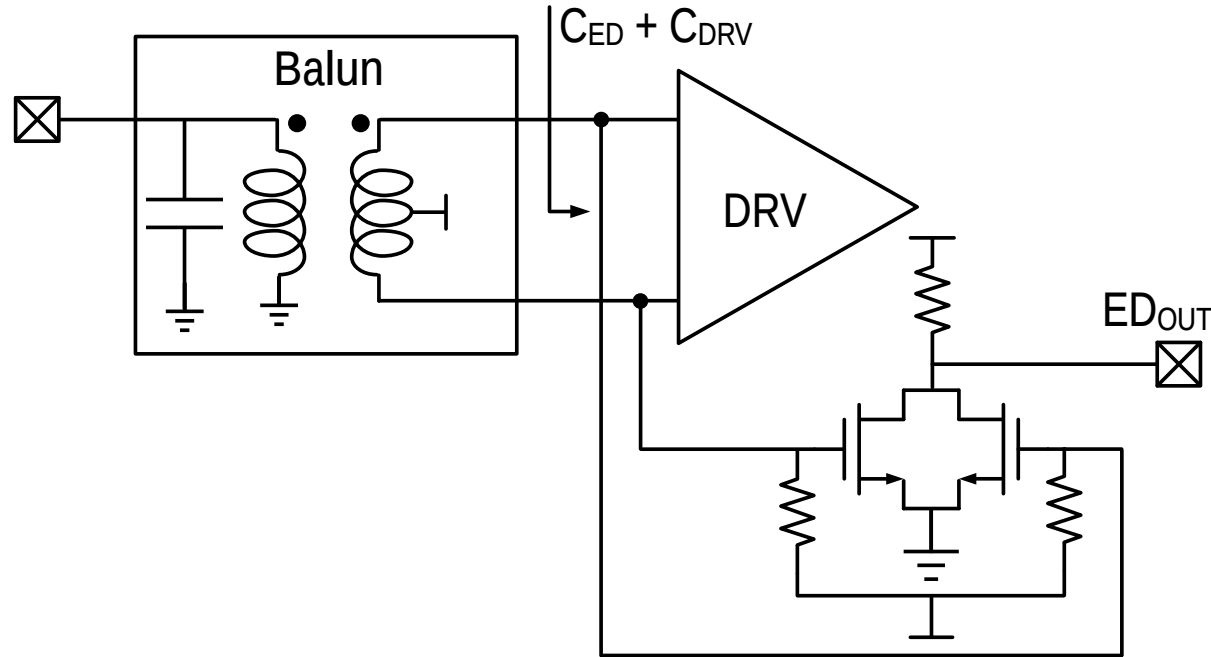
Adaptive Biasing Circuit



- Two stages: envelope detector (ED) + buffer
- ED: extract signal envelope
- Buffer: drive large parasitic capacitance of power cells



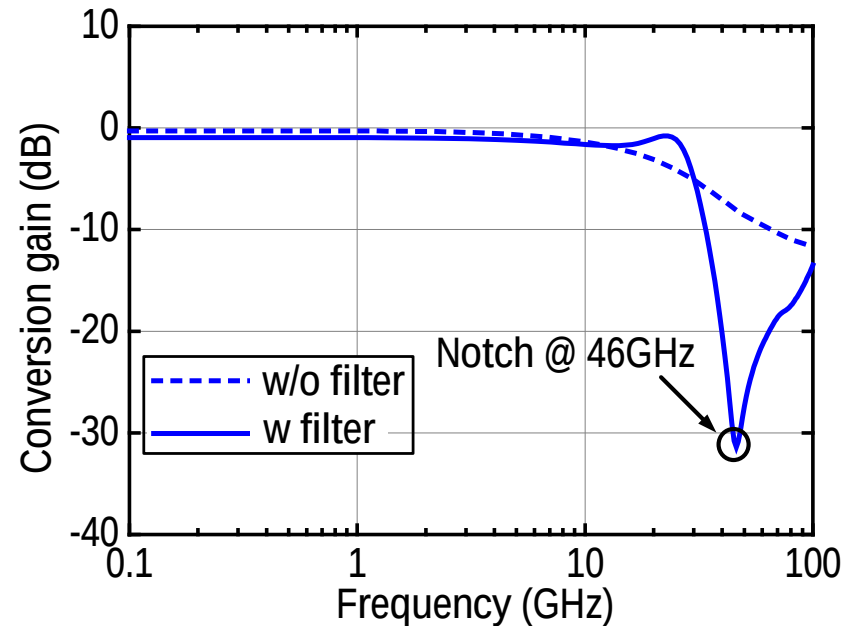
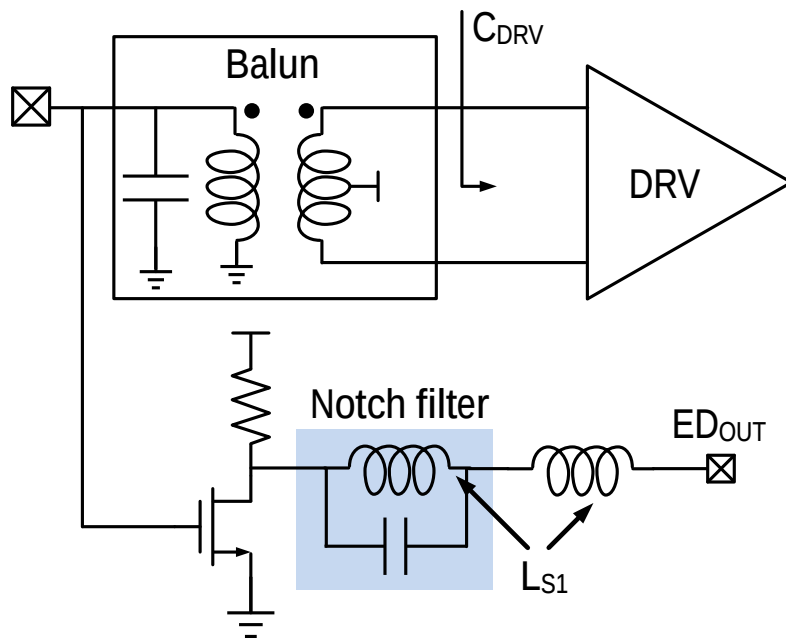
Envelope Detector (ED)



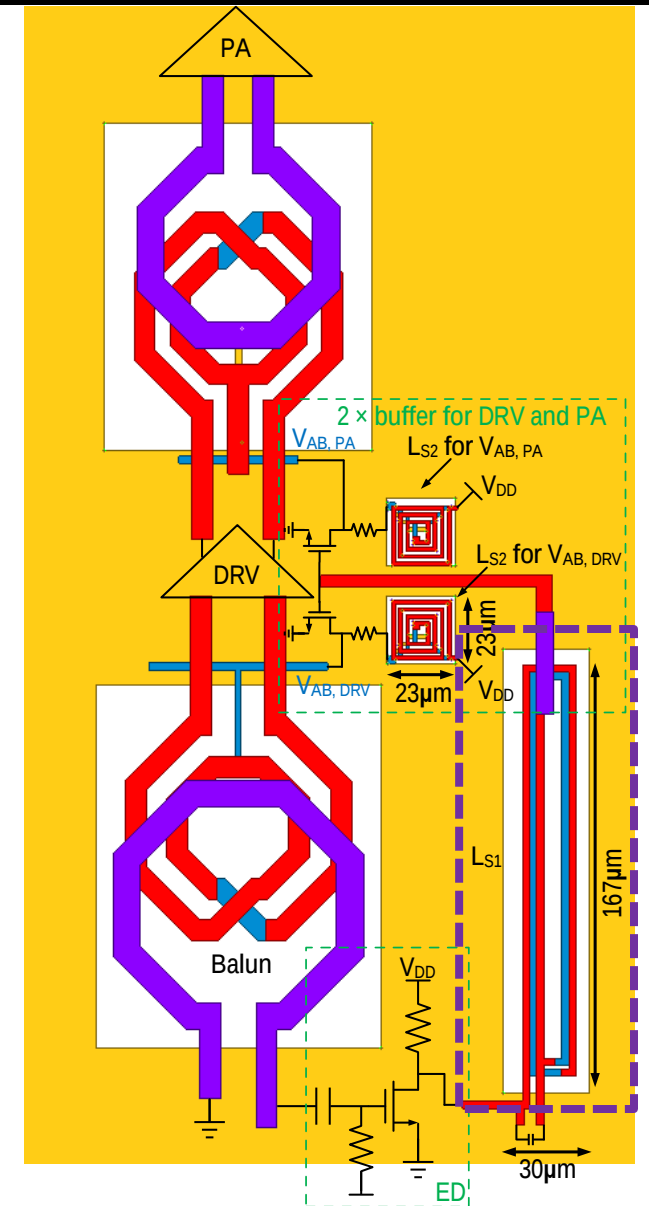
- 😊 ● Fundamental leakage rejection
- ☹ ● Large $C_{ED} + C_{DRV} \rightarrow \sim 1\text{dB}$ extra balun loss and narrower BW

- 😊 ● Naturally absorb C_{ED} at balun input
- ☹ ● Severe fundamental leakage at ED_{out}

Single-Ended ED with a Notch Filter

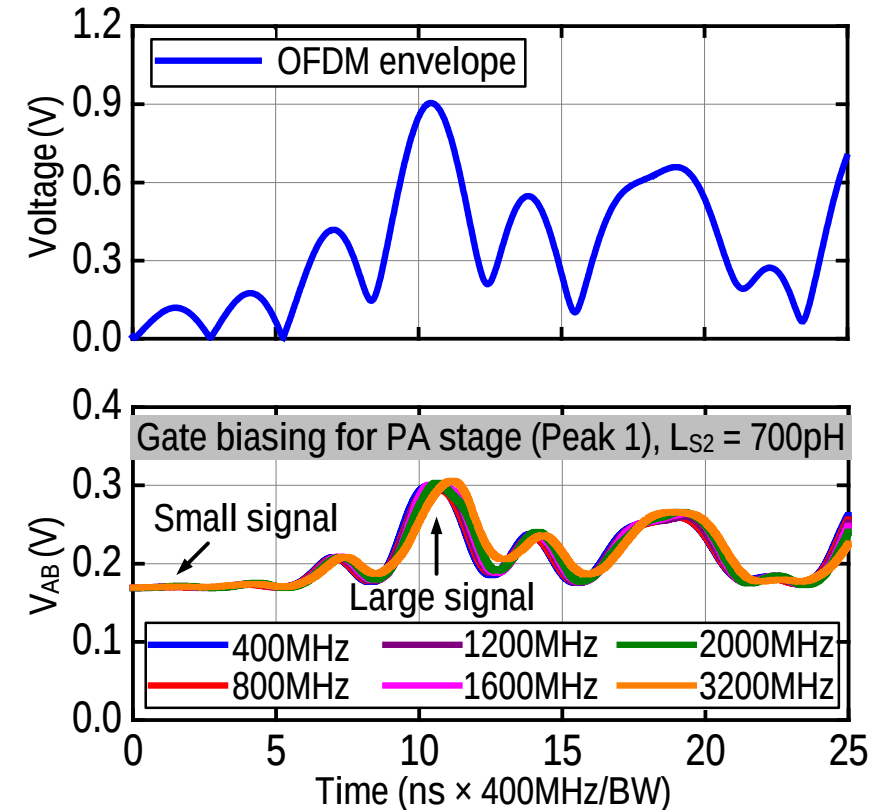
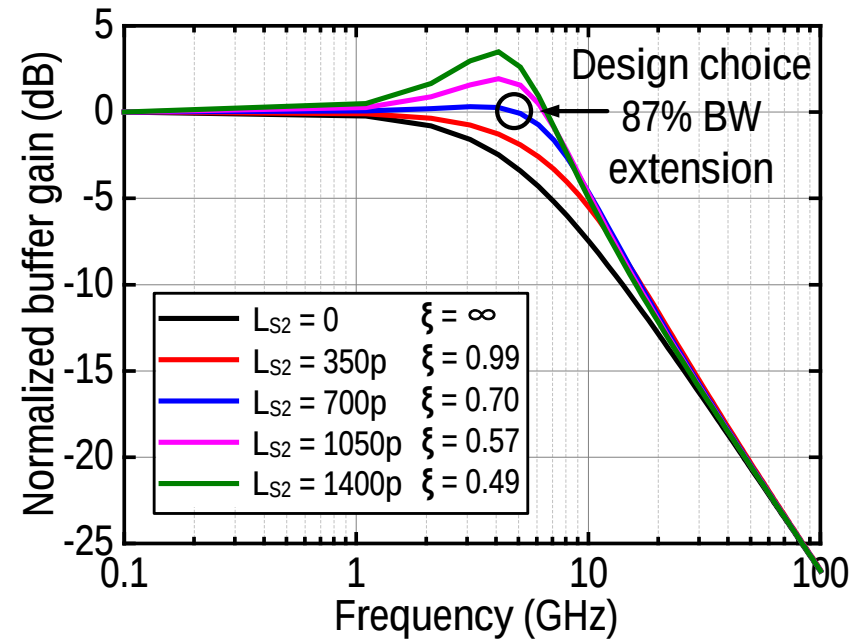
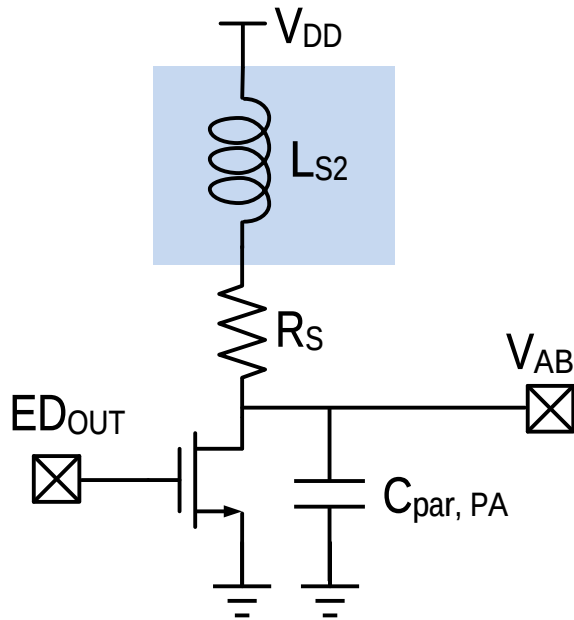


- Naturally absorb C_{ED} at balun input
- Fundamental leakage rejection by notch filter
- Fit L_{S1} into existing routing space between ED and buffer
→ negligible area overhead





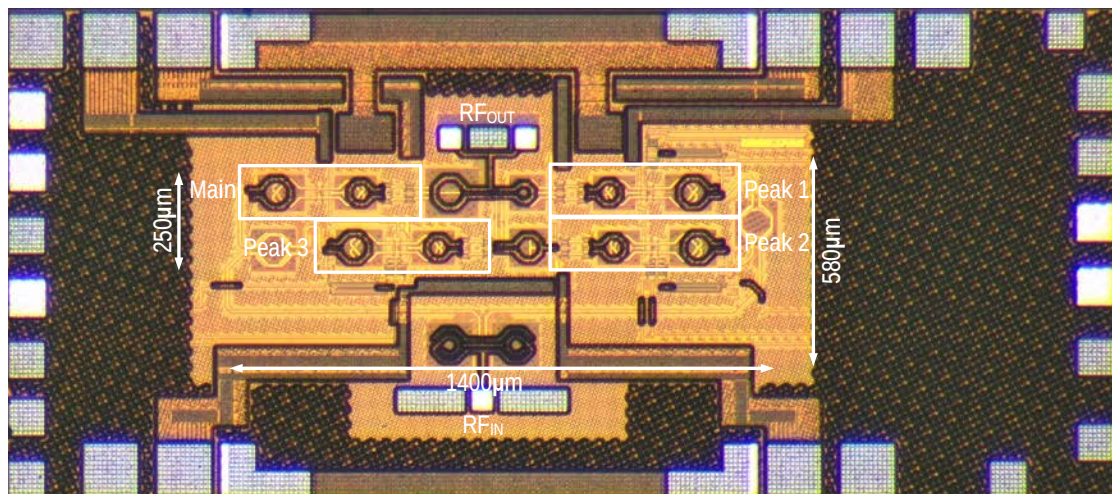
Shunt Peaking to Increase Buffer Bandwidth



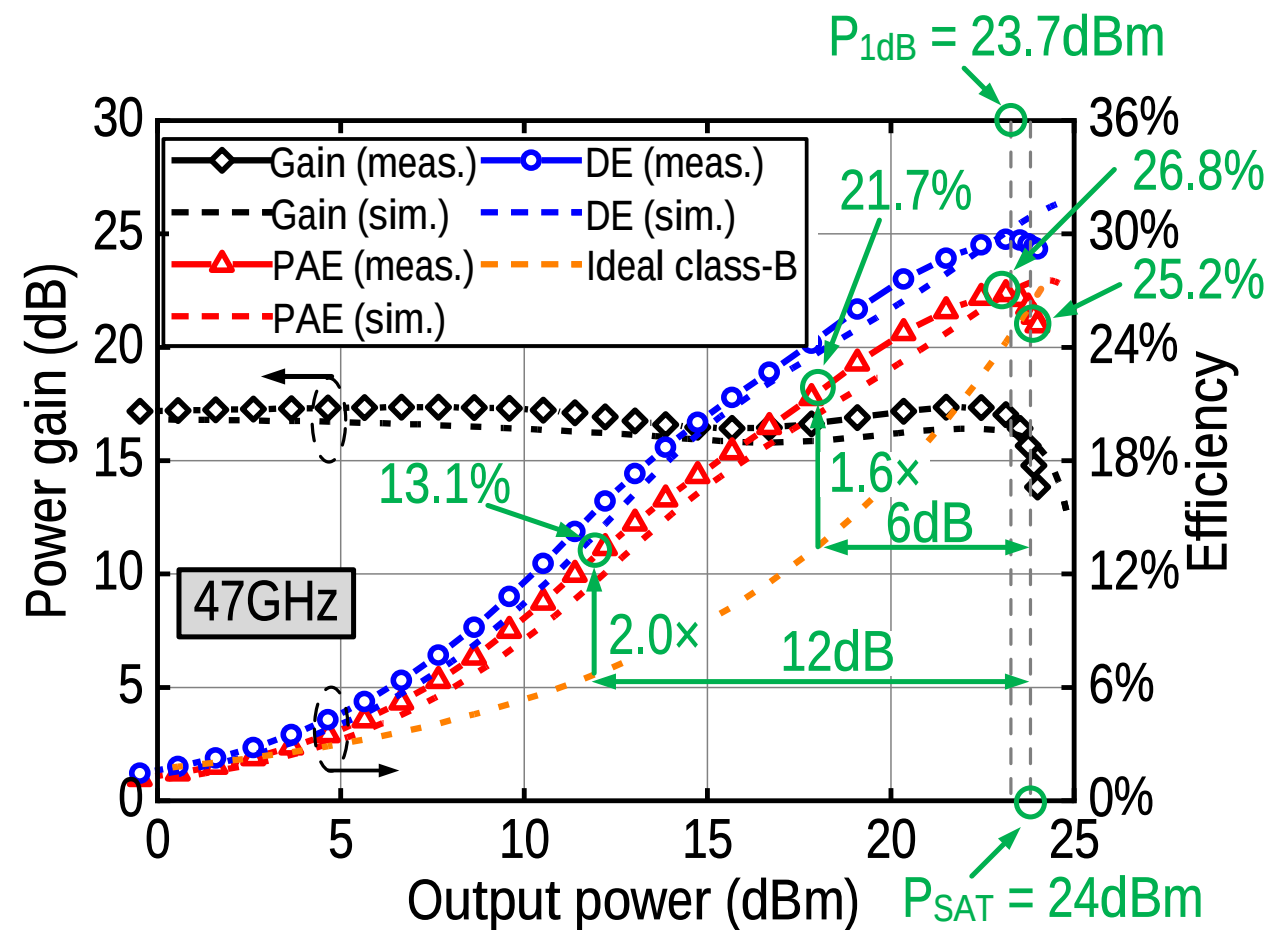
- $L_{S2}=700\text{p}$ ($\xi = 0.7$) $\rightarrow$ 87% BW extension
- Simulated transient waveform of V_{AB} under 400-3200MHz OFDM modulation $\rightarrow$ error remains small up to 2000MHz BW



47-GHz Four-Way Doherty PA: Measurement Results



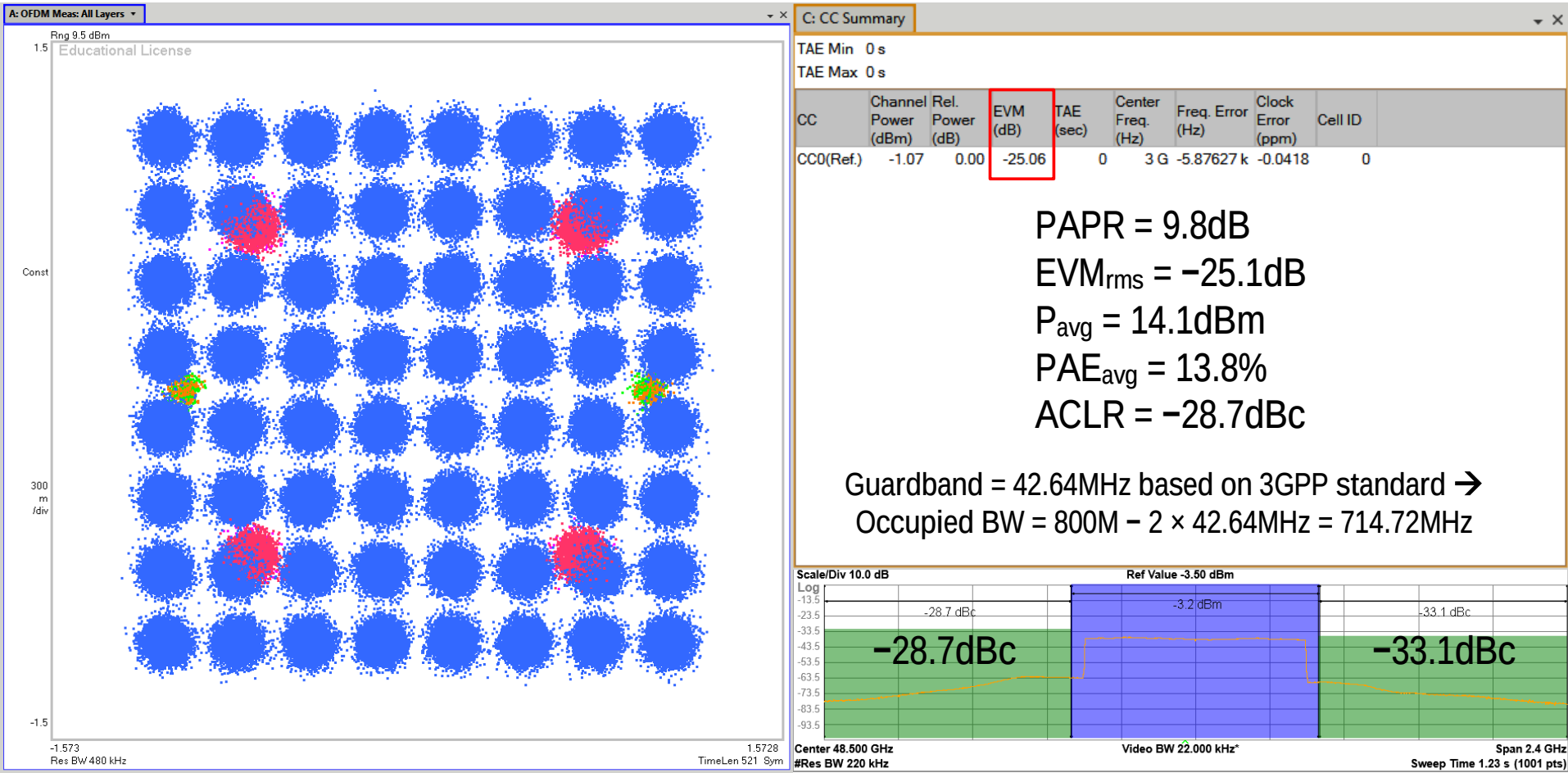
- GlobalFoundries 45nm CMOS SOI process
- Active area: $1.4 \times 0.58 \text{ mm}^2 = 0.81 \text{ mm}^2$





47-GHz Four-Way Doherty PA: Measurement Results

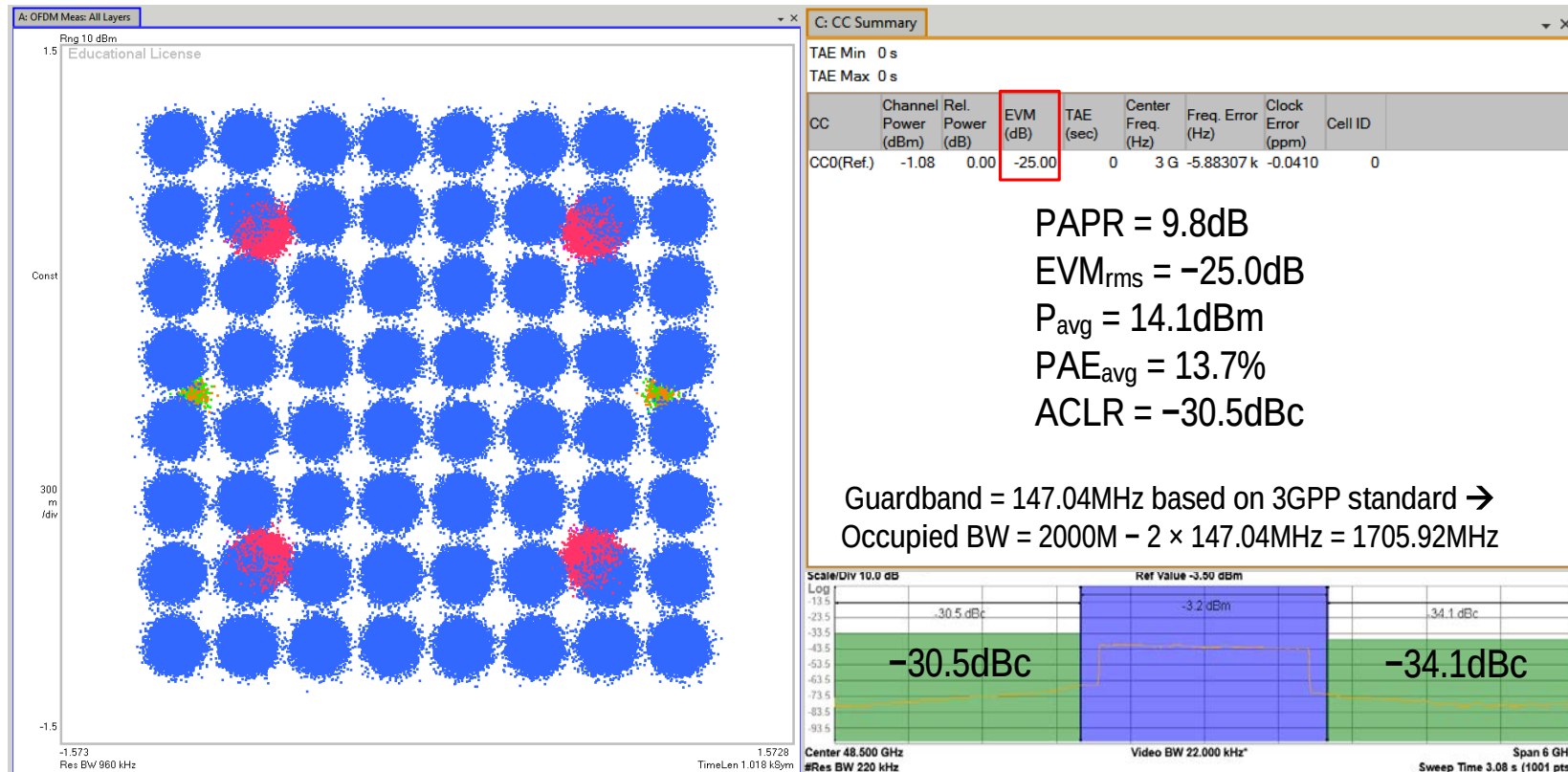
800MHz 5G NR FR2 1-CC 64-QAM OFDM Signal





47-GHz Four-Way Doherty PA: Measurement Results

2000MHz 5G NR FR2 1-CC 64-QAM OFDM Signal



- High-speed envelope tracking capability → 1st silicon PA demonstration of 2000MHz channel BW for 5G NR OFDM along with back-off efficiency enhancement up to 12dB back-off.



47-GHz Four-Way Doherty PA: Performance Comparison

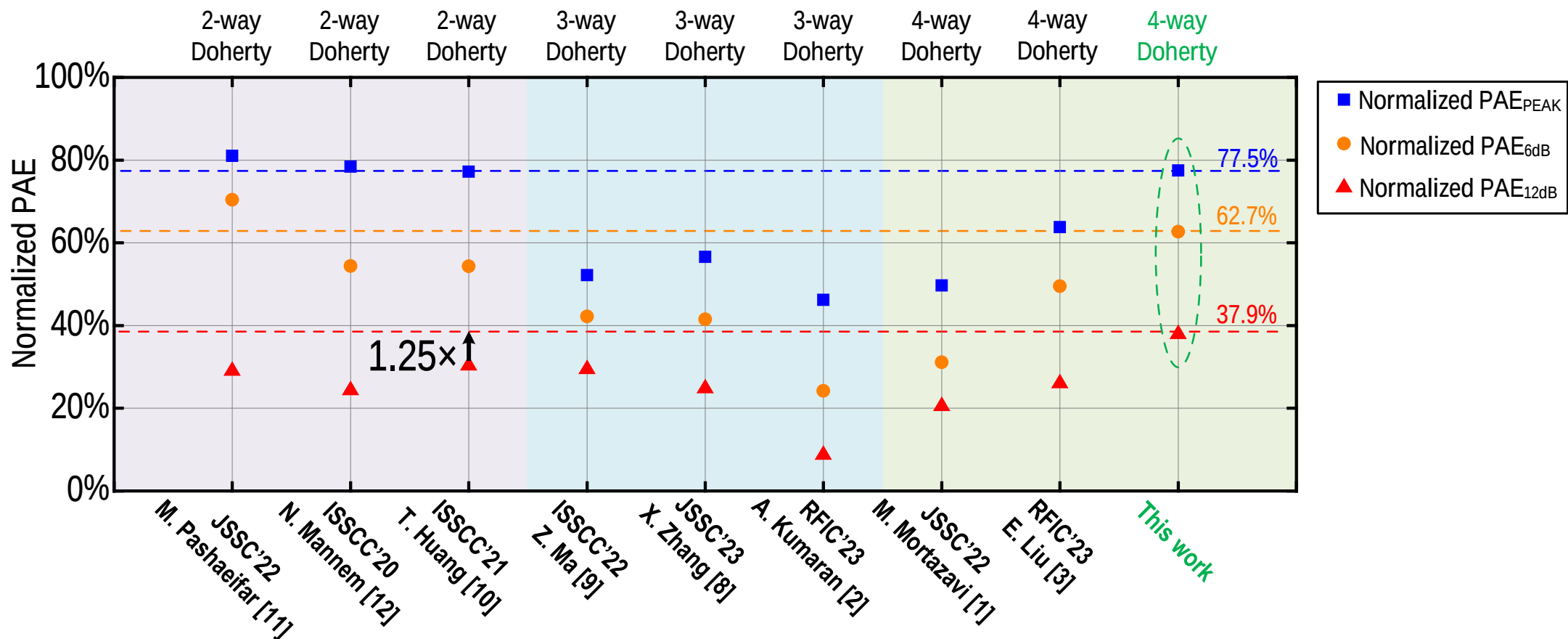
Specifications	This Work	M. Mortazavi JSSC'22	E. Liu RFIC'23	A. Kumaran RFIC'23	X. Zhang JSSC'23	Z. Ma ISSCC'22	T. Huang ISSCC'21
Architecture	Transformer-based 4-way Doherty	Digital polar + 4-way Doherty	Coupler-based 4-way Doherty	Balun-first 3-way Doherty	Coupled-inductor -based 3-way Doherty	Transformer- based 3-way Doherty	Continuous- coupler-based 2-way Doherty
Supply (V)	2 (PA), 1 (DRV)	1 (PA)	2 (PA), 1 (DRV)	1.1 (PA and DRV)	1.8 (PA), 1 (DRV)	2.4 (PA), 1.2 (DRV)	2 (PA), 1 (DRV)
Frequency (GHz)	47	29.5	40	26	38	28	47.5
Gain (dB)	17.1	N/A	11.8	20	15	16.1	12.5*
P _{SAT} (dBm)	24.0	18.7	22.7	20.7	18.9	25.5	21.7
P _{1dB} (dBm)	23.7	N/A	22.5	N/A	18.4	24.3	20.9
PAE _{PEAK}	26.8%	24% / 36% [†]	26.2%*	22.3%	23.3%	25.2%	26.7%
PAE @ P _{1dB}	26.3%	N/A	26.2%	N/A	23.0%	24.4%	26%
PAE _{6dB}	21.7%	15% / 33% [†]	20.6%*	11.7%	17.1%	20.4%	18.8%
PAE _{12dB}	13.1%	10% / 22% [†]	10.9%*	4.2%	10.2%*	14.2%	10.5%*
Normalized PAE _{PEAK} [#]	77.5%	49.7% / 74.5% [†]	63.6%	46.2%	56.6%	52.2%	77.2%
Normalized PAE _{6dB} [#]	62.7%	31.1% / 68.3% [†]	50.0%	24.2%	41.5%	42.2%	54.3%
Normalized PAE _{12dB} [#]	37.9%	20.7% / 45.5% [†]	26.4%	8.7%	24.8%	29.4%	30.3%
Technology	45nm CMOS SOI	40nm CMOS	45nm CMOS SOI	40nm CMOS	45nm CMOS SOI	55nm CMOS	45nm CMOS SOI

* Graphically estimated

[†]Last-stage drain efficiency



47-GHz Four-Way Doherty PA: Performance Comparison

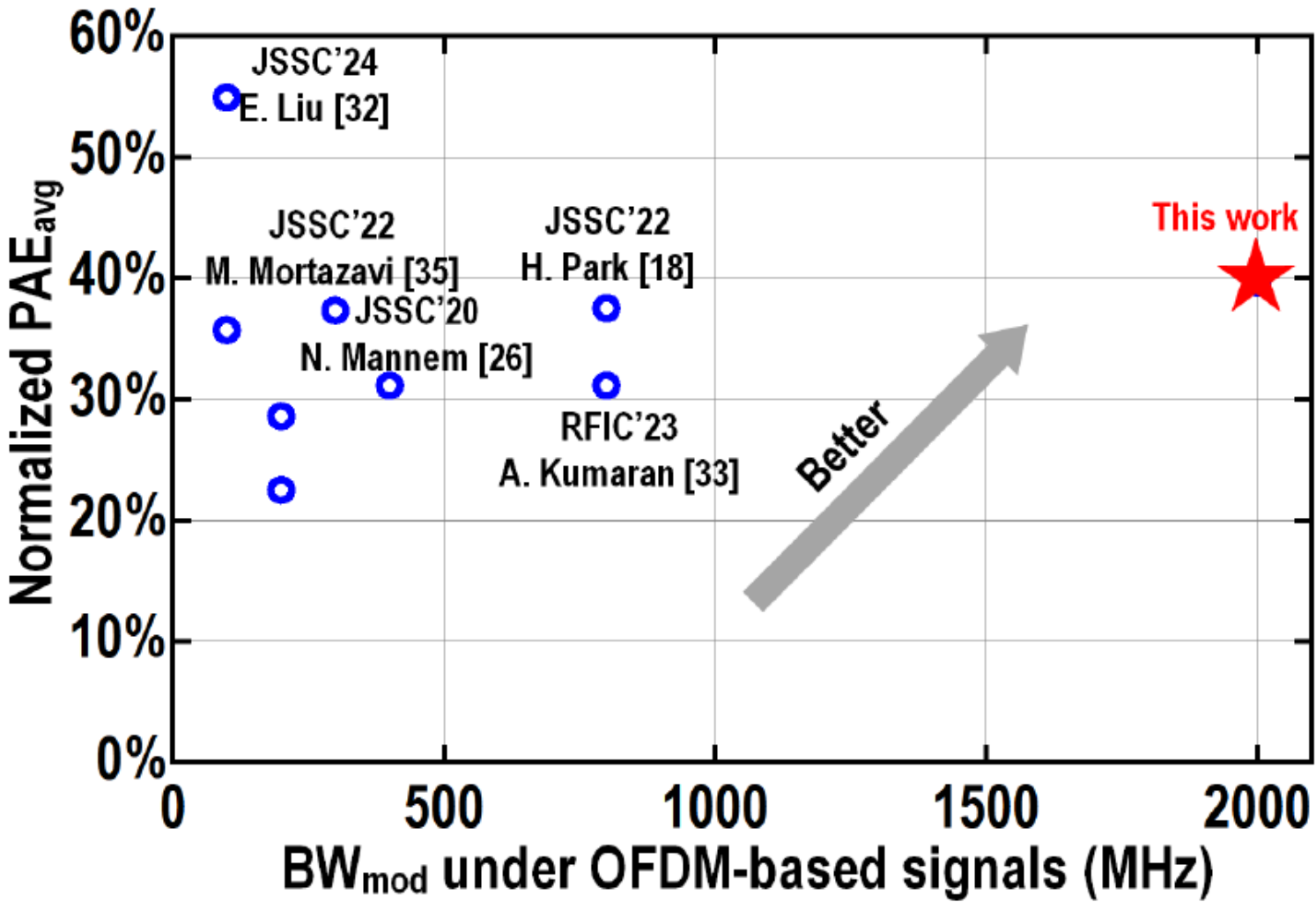


- Normalized PAE = measured PAE / highest PAE_{PEAK} reported at each PA's operating frequency
- Normalized PAE_{PEAK} = 26.8%/34.6% = 77.5% in our work

47-GHz Four-Way Doherty PA: Performance Comparison



Specifications	
Architecture	Tr
Supply (V)	
Frequency (GHz)	
P _{SAT} (dBm)	
P _{1dB} (dBm)	
Modulation Scheme	5G NR 64-QAM
Modulation BW (Hz)	800
PAPR (dB)	9.1
P _{avg} (dBm)	14.1
PAE _{avg}	13.8
Normalized PAE _{avg} [#]	39.5
EVM _{rms} (dB)	-25.2
DPD	
Core Area (mm ²)	



	T. Huang ISSCC'21
Architecture	Continuous-coupler-based 2-way Doherty
Configuration	2 (PA), 1 (DRV)
Frequency (GHz)	47.5
P _{SAT} (dBm)	21.7
P _{1dB} (dBm)	20.9
Modulation Scheme	5G NR 1-CC 64-QAM OFDM
Modulation BW (Hz)	200M
PAPR (dB)	9.64
P _{avg} (dBm)	7.4
PAE _{avg}	7.8%
Normalized PAE _{avg} [#]	22.5%
EVM _{rms} (dB)	-25.2
DPD	No
Core Area (mm ²)	0.62

* Graphically estimated



- Introduction to Passive Networks Design
- Three Transceiver Building Block Design Examples
 - LNA with an Octave Bandwidth
 - Doherty PA with Efficiency Enhancement up to 12-dB Back-off
 - 100-GHz LO Generator with >190 dBc/Hz FoM
- Conclusions



Hao Guo

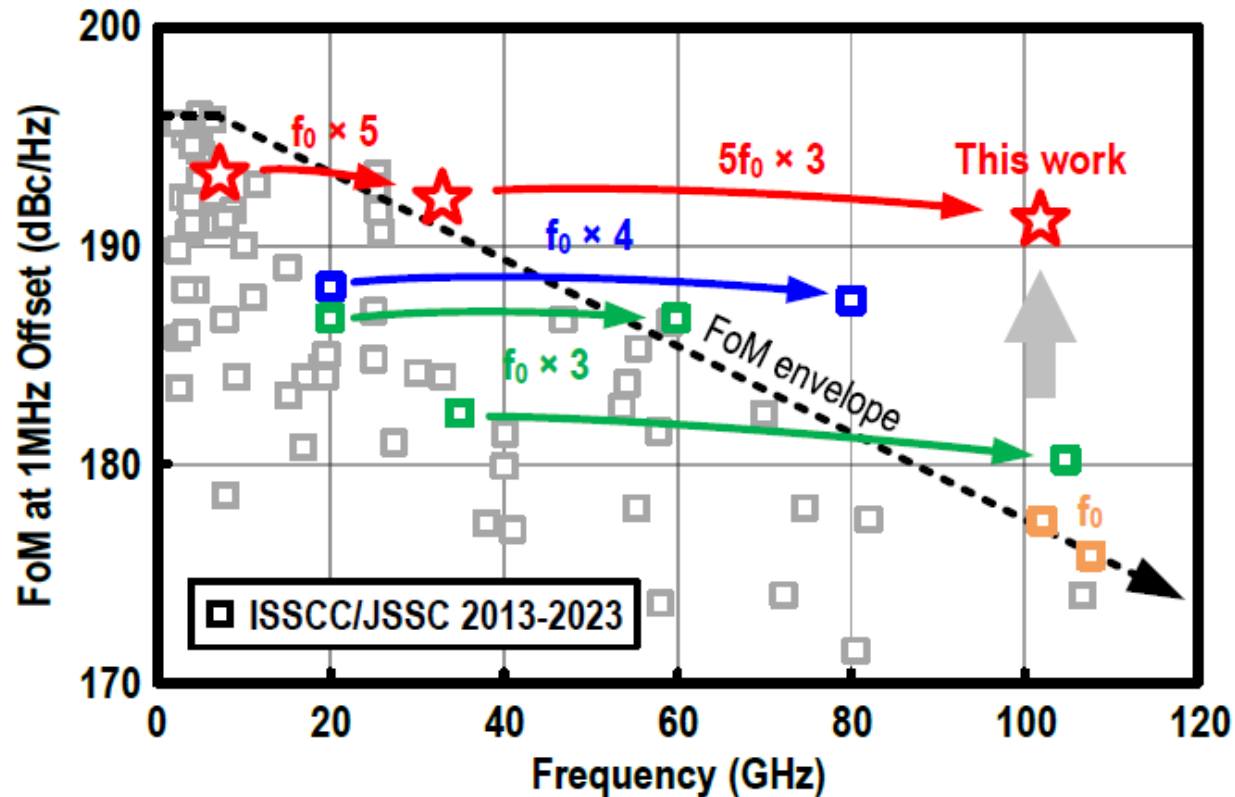
H. Guo and T. Chi, "[An 86.5-105.6GHz LO Generator with Cascaded Implicit Frequency Quintupling and Tripling Achieving -107.7dBc/Hz Phase Noise and 191. 2dBc/Hz FoM at 1MHz Offset,](#)" CICC 2023.



LO Generator Figure of Merit (FoM)

$$\text{FoM} = |\text{PN}| + 20\log_{10}(f_0/\Delta f) - 10\log_{10}(P_{\text{DC}}/1\text{mW})$$

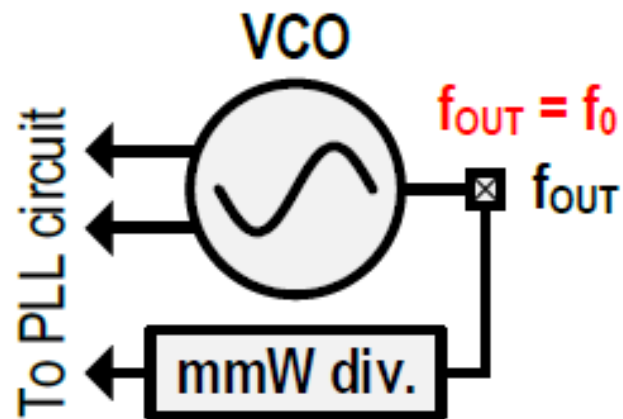
$$\text{FoM}_T = \text{FoM} + 20\log_{10}(\text{TR}/10)$$



- FoM remains flat to ~7 GHz
- Then decreases by ~20 dB/dec
- Best achievable FoM is 180 dBc/Hz at 100+ GHz



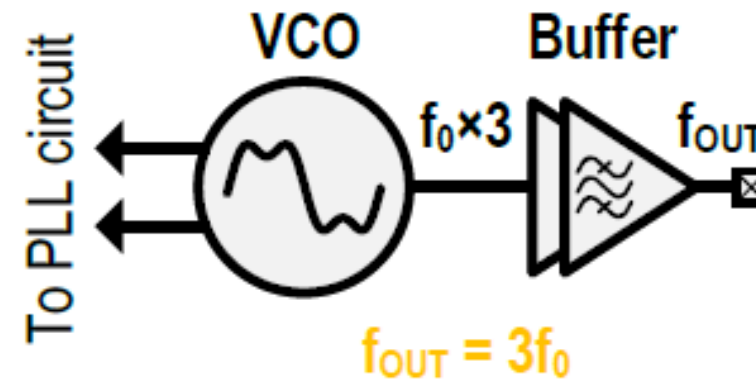
LO Generator Solutions at 100+ GHz



❌ Require W-band VCO

→ FoM↓↓, TR↓↓

❌ May require a mmW divider



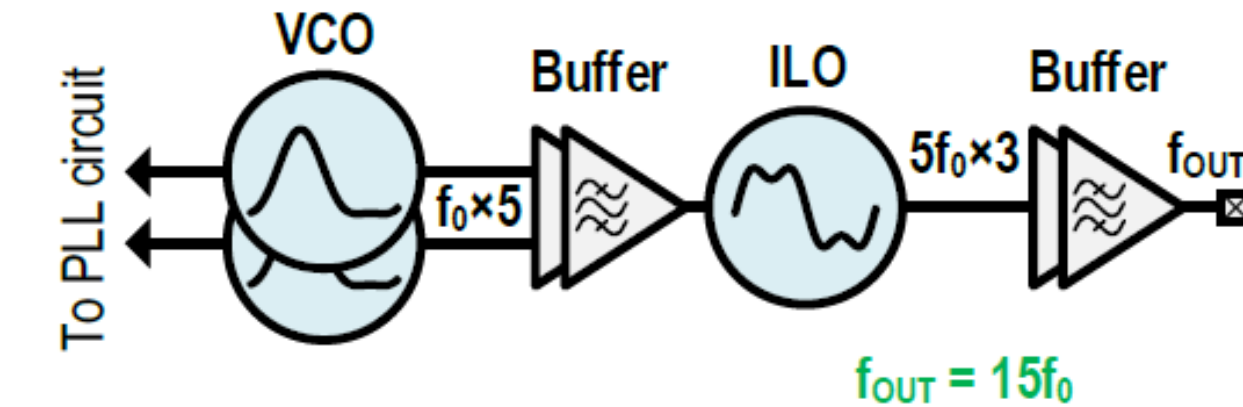
❌ Require mmW VCO

→ FoM↓, TR↓

❌ Strong $V_{D,3f_0}$ compromises FoM → FoM↓



Proposed Frequency Plan for 100+ GHz LO Generator



$$FoM_{VCO} \uparrow - 10\log_{10}\left(1 + \frac{P_{DC,ILO} \downarrow + P_{DC,Buffer} \downarrow}{P_{DC,VCO}}\right) = FoM_{Overall} \uparrow$$

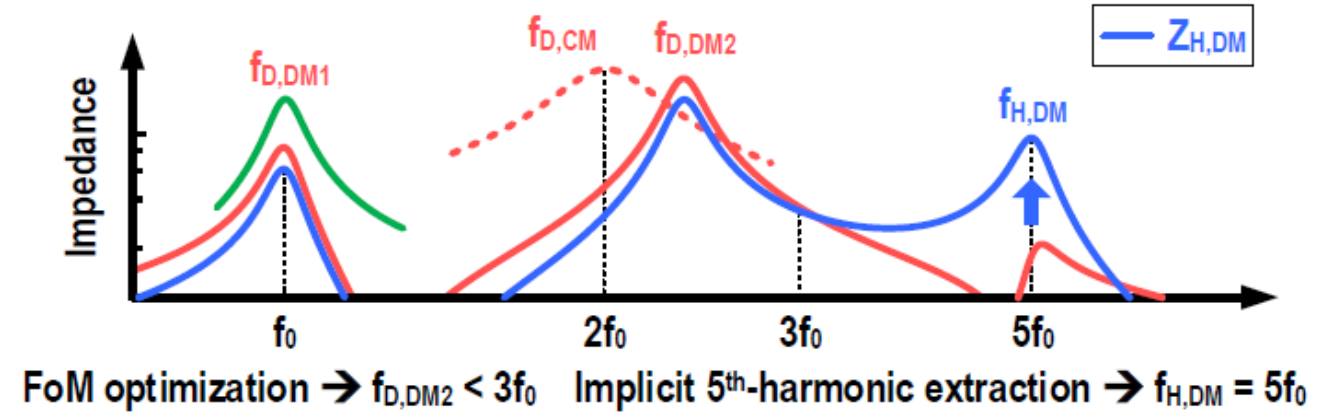
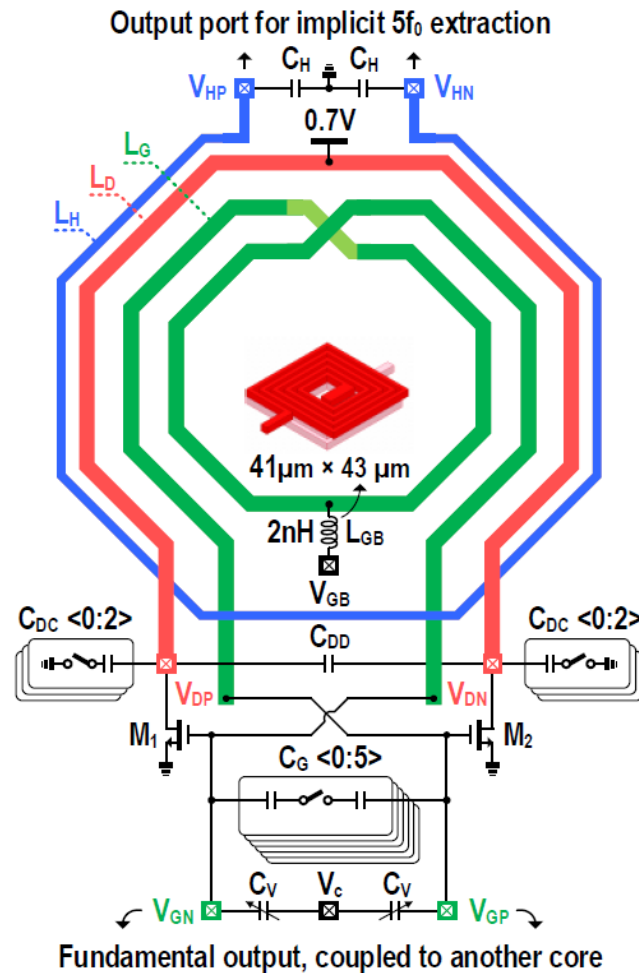
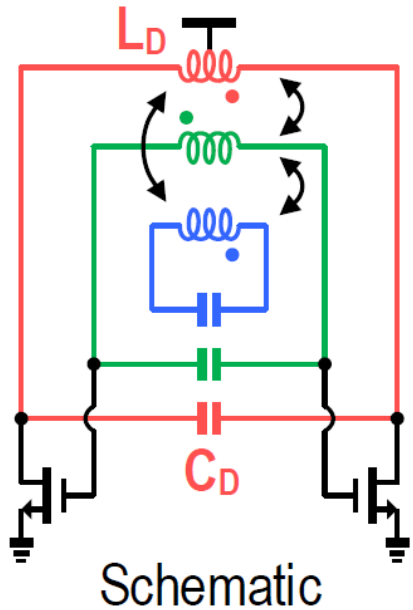
VCO ☒ Implicit harmonic extraction without FoM degradation $\rightarrow FoM_{VCO} \uparrow, 5 \times f_0$

ILO ☒ Extended locking range $\rightarrow$ Robust locking, $P_{DC,Buffer} \downarrow$

☒ Enhanced 3rd harmonic through 2nd harmonic control $\rightarrow P_{DC,Buffer} \downarrow$

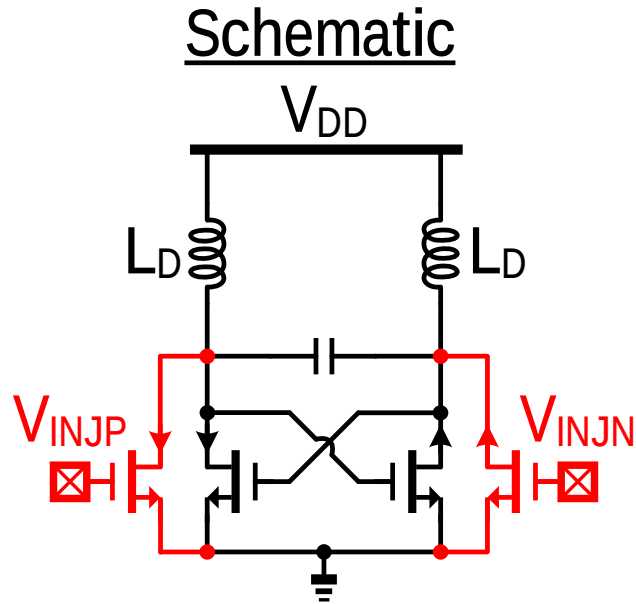
H. Guo and T. Chi, "[An 86.5-105.6GHz LO Generator with Cascaded Implicit Frequency Quintupling and Tripling Achieving -107.7dBc/Hz Phase Noise and 191.2dBc/Hz FoM at 1MHz Offset](#)," CICC 2023.

Fundamental Oscillator Design at 7 GHz

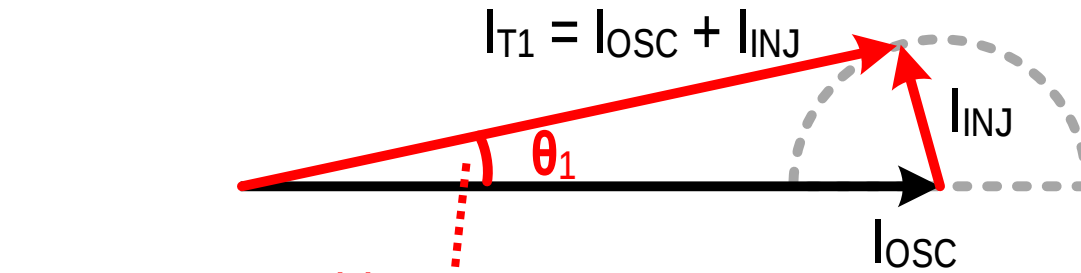


- Coupled L_D and $L_G \rightarrow$ voltage gain from D to G $\rightarrow$ boost effective $g_m \rightarrow$ improve PN
- L_D designed to achieve high impedance at $2f_0 \rightarrow$ improve PN and reduce $1/f^3$ PN corner
- L_H achieves implicit $5f_0$ harmonic extraction

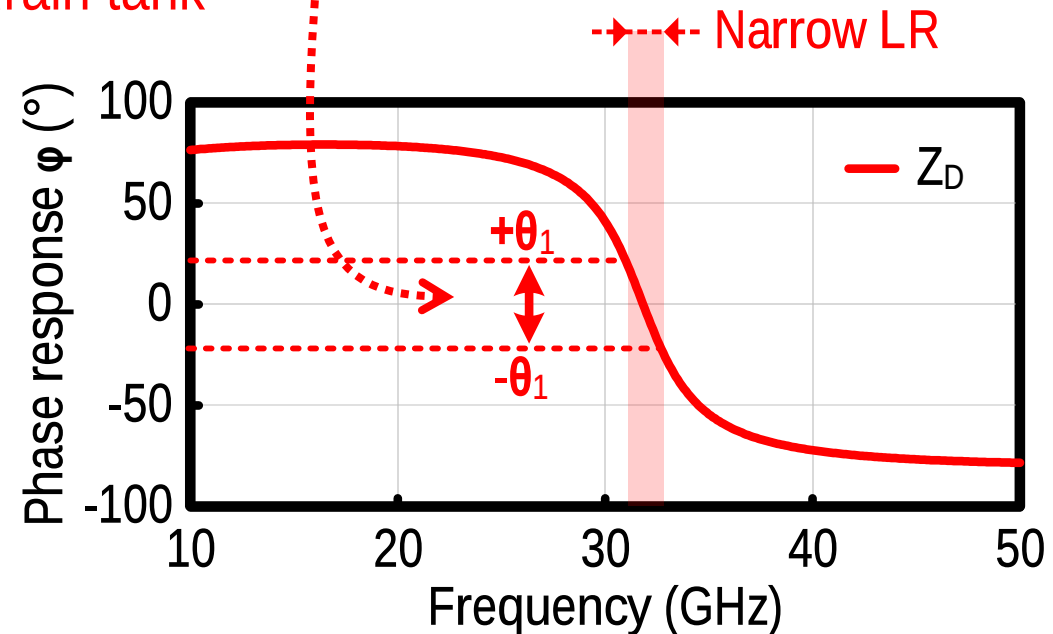
Conventional Injection Locking Oscillator (ILO)



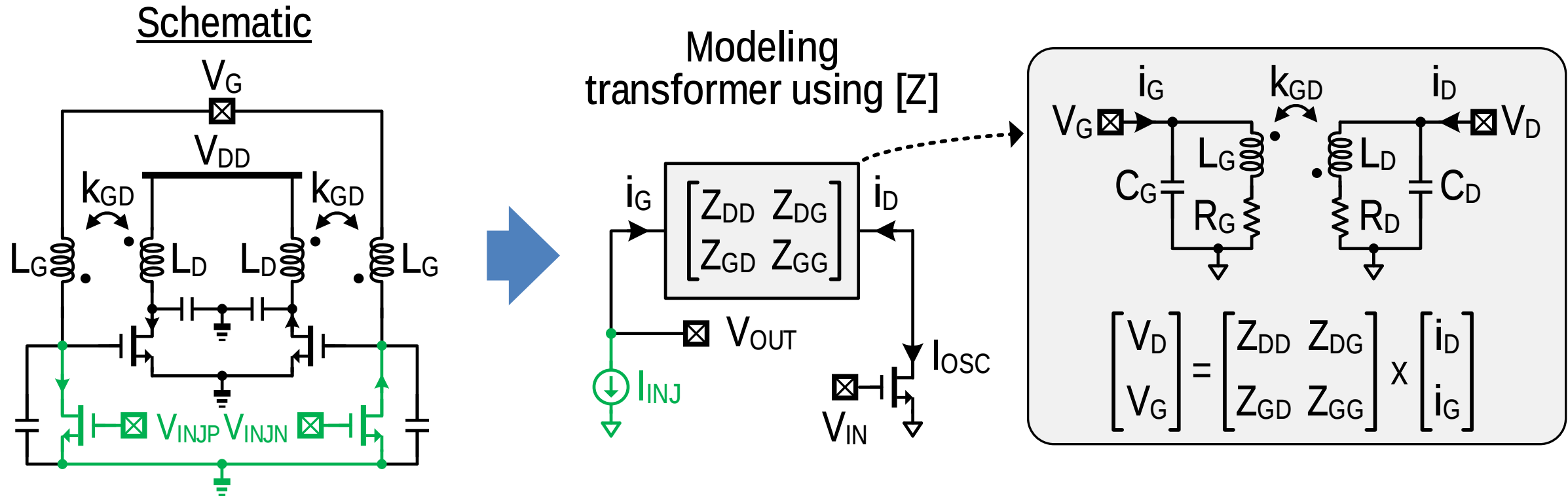
- Key to increase locking range: increase effective I_{INJ}



Compensated by
drain tank



Proposed Gate Injection Scheme Using a Transformer

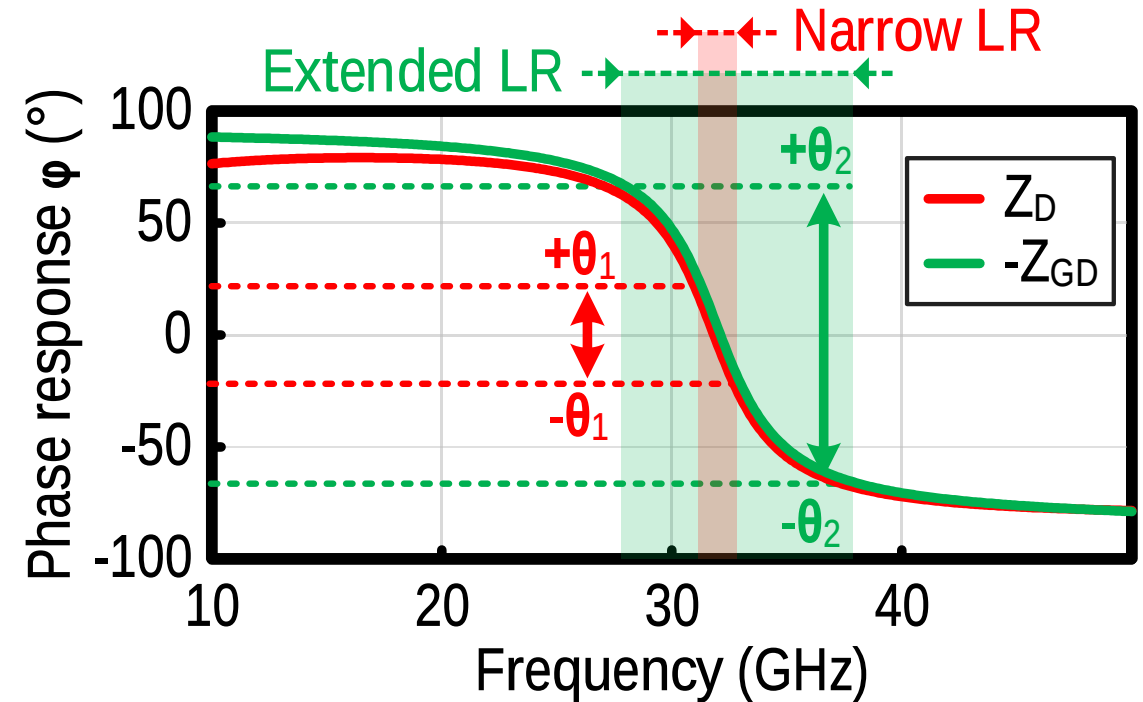
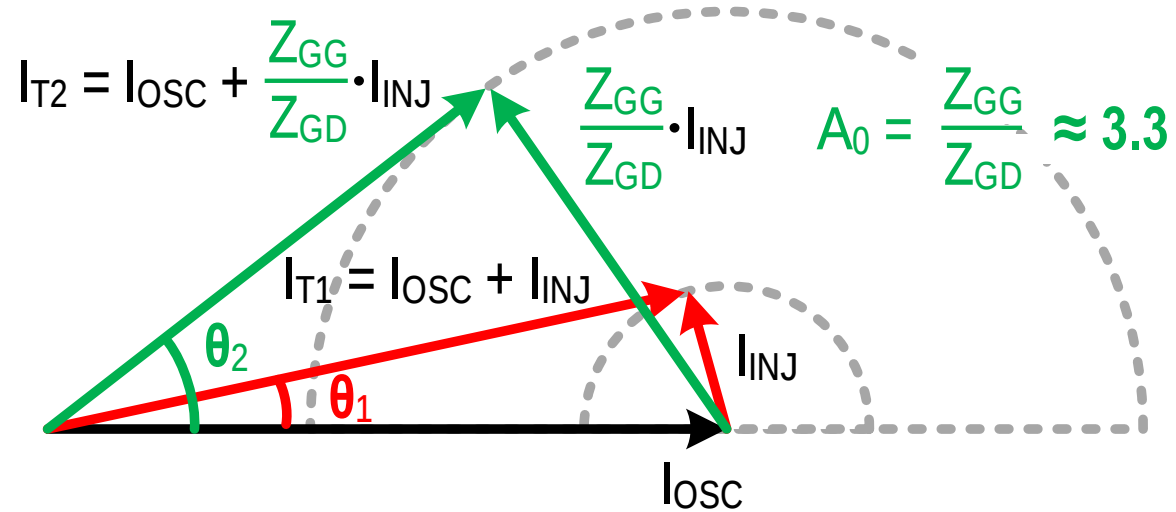


- Current gain from gate to drain $\rightarrow I_{INJ}$ is boosted by a factor of $A_0 = Z_{GG}/Z_{GD}$



Proposed Gate Injection Scheme Using a Transformer

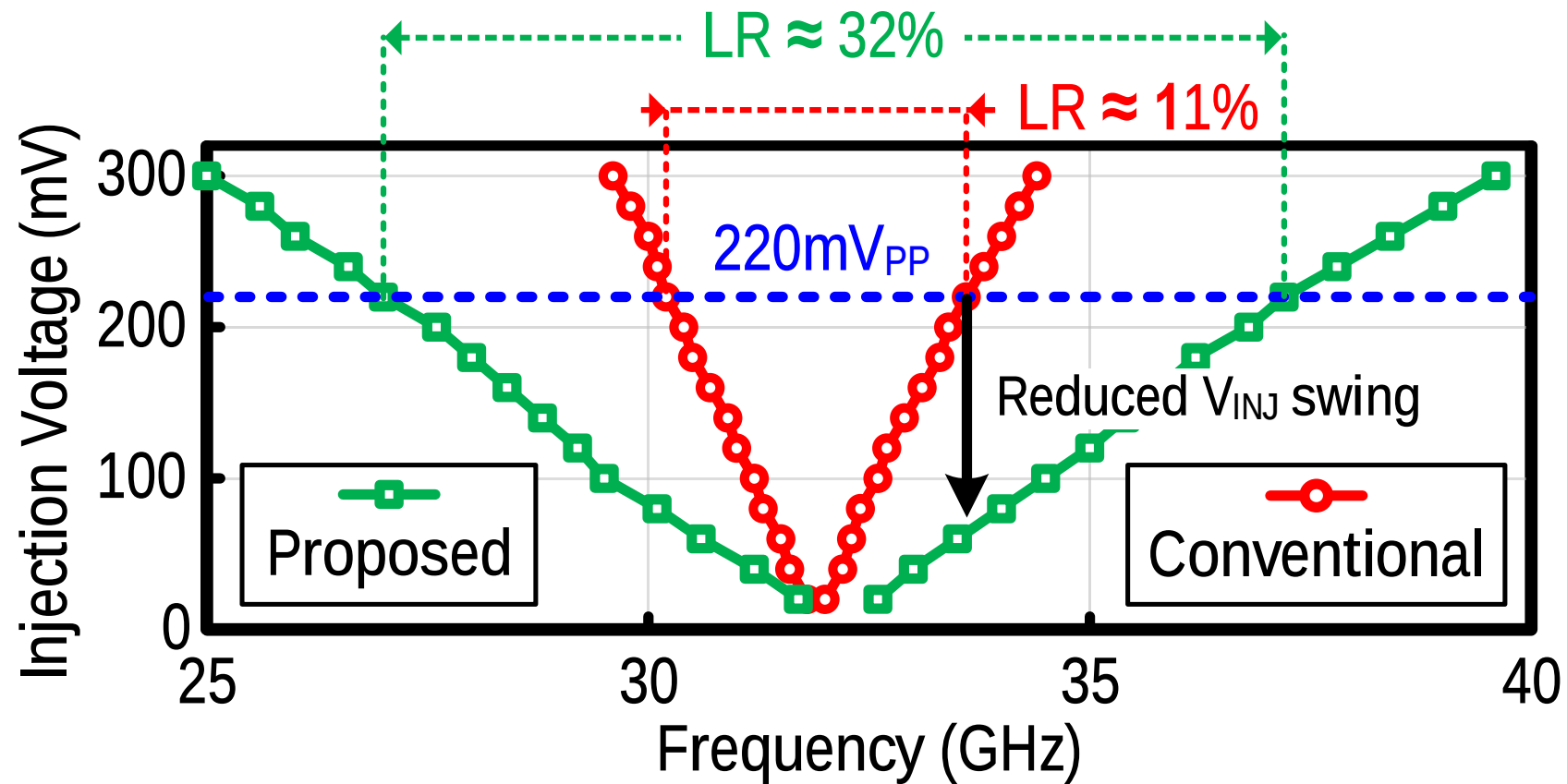
Phasor diagram



😊 Effective I_{INJ} boosted by $Z_{GG}/Z_{GD} \rightarrow$ Extended locking range

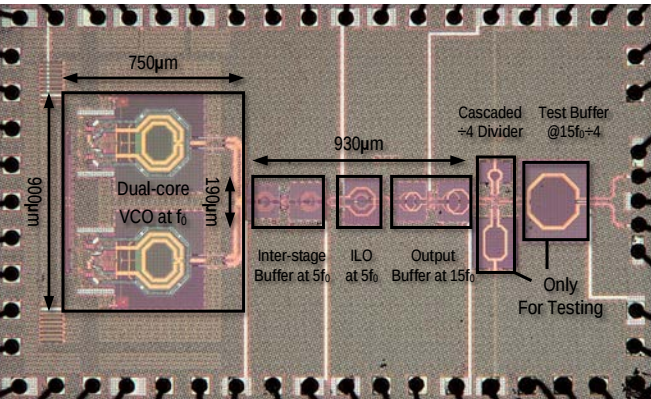


Locking Range Comparison

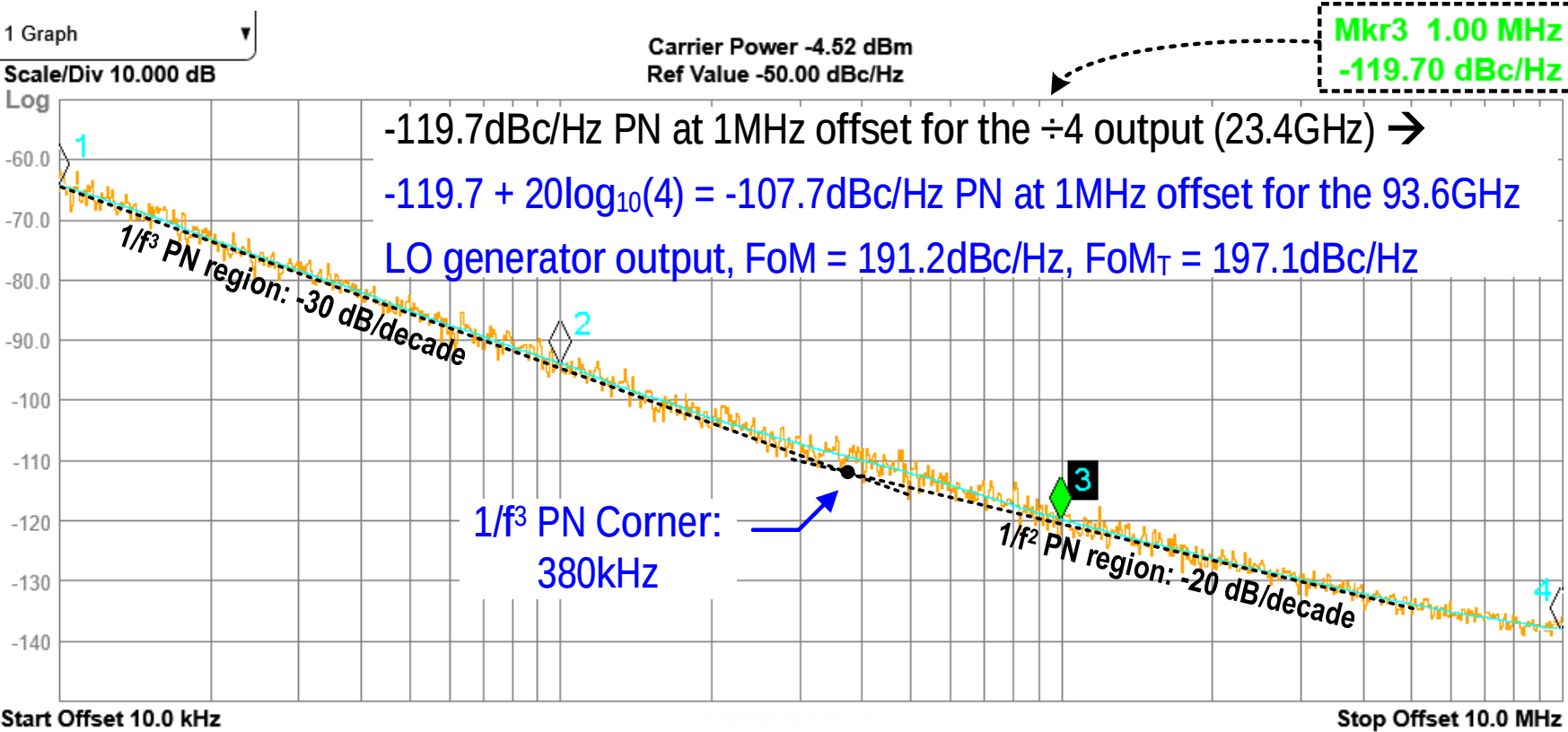




100-GHz LO Generator: Measurement Results



GlobalFoundries 22nm
CMOS SOI



3 Marker Table

	Mode	Trace	X	Y	Function	Function Width	Function Value
1	N	2	10.0 kHz	-64.16 dBc/Hz			
2	N	2	100 kHz	-93.69 dBc/Hz			
3	N	2	1.00 MHz	-119.70 dBc/Hz			
4	N	2	10.0 MHz	-138.00 dBc/Hz			



100-GHz LO Generator: Performance Comparison

	This Work	ISSCC'21	JSSC'19	RFIC'22	JSSC'17	ISSCC'23
Frequency Generation	$f_0 \times 5 \times 3$	f_0	f_0	$f_0 \times 2$	$f_0 \times 4$	$f_0 \times 3$
Frequency (GHz)	86.5 to 105.6	102	82 to 107.6	76 to 82	70 to 81	83.3 to 104.7
Tuning Range (%)	19.9%	3%	27%	7.2%	15%	20.9%
PN (dBc/Hz) @100kHz Δf	-76.9 to -83.9	N/A	N/A	-71.7 * to -75.4	N/A	N/A
PN (dBc/Hz) @1MHz Δf	-102.7 to -108.6	-76.8 *	-86 to -90 &	-101 * to -109.8	-106.5	-98 to -103.1
FoM (dBc/Hz) @1MHz Δf	187.4 to 191.2	167.9	175.8 to 177.5 &	182 * to 188.3	187.5	180.2 to 183.5
FoM _T (dBc/Hz) @1MHz Δf	193.3 to 197.1	157.4	184.4 to 186.1 &	179.2 to 185.5	191	186.6 to 189.9
1/f ³ PN Corner (kHz)	220 to 620	N/A	N/A	300	N/A	630 to 1100
Supply (V)	0.7(VCO + Buffers) /0.4 (ILO)	1.2/0.8	0.8	1	1.2	1.2/0.7
Power (mW)	37.5 to 40.3	8	12	102.7	50	63.7 to 65.5
Core Area (mm ²)	0.85	0.01 *	0.06 *	0.09	0.6	0.42
Technology	22nm CMOS SOI	65nm CMOS	65nm CMOS	65nm CMOS	55nm BiCMOS	65nm CMOS

FoM = -PN + 20log₁₀(f₀/Δf)-10log₁₀(P_{DC}/1mW) FoM_T = FoM + 20log₁₀(TR/10) * Graphically estimated

& Only 10MHz data are reported. The metrics at 1MHz are estimated by PN_{@1MHz Δf} = PN_{@10MHz Δf} + 20dB



- The role of passive network designs is becoming increasingly important
 - Three design examples of LNA, PA, and LO generator
 - Performance improvement + new functionalities
- Transformer is a powerful tool
 - Equivalent lumped circuit model

Acknowledgment



- GlobalFoundries for chip fabrication
- Sponsors: NSF, Samsung Research America, Keysight
- Members of Rice Integrated Systems and Electromagnetics (RISE) Lab, especially Yaolong Hu, Xiaohan Zhang, and Hao Guo

Thank you!

<https://chilab.info/>

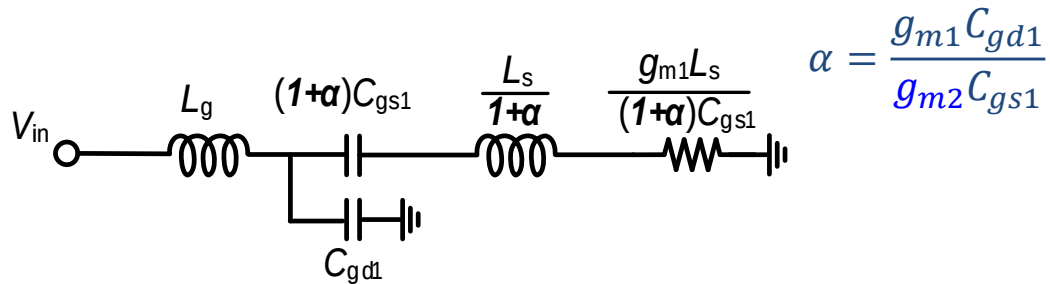
Backup Slides





Octave-Bandwidth LNA: Dual-Resonant Input Matching

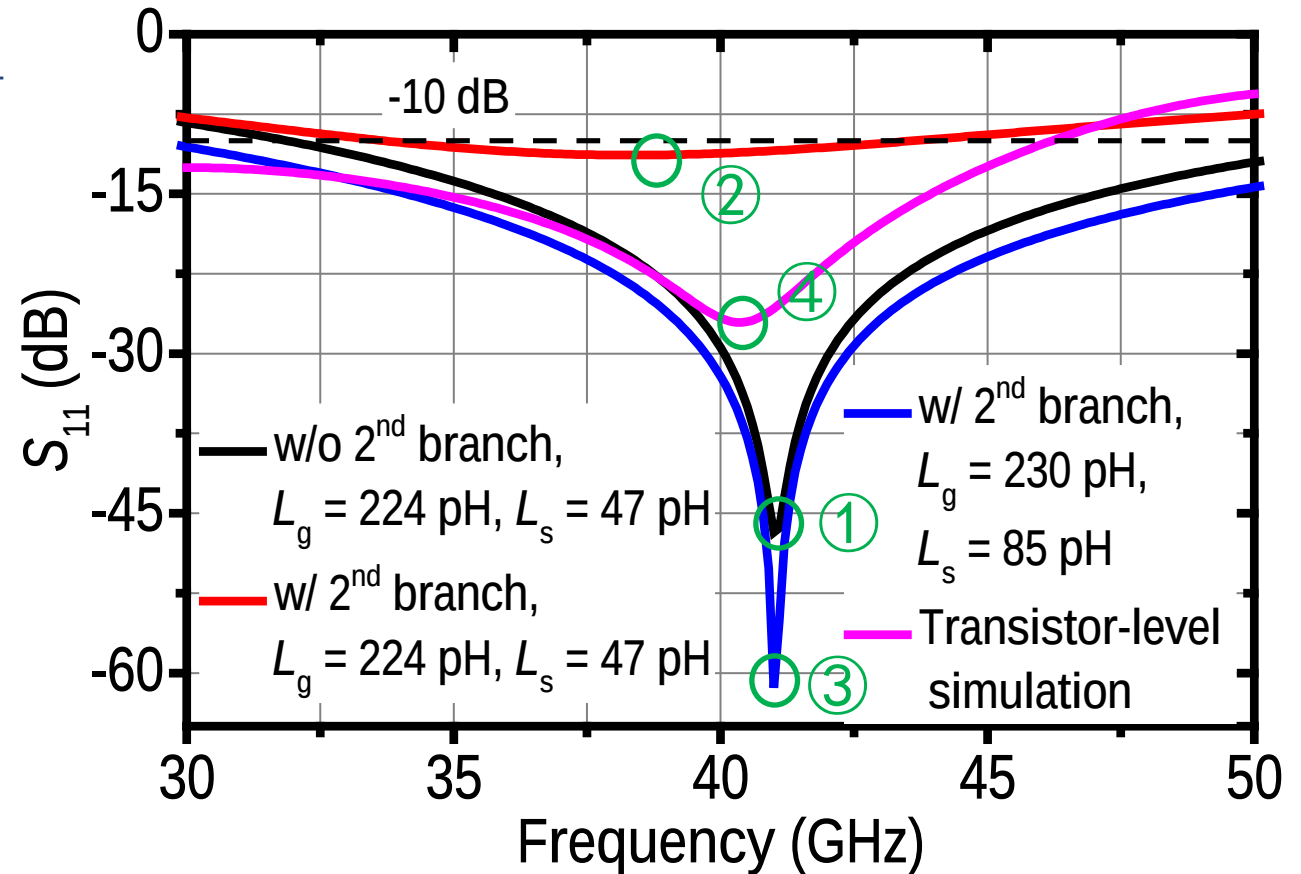
- Design Procedure: rigorous approach



$\text{Im}\{Z_{in}\} \sim 0$ at $\omega_H \rightarrow$ solve L_g

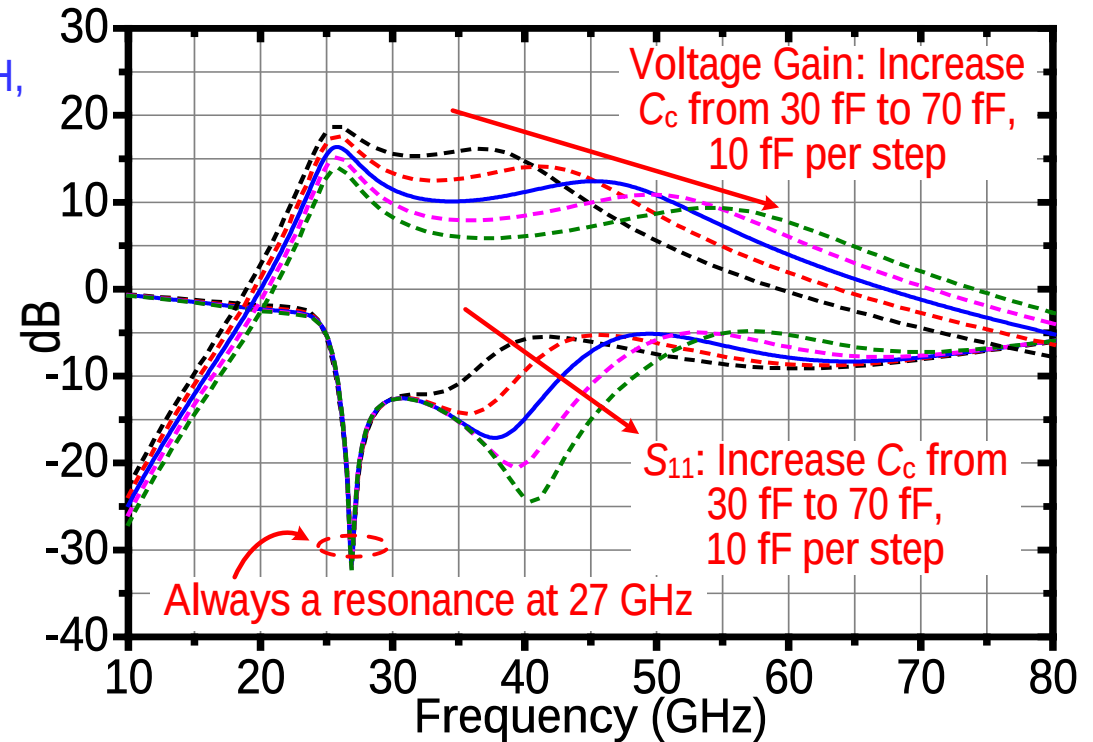
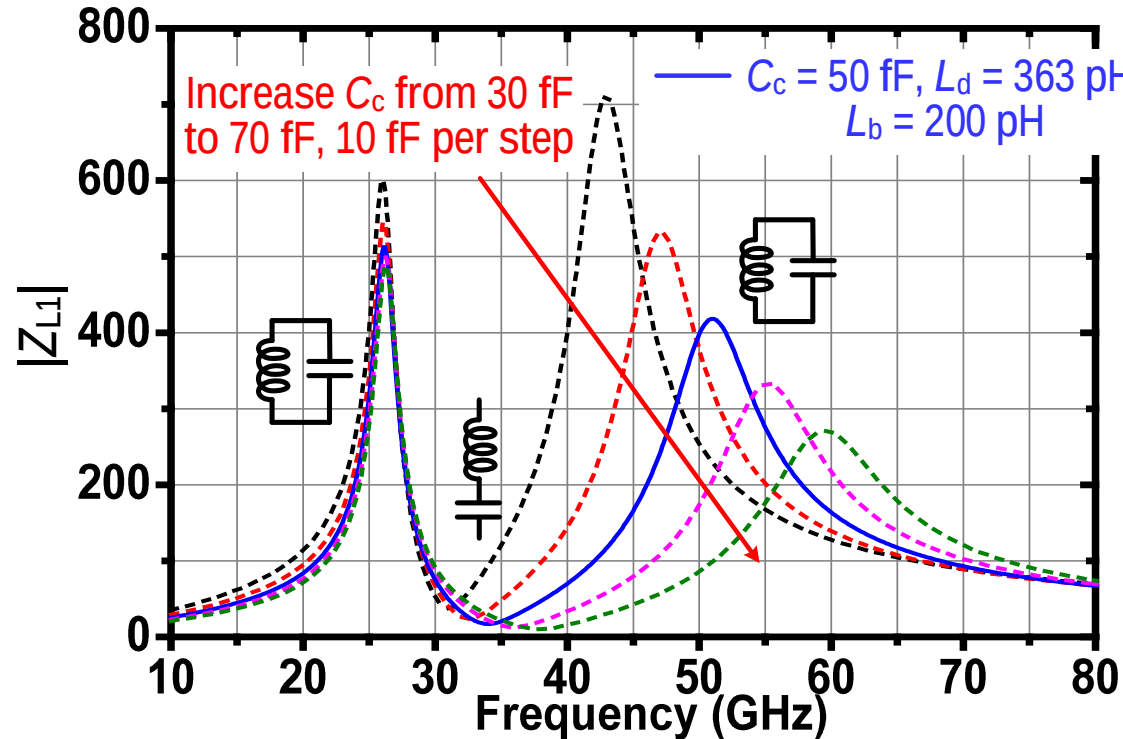
$\text{Re}\{Z_{in}\} \sim 50 \Omega$ at $\omega_H \rightarrow$ solve L_s

- 3 variables (L_g , L_s , and g_{m2}) and two equations $\rightarrow$ a family of solutions



Octave-Bandwidth LNA: Dual-Resonant Input Matching

• Design Procedure

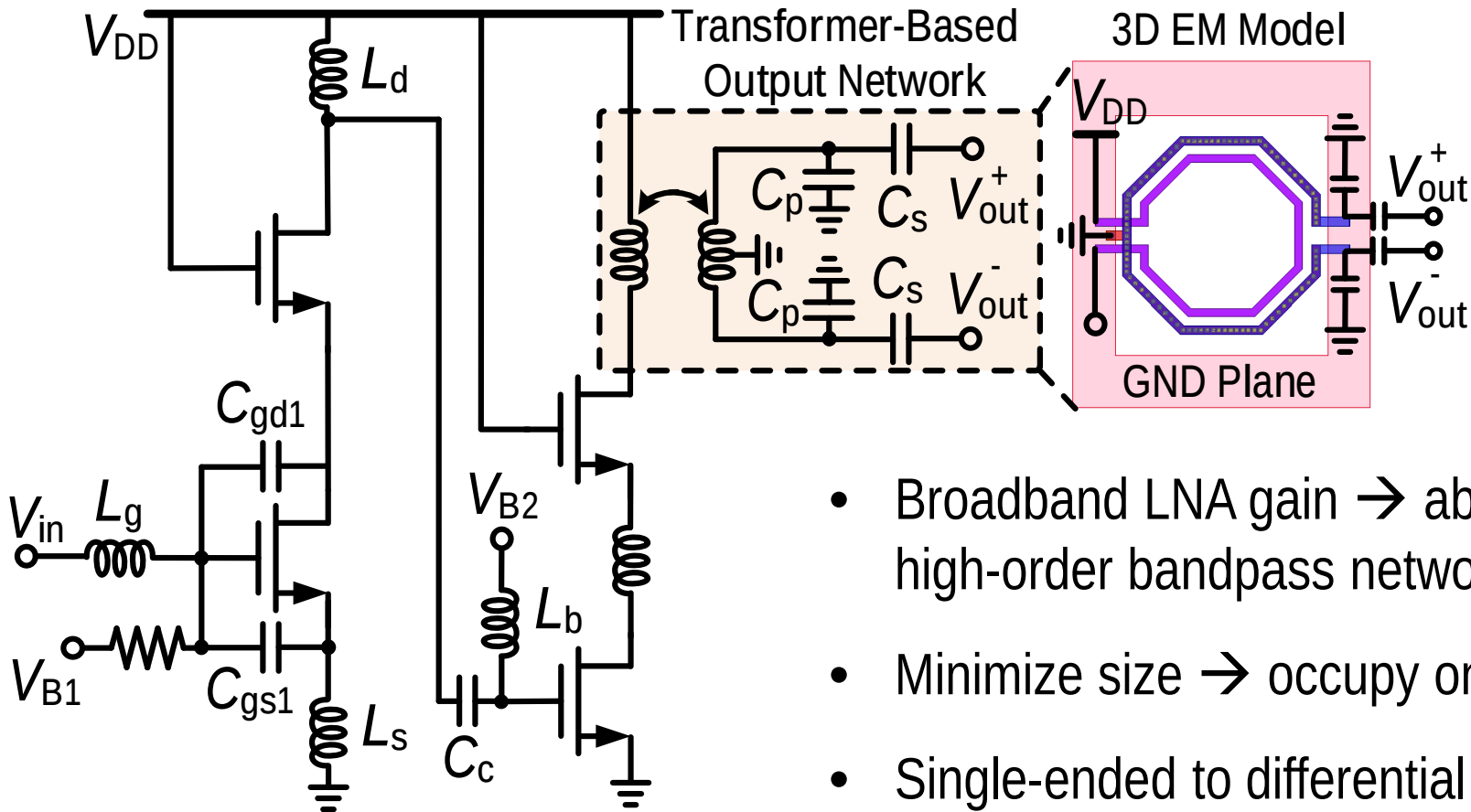


- C_c is small $\rightarrow |Z_{L1}|$ is not low impedance around $\omega_H \rightarrow S_{11}$ dual-resonance disappears;
- C_c is large $\rightarrow 2^{\text{nd}}$ gain shape move to higher frequency $\rightarrow$ overall LNA BW shrinks;
- $C_c = 50$ fF, $L_d = 363$ pH, and $L_b = 200$ pH $\rightarrow$ staggered gain tuning to extend the BW;



Octave-Bandwidth LNA: Output Network

- Motivation: Broadband Output Network to Extend 3-dB Gain Bandwidth



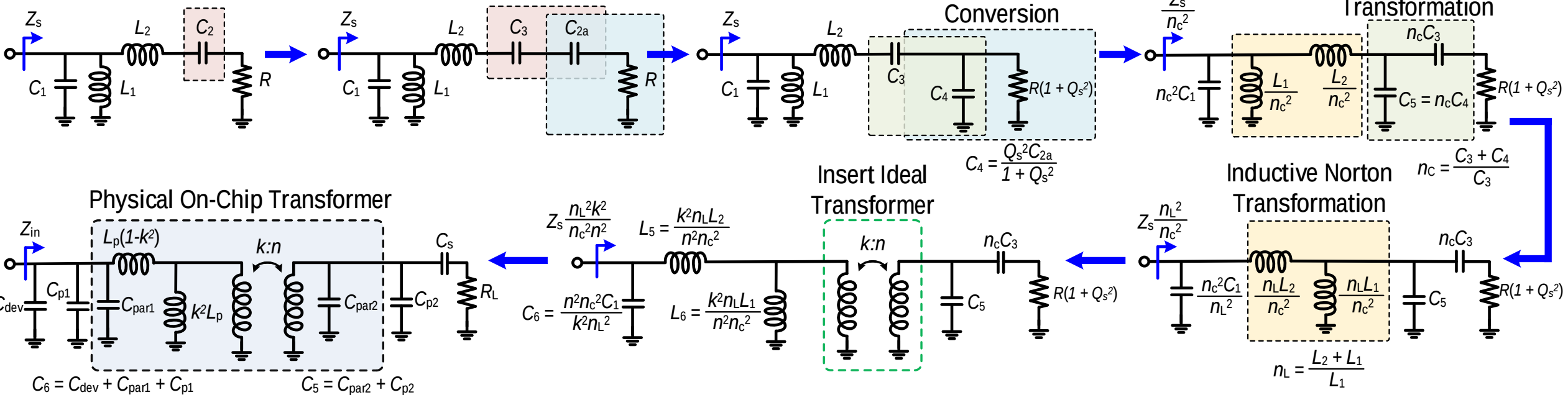
- Broadband LNA gain $\rightarrow$ absorb parasitic cap into a high-order bandpass network
- Minimize size $\rightarrow$ occupy one inductor footprint
- Single-ended to differential conversion



Octave-Bandwidth LNA: Output Network

- Two-Step Norton Transformation

Canonical 2nd-Order Bandpass



- Key: Realize 2nd order band pass filtering with single on-chip transformer
- Series capacitor at the secondary



Octave-Bandwidth LNA: Output Network

- Design Flow

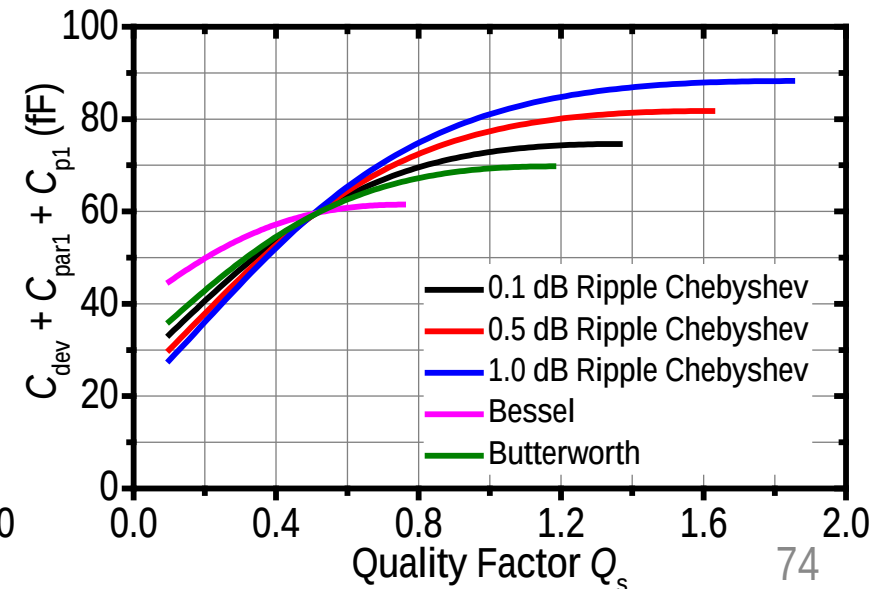
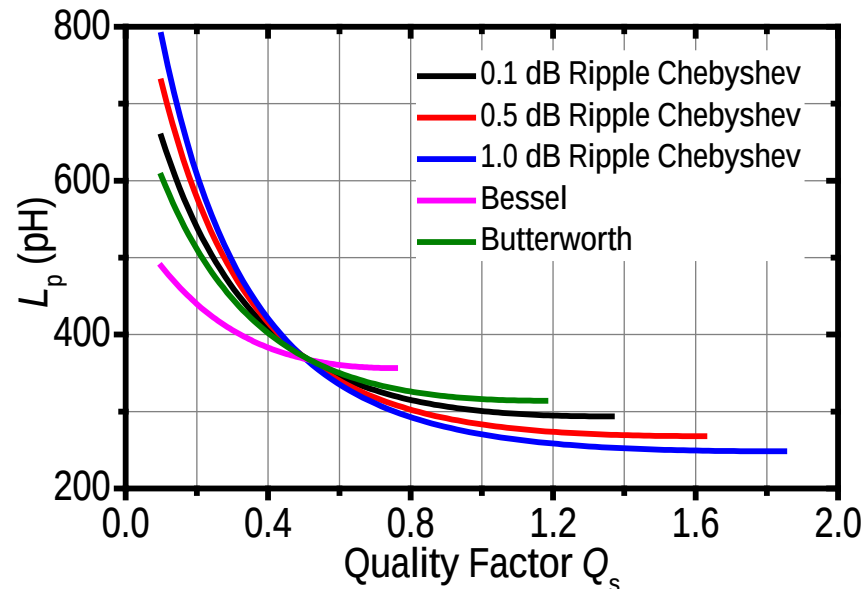
Step 1: Specify center frequency ω_0 , fractional bandwidth Δ and bandpass prototype (g_1, g_2)

Step 2: Determine required k , upper bound of quality factor $Q_s \leq g_2/\Delta$

Step 3: Calculate L_p , C_{par1} , C_{par2} , C_s , and Z_{in} vs. Q_s (low Q_s preferred)

Simulation Condition: $\Delta = 0.6$, $\omega_0 / 2\pi = 34.0$ GHz, $\omega_1 / 2\pi = 25.6$ GHz, $\omega_2 / 2\pi = 46$ GHz, $R_L = 100 \Omega$, $n = 1.0$

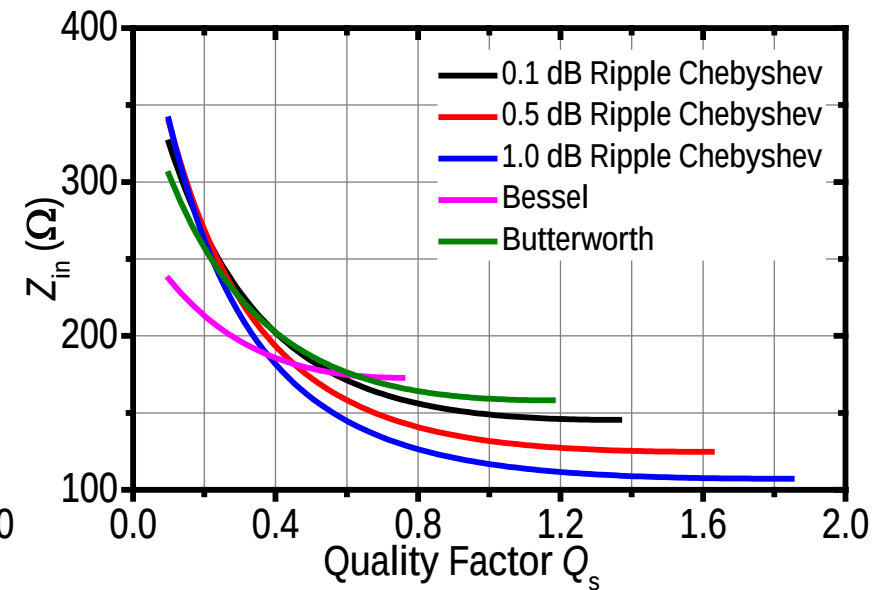
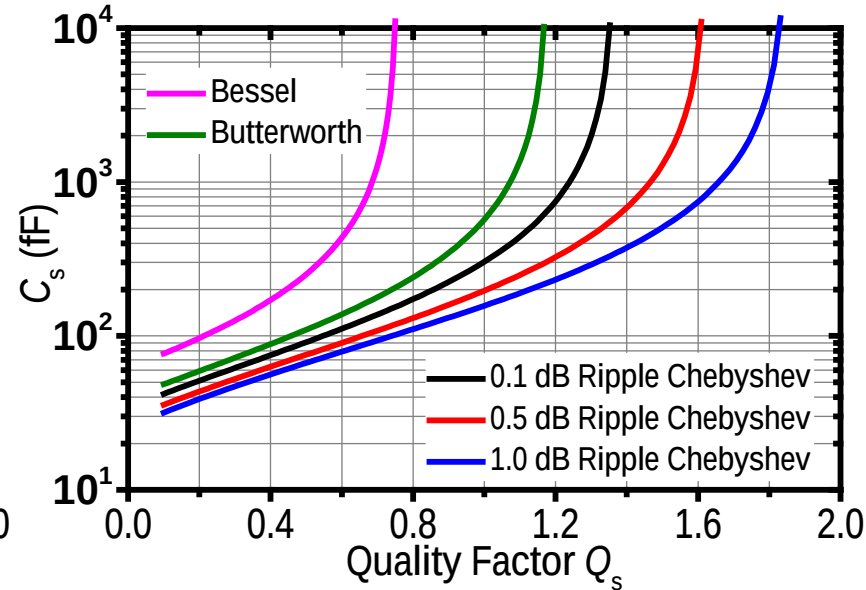
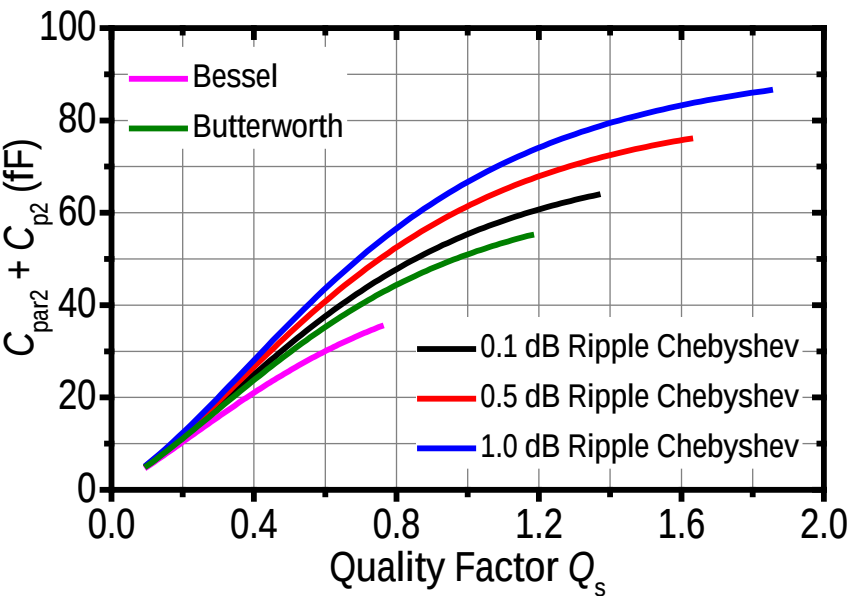
Network Prototype	k
0.1 dB Ripple Chebyshev	0.49
0.5 dB Ripple Chebyshev	0.47
1.0 dB Ripple Chebyshev	0.46
Bessel	0.61
Butterworth	0.51





Octave-Bandwidth LNA: Output Network

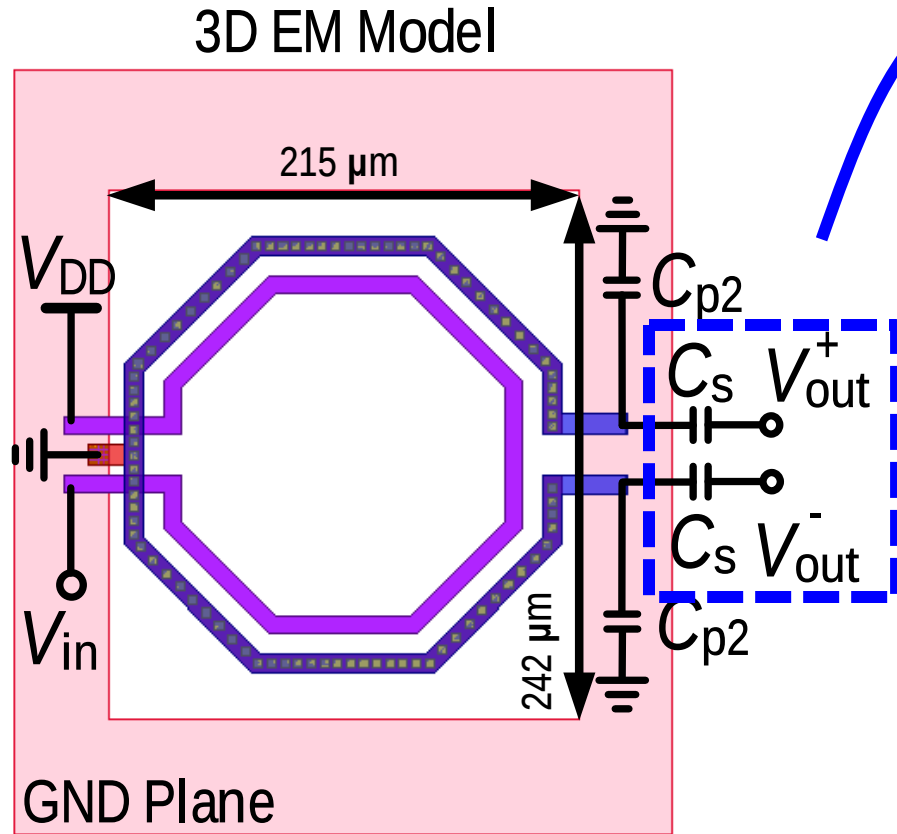
- Design Flow



Step 4: Based on calculations in Step 3, design a physical transformer that can satisfy all the parameters; if not, go back to Step 1/2, increase Q_s or revise g_1 , g_2 , or Δ

Y. Hu and T. Chi, "[A Systematic Approach to Designing Broadband Millimeter-Wave Cascode Common-Source With Inductive Degeneration Low Noise Amplifiers](#)," T-CAS I 2023.

Octave-Bandwidth LNA: Output Network



- Minimal ripple vs. commonly used magnetically coupled resonators

